

Q – BANK

April 2017

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PC-1 Exams

BSOF Final Exam 06/07 Batches (Same exam)

1. Which of the following statement **is not** correct concerning circulatory system?
 - A. Capillaries are always located at the end of arterioles and beginning of venules
 - B. Exchange of substances with ECF occurs at the capillary bed
 - C. Arteries in the retina are true terminal arteries
 - D. Pulmonary veins are atypical veins
 - E. Lymphatic plexuses are absent in some tissues of the body
 2. Which of the following shows the appropriate respective section thickness for light and electron microscope?
 - A. 5-10 μm and 40-70 nm
 - B. 40-70 nm and 5-10 μm
 - C. 25-50 μm and 100-200 nm
 - D. 100-200 nm and 25-50 μm
 - E. 25-50 mm and 100-200 μm
 3. Which of the following is a commonly used clearing agent in tissue section preparation for light microscopy?
 - A. Ethanol
 - B. Methanol
 - C. Paraffin wax
 - D. Formalin
 - E. Xylene
 4. In hematoxylin and eosin method of the general tissue section staining, hematoxylin
 - A. Acts like an acidic dye
 - B. Stains best the ribosomes
 - C. Stains blue black
 - D. A and B
 - E. B and C
 5. Which one of the following is the correct order of tissue block preparation from an autopsy tissue?
 - A. Dehydration, fixation, clearing, infiltration, & embedding
 - B. Dehydration, clearing, infiltration, fixation & embedding
 - C. Fixation, dehydration, clearing, infiltration, & embedding
 - D. Fixation, dehydration, infiltration, clearing & embedding
 - E. Clearing, dehydration, fixation, infiltration, & embedding
- Ans = C (Mnemonics - **F**ind **DC** In **E**ast)
6. Among the different lenses in bright field microscope, which one functions in resolution of a tissue image?
 - A. Objective lens
 - B. Ocular lens
 - C. Condenser lens
 - D. A and B
 - E. B and C

7. To focus a blurred tissue image seen under a bright field microscope, you normally need to:
- A. Change the objective lens to the appropriate one
 - B. Move the stage carrying the tissue slide up and down**
 - C. Move the objective lens up or down
 - D. Move the condenser lens up or down
 - E. Move the ocular lens up or down
8. A man having a total body weight of 90 kg could have _____ liter of plasma water.
- A. 54.00**
 - B. 18.00
 - C. 4.70
 - D. 3.86
 - E. 28.25
9. Which of the following doesn't characterize osmosis?
- A. Occurs if concentration variation is there between compartments
 - B. Water molecules move from high to low water molecules
 - C. Water molecules move from hypotonic to hypertonic solution
 - D. Sustained until isotonic solution is built
 - E. Energy drives water from high to low water molecules**
10. Which of the following is true about fluid exchange between compartments?
- A. The more the tissue colloid osmotic pressure, the less the fluid that gets to the tissue
 - B. The more the blood osmolarity, the more the fluid that gets into the tissue
 - C. Cell in the hypotonic solution loses water from intracellular compartment**
 - D. The more the capillary permeability, the less the fluid that gets into the tissue
 - E. Osmotic pressure doesn't affect the fluid exchange between compartments.
11. Which of the following is true about the organization of the human body?
- A. Only one type of tissue is found in one organ
 - B. Cells in one type of tissue are different in function
 - C. Unrelated organs with function form one system
 - D. Atoms are functional components of the body**
 - E. Human body is merely the sum of its parts
12. Select the one which is true about the cells
- A. Prokaryotic cells have organelles in different compartments
 - B. Membrane bounded organelles are found in prokaryotic cells
 - C. More than one chromosomes are found in one prokaryotic cells
 - D. Organelles are found in the different compartments in eukaryotic cells**
 - E. The outer and the inner membranes of eukaryotic cells are identical
13. Which of the following is true about the integral proteins in the cell membrane?
- A. Easily removed without destroying the cell membrane
 - B. Serve as ion channels and receptors**
 - C. Found in the inner membrane facing towards the cytoplasm
 - D. Do not cross the whole cell membrane
 - E. Do not penetrate the hydrophobic core of the membrane
14. Which of the following is not the function of the membrane proteins?
- A. Signal transduction

- B. Communication
 - C. Recognition
 - D. Transportation
 - E. None of the above
15. Why does the nucleus control cell activities?
- A. Because it's found at the center of the cytoplasm
 - B. Because it is the largest organelle
 - C. Because it contains genetic material
 - D. Because it has double membrane
 - E. Because it has a nuclear envelope
16. The plane that separates the body into unequal right and left sides is
- A. Transverse plane
 - B. Midsagittal plane
 - C. Parasagittal plane
 - D. Frontal plane
 - E. Coronal plane
17. The only bone that doesn't articulate with any other bone is:
- A. Sternum
 - B. Nasal bone
 - C. Vomer
 - D. Hyoid bone
 - E. None of the above
18. Which of the following bone is not part of appendicular skeleton?
- A. Scapula
 - B. Hip bone
 - C. Clavicle
 - D. Humerus
 - E. Sacrum
19. A unique fibrous type of joint between a tooth and its socket is called:
- A. Suture
 - B. Syndesmoses
 - C. Gomphosis
 - D. Symphysis
 - E. None of the above
20. Which of the following is not a factor influencing synovial joint stability?
- A. The nature of the articular surfaces
 - B. The number and positioning of the ligaments
 - C. The tone and strength of the muscles acting upon the joint
 - D. The type of the joint
 - E. None of the above
21. A type of synovial joint where a cylindrical shaped projection of one bone fits into a trough shaped surface of another bone with movement occurring within a single plane is called:
- A. Plane
 - B. Hinge

- C. Condylloid
 - D. Saddle
 - E. None of the above
22. Which one describes a condition in which independent bones fuse with other bones during development and remain as part of that bone?
- A. Pressure epiphysis
 - B. Traction epiphysis
 - C. Apophysis
 - D. Atavistic epiphysis
 - E. None of the above
23. The following are connective tissue layers around a skeletal muscle. Identify the correct order from internal to external:
- A. Endomysium – perimysium – epimysium
 - B. Epimysium – perimysium – endomysium
 - C. Perimysium – endomysium – epimysium
 - D. Endomysium – epimysium – perimysium
 - E. None of the above
24. The ability of muscle tissue to return to original shape and length after contraction and stretching is known as:
- A. Extensibility
 - B. Excitability
 - C. Contractility
 - D. Elasticity
 - E. All of the above
25. The muscle fiber arrangement that we find around the openings of the body is:
- A. Convergent
 - B. Parallel
 - C. Sphincter
 - D. Pennate
 - E. All of the above
26. All of the following cranial nerves carry sensory, motor and parasympathetic fibers except:
- A. Facial nerve
 - B. Oculomotor nerve
 - C. Glossopharyngeal nerve
 - D. Vagus nerve
 - E. None of the above
27. A membrane which covers body cavities confined to the internal surface of the body is:
- A. Cutaneous membrane
 - B. Meningeal membrane
 - C. Serous membrane
 - D. Mucus membrane
 - E. All of the above
28. The membrane of which of the following organelle is continuous with the outer nuclear membrane?

- A. Centrioles
 - B. Microtubule
 - C. Golgi apparatus
 - D. Rough endoplasmic reticulum
 - E. Smooth endoplasmic reticulum
29. Vesicles produced from rough endoplasmic reticulum will be transported to which organelle?
- A. Smooth endoplasmic reticulum
 - B. Lysosome
 - C. Peroxisome
 - D. Golgi apparatus
 - E. Microtubule
30. Which of the following is **not** a function of smooth endoplasmic reticulum?
- A. Synthesis of steroid hormones
 - B. Detoxification
 - C. Storage and release of calcium ions in the striated muscle cells
 - D. All of the above
 - E. None of the above
31. Which of the following is wrong about microtubules?
- A. Are made from 13 protofilaments arranged circumferentially
 - B. Could increase in length but not in width
 - C. Transport vesicles within their luminal space from one region to another region in the cell
 - D. All of the above
 - E. None of the above
32. Regarding the nucleolus, which of the following statements is true?
- A. Is acidophilic
 - B. Is the site of mitochondrial synthesis
 - C. Is membrane bound region within the nucleus
 - D. Is made by regions of chromatin contributed by different chromosomes
 - E. Is not visible under the light microscope
33. Organelles made by stalk of 3-10 discrete flattened and slightly curved bag-like channels or cisternae are:
- A. Mitochondria
 - B. Smooth endoplasmic reticulum
 - C. Peroxisome
 - D. Golgi bodies
 - E. Microtubules
34. During the cell cycle,
- A. Duplication of centrioles begins and completes within the G_1 phase
 - B. The S phase is the longest and most variable phase
 - C. Daughter cells grow to the size of the parent cell during G_1 phase
 - D. Synthesis of RNA and proteins occurs mainly during the S phase
 - E. Replications of the chromosomes occurs during G_2 phase
35. Which of the following is true about the transportation process across the cell membrane?
- A. Conformational change is the property of permeases

- B. Energy brings conformational changes on the leaky ion channels
 - C. ATP is the direct energy in secondary active transport
 - D. All carrier mediated transport is active
 - E. Binding site on the channels are specific
36. All of the following increase the diffusion rate of a substance across the cell membrane except one:
- A. The more the concentration gradient
 - B. The more the permeability of the membrane
 - C. The more the temperature of the substance
 - D. The more the thickness of the membrane
 - E. The smaller the size of the substance
37. Which of the following is true about the Van Hoff's law?
- A. The more the number of osmotically active particles, the less the osmotic pressure
 - B. The number of the osmotically active particle for non-ionized chemical is one
 - C. The osmotically active particle of urea is greater than that of sodium chloride
 - D. Addition of calcium chloride into a solution reduces the osmotic pressure of the solution
 - E. The osmotic pressure of a solution with calcium chloride is less than sodium chloride with the same concentration

Given a solution of 0.4%Na₂PO₄ with the following atomic weight: Na⁺ = 23, P = 31, O = 16. Then answer the following questions:

38. Osmolarity of the solution
- A. 0.1135 osm/L
 - B. 0.0850 osm/L
 - C. 0.0567 osm/L
 - D. 0.0650 osm/L
 - E. 0.0485 osm/L
39. Osmotic pressure of the solution
- A. 1550.7 mmHg
 - B. 1640.5 mmHg
 - C. 5475.1 mmHg
 - D. 1563.5 mmHg
 - E. None of the above
40. Which of the following is not the property of the simple diffusion through passive channels?
- A. Transport maximum
 - B. Dissipate potential
 - C. Leaky ion channels
 - D. Hydrophilic open field proteins
 - E. No energy needed
41. Which of the following is true about the active transport across the cell membrane?
- A. The source of energy for 2^o active transport are indirectly ions
 - B. Calcium influx into the sarcoplasmic reticulum is 2^o active transport
 - C. Na⁺-K⁺ Exchange in the cardiac cell is direct active transport
 - D. Vesicular bulk transport is passive transportation process

- E. Pinocytosis is non-energy mediated transport
42. Which of the following is not the function of the $\text{Na}^+\text{-K}^+$ pump?
- A. Prevent intracellular edema
 - B. Create sodium diffusion gradient
 - C. Important for 2° active transport
 - D. Creating membrane potential difference
 - E. Avoid sodium gradient across the cell membrane
43. What happens to the rate of the transportation of a substance across the cell membrane in the facilitated diffusion after transport maximum is achieved at a given time?
- A. Constant
 - B. Increased
 - C. Reduced
 - D. Slightly reduced
 - E. No transport maximum in facilitated diffusion
44. Which of the following is true about body fluid?
- A. More water is removed by the kidneys during exercise
 - B. No water is lost from the body without regulation
 - C. Protein in the urine brings tissue edema
 - D. The more the filtration coefficient is the less the filtration rate
 - E. Capillary hydrostatic pressure gets high across the capillary
45. Which of the following is true about the composition of the intracellular and extracellular fluid compartments?
- A. Intracellular osmolarity is greater than the extracellular
 - B. More magnesium is outside the cell than the inside
 - C. More proteins are found inside the cell than the outside
 - D. Potassium is the extracellular cation
 - E. More water is found outside the cell than the inside
46. Which of the following electrolyte is found more in the blood vessels than in the tissue?
- A. Calcium
 - B. Chloride
 - C. Sulfate
 - D. Bicarbonate ion
 - E. Phosphate
47. Valves in the veins were discovered by:
- A. Hieronymous Fabricius
 - B. William Harvey
 - C. Andreas Vesalius
 - D. Leonardo da Vinci
 - E. Hippocrates
48. Which one is not true about Anatomical Position?
- A. Body erect
 - B. Head, eyes, and toes directed forward
 - C. Feet together
 - D. Arms at side with palms facing backward

- E. None of the above
49. Flexion and extension movements are possible through:
- A. Sagittal axis
 - B. Horizontal axis
 - C. Vertical axis
 - D. Longitudinal axis
 - E. None of the above
50. Which one of the following movement is different from the others?
- A. Protrusion
 - B. Flexion
 - C. Extension
 - D. Abduction
 - E. Hyperextension
51. Which one of the following bone markings is not related to joints?
- A. Head
 - B. Facet
 - C. Condyle
 - D. Foramen
 - E. None of the above
52. All of the following statements are true about sesamoid bones except:
- A. Prevent excessive wear and tear
 - B. Helps tendons to glide over bony surfaces
 - C. Alter the direction of pull of the tendons
 - D. Decrease mechanical advantage in the joints of long bones
 - E. None of the above
53. Which one is a correct order in structural organization of skeletal muscle from cell to the whole muscle?
- A. Fascicle – muscle fiber – muscle
 - B. Muscle fiber – fascicle – muscle
 - C. Muscle – muscle fiber – fascicle
 - D. Muscle fiber – muscle – fascicle
 - E. None of the above
54. The amino acid which can serve as proton donor and acceptor is:
- A. Glycine
 - B. Proline
 - C. Histidine
 - D. Isoleucine
 - E. Tryptophan
55. A semi-essential amino acid that is required especially during pregnancy is:
- A. Methionine
 - B. Valine
 - C. Arginine
 - D. Leucine

- E. Tyrosine
56. The most common type of amino acid that is frequently present in β -turns is:
- A. Lysine
 - B. Glycine
 - C. Valine
 - D. Isoleucine
 - E. Tyrosine
57. The electrostatic bond of side chains to form specific shape of a protein is described by:
- A. Primary structure
 - B. Secondary structure
 - C. Tertiary structure
 - D. Quaternary structure
58. Which is true regarding the folding of large polypeptides from polyribosome complexes?
- A. Folding starts in the ribosome
 - B. Its native folding is ATP dependent
 - C. It is a random and spontaneous process
 - D. It has high tendency for aggregation
 - E. All of the above
59. The role of molecular chaperones includes the following except:
- A. They shield interactive surface of non-native polypeptides
 - B. Mostly expressed constitutively
 - C. Recognize hydrophilic residues
 - D. They can reverse intramolecular misfolding
 - E. Facilitate assembly/disassembly of multiprotein complexes
60. Chemical denaturing agents of proteins include:
- A. Ionizing radiation
 - B. Extreme pH
 - C. Heat
 - D. Vigorous shaking
 - E. All of the above
61. The level of protein structure disrupted during denaturation include the following except:
- A. Primary structure
 - B. Secondary structure
 - C. Tertiary structure
 - D. Quaternary structure
 - E. Both C and D
62. Organic compounds usually affect the protein structure by:
- A. Affecting hydrogen bonding
 - B. Disrupting hydrophobic interactions
 - C. Forming ionic bonding
 - D. Protonating key amino acid residues
 - E. Breaking peptide bonds
63. Which one is not true about denaturation of DNA?
- A. Covalent bonds break

- B. Physical changes occur because of high temperature
 - C. Hydrogen bonding gets disrupted
 - D. A-T base pair is more resistant than G-C base pair
 - E. Ionic interactions dissociate
64. Which **is not** correctly matched?
- A. Caudal/inferior
 - B. Ventral/anterior
 - C. Dorsal/posterior
 - D. Cranial/superior
 - E. None of the above
65. Wrong about thalidomide?
- A. Amelia is a teratologic effect of thalidomide
 - B. Influenced the study of teratology by causing birth defects
 - C. Used as antinauseant to pregnant women, resulting no birth defects
 - D. Affects the bones of the limbs
 - E. None of the above
66. Which one is not part of a uterine tube?
- A. Isthmus
 - B. Infundibulum
 - C. Ampulla
 - D. Cervix
 - E. Intrauterine part
67. Which of the following is not found in the pelvis?
- A. Testes
 - B. Prostate gland
 - C. Seminal vesicles
 - D. Ductus deferens
 - E. Ejaculatory duct
68. Which one is not found in the pelvis?
- A. Cervix
 - B. Vagina
 - C. Ovary
 - D. Uterus
 - E. None of the above
69. Wrong about primordial germ cells (PGCs)?
- A. PGCs arise from the epiblast of the embryo
 - B. During their migration to the gonads, they increase their number with mitotic division
 - C. Enter into meiotic division during the 4-5th weeks of development
 - D. Formed during the 1st week of development
 - E. Teratomas are tumors from abnormal development of PGCs
70. Wrong match about the stages of mitosis?
- A. Metaphase → double structured chromosomes become visible
 - B. Telophase → chromosomes uncoil and elongate
 - C. Anaphase → centromere divides

- D. Metaphase → chromosomes migrate to opposite poles
 - E. Prophase → condensation of chromosomes
71. Wrong about the process of cross over?
- A. A chiasma is formed at the point of interchange of chromatid segments of homologous chromosomes
 - B. It occurs at meiosis II
 - C. Results in genetic variability
 - D. Random distribution of homologous chromosomes to the daughter cells
 - E. None of the above
72. Wrong about the process of cross over?
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Biochem module Final (07 Batch)

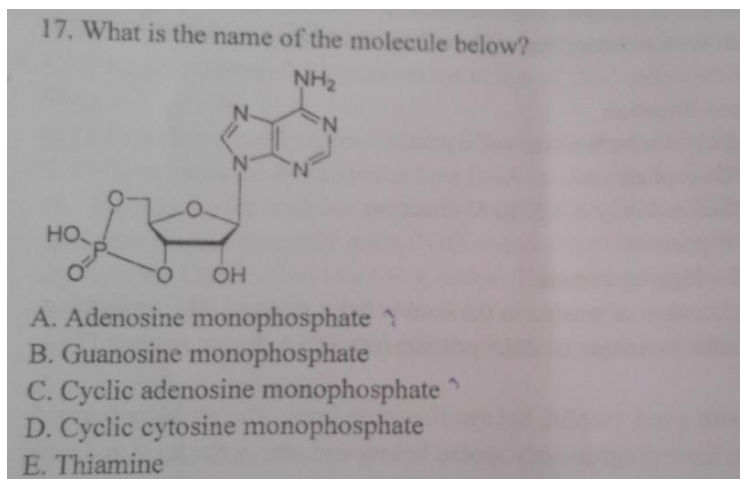
1. A compound normally used to conjugate bile acids is:
 - A. Serine
 - B. Glycine
 - C. Glucuronic acid
 - D. Fatty acid
2. For extramitochondrial fatty acid synthesis, acetyl CoA may be obtained from:
 - A. Citrate
 - B. Isocitrate
 - C. Oxaloacetate
 - D. Succinate
- Question is not correct. Pyruvate should have been in the choices.
3. Release of free fatty acids from adipose tissue is increased by all of the following except:
 - A. Glucagon
 - B. Epinephrine
 - C. Growth hormone
 - D. Insulin
4. Refsum's disease results from a defect in the following pathway except:
 - A. Alpha-oxidation of fatty acids
 - B. Beta-oxidation of fatty acids
 - C. Gamma-oxidation of fatty acids
 - D. Omega-oxidation of fatty acids
5. In adipose tissue, glycerol-3-phosphate required for the synthesis of triglyceride comes mainly from:
 - A. Hydrolysis of pre-existing triglycerides
 - B. Hydrolysis of phospholipids
 - C. Dihydroxyacetone phosphate formed in glycolysis
 - D. Free glycerol

6. Which of the following reaction in the biosynthesis of porphyrins is the rate-controlling step?
 - A. δ -Amino-levulinate (ALA) synthase reaction
 - B. Phosphobilinogen synthase reaction
 - C. Uroporphyrinogen synthase reaction
 - D. Decarboxylation of Uroporphyrinogen III
 - E. Conversion of protoporphyrin IX to protoheme IX
7. The two enzymes that catalyze the bulk of the nitrogen flow from amino acids to ammonia are:
 - A. Transaminases and dehydratases
 - B. Transaminases and glutamate dehydrogenase
 - C. Transaminases and amino acid oxidases
 - D. Dehydratases and glutamate dehydrogenase
 - E. Dehydratases and amino acid oxidases
8. Maple syrup urine disease is due to a deficiency of:
 - A. The enzyme that specifically transaminate the branched-chain amino acids
 - B. The branched chain α -keto acid dehydrogenase that converts branched chain amino acids to acyl coenzyme A (acyl CoA) esters and carbon dioxide
 - C. The mitochondrial enzyme system that oxidizes acyl CoA esters derived from the branched chain amino acids
 - D. Mitochondrial CoA required for oxidation of branched chain amino acids
 - E. Amino acid transport systems
- ✓ Note: Maple syrup urine disease is a deficiency of (branched chain) α -keto acid dehydrogenase.
9. Congenital deficiency of argininosuccinic acid synthetase leads to:
 - A. Argininoosuccinicacidaemia
 - B. Argininoosuccinic aciduria
 - C. Citrullinaemia
 - D. Hyper-ornithinaemia
 - E. Hyperammonaemia type I
10. Which of the following is a donor of labile methyl group?
 - A. Methionine
 - B. S-Adenosylmethionine
 - C. Histidine
 - D. Methymalonyl CoA
 - E. Tetrahydrofolate
11. Tetrahydrobiopterin (BH₄) is used in the hydroxylation reaction of:
 - A. Phenylalanine
 - B. Tyrosine
 - C. Tryptophan
 - D. Alanine
 - E. Methionine

This question has an error – both phenylalanine, tyrosine, and tryptophan hydroxylation needs BH₄. But choose phenylalanine if it comes on exam – that's the initial reaction.

12. Ammonia is transported from muscles to liver mainly in the form of:
 - A. Free ammonia

- B. Glutamine
 - C. Asparagine
 - D. Alanine**
 - E. Methionine
13. The carbon atom of urea is provided by:
- A. Carbon dioxide**
 - B. Aspartate
 - C. Ornithine
 - D. Citrulline
 - E. None of the above
14. Which type of porphyrin is present in heme?
- A. Uroporphyrin
 - B. Protoporphyrin I
 - C. Coproporphyrin
 - D. Protoporphyrin IX**
 - E. Hydroxymethylbilane
15. All of the following statements about phenylketonuria are correct except:
- A. Phenylalanine cannot be converted to tyrosine
 - B. Urinary excretion of phenylpyruvate and phenyl-lactate is increased
 - C. It can be controlled by giving a low phenylalanine diet
 - D. It leads to decreased synthesis of thyroid hormones**
 - E. Phenylketonuria results from deficiency of Phenylalanine hydroxylase
16. The study of the micro-organisms in and on our body, is known as:
- A. Metabolomics
 - B. Microbiomics**
 - C. Virobacteromics
 - D. Genomics
 - E. Pathomics
17. SKIP IT! The answer is C – i.e. cAMP



18. Which 5-carbon sugar is a component of DNA?
- A. Ribose

- B. 3'-deoxyribose
 - C. 5'-deoxyribose
 - D. 2',3'-deoxyribose
 - E. 2'-deoxyribose
19. Which of the following is NOT a feature of DNA?
- A. The DNA molecular structure contains major and minor grooves along its length
 - B. Each DNA strand has free hydroxyl group at the 3'-end
 - C. Each DNA strand has a free hydroxyl group at the 5'-end
 - D. Guanine forms base-pairs with cytosine
 - E. The backbone of the DNA molecule is negatively charged
- ✓ The hydroxyl group at 5' end is not free.
20. Which is true about the *de novo* pathway of pyrimidine biosynthesis?
- A. The pyrimidine ring is completed before it is added to the ribose sugar unit
 - B. The pyrimidine ring is made sequentially while it is attached to the ribose sugar
 - C. Phosphoribosyl pyrophosphate (PRPP) is not used in the *de novo* pyrimidine synthesis pathway.
 - D. The pyrimidine ring is added first to deoxyribose sugar, which is later converted to ribose
 - E. There are no known genetic diseases involving abnormal enzymes of pyrimidine synthesis
21. When we say that DNA replication is semiconservative, we mean that:
- A. Each of the two "parent" DNA strands is digested into fragments and these fragments are used to make new DNA molecules.
 - B. Both parent DNA strands stay together in one "daughter" DNA molecule and two completely newly synthesized strands are used to form the other "daughter" DNA molecule.
 - C. The two DNA strands unwind and new DNA is synthesized on each strand, producing two "daughter" DNA molecules, each with an intact strand from the parent DNA molecule and each with a newly synthesized strand
 - D. DNA is synthesized only in one direction
 - E. DNA is replicated only during certain parts of the cell cycle
22. Which is NOT true about DNA replication?
- A. Each new DNA strand is synthesized only in 5' to 3' direction
 - B. DNA replication requires RNA primers
 - C. Okazaki fragments occur on the lagging strands
 - D. DNA topoisomerase allows relaxation of tension in the double helix during DNA replication
 - E. DNA polymerase-gamma initiates extension of RNA primers into DNA during nuclear DNA replication
23. A patient comes to see you with red, painful, hot swollen knee joint. The problem began 5 days ago, and the redness and pain have progressively spread below and above the knee and now the patient has a fever. You are particularly concerned that the patient has:
- A. Acute gouty arthritis
 - B. Acute Pseudogout
 - C. A fractured femur
 - D. Cellulitis and possible infectious arthritis

- E. Chronic osteoarthritis
24. Which of the following karyotypes is NOT an aneuploidy?
- A. 47, XXY
 - B. 47, XYY
 - C. 45, X
 - D. 46, XX, 5p-
 - E. 48, XXXY
25. Features of Lesch-Nyhan syndrome include all of the following except which one?
- A. A deficiency of the enzyme, uricase
 - B. Episodes of gouty arthritis
 - C. Recurrent kidney stones
 - D. Self-injurious behavior
 - E. Formation of chronic gouty arthritis
26. Which is true about Philadelphia chromosome?
- A. It occurs in over 95% of patients with acute lymphoblastic leukemia (ALL)
 - B. It involves over-activation of a gene encoding a tyrosine kinase, resulting in uncontrolled cell growth, leading to leukemia
 - C. It does not occur in chronic myeloid leukemia (CML)
 - D. It is due to an aneuploid chromosome disorder
 - E. It involves the formation of multiple copies of a gene encoded on chromosome 21
27. Which of the following is a correct statement?
- A. DNA polymerase-gamma replicates nuclear DNA on the lagging strand
 - B. DNA polymerase-alpha synthesizes DNA only on the leading strand
 - C. DNA polymerase-delta requires the action of DNA polymerase-alpha before it can make new DNA
 - D. DNA polymerase-epsilon initiates DNA synthesis directly from RNA primers
 - E. DNA polymerase-delta synthesizes DNA only on the leading strand
28. The codons for arginine are AGG, AGA, CGU, CGA, and CGG. The gene encoding a particular protein normally has a CGG codon within its sequence, but a mutation occurs and converts the CGG codon to a CGA codon. This is an example of:
- A. A non-sense mutation
 - B. A mutation causing an amino acid substitution in the protein sequence
 - C. A frame shift mutation
 - D. A deletion mutation
 - E. A silent mutation
29. Which enzyme removes an abnormal base from a mutated DNA molecule by breaking the bond between the abnormal base and the deoxyribose sugar during base-excision repair?
- A. DNA exonuclease
 - B. DNA helicase
 - C. DNA topoisomerase
 - D. DNA glycosylase
 - E. DNA polymerase
30. Examples of diseases that are caused by defects in DNA repair mechanisms include all of the following **except** which one:

- A. Xeroderma pigmentosum
 - B. Ataxia telangiectasia
 - C. Werner syndrome (adult progeria)
 - D. Hereditary nonpolyposis colorectal cancer (HNPCC)
 - E. Lesch-Nyhan syndrome
31. When there is extensive damage to DNA due to mutations, a cell may undergo various processes in response to this damage. Which of the following **is not** a common response of a cell to extensive DNA damage?
- A. The cell may fuse with another cell and form a hybrid cell and reinstate a normal genome
 - B. The cell may undergo programmed cell death (apoptosis)
 - C. The cell may become “senescent” and stop dividing
 - D. The cell may repair the DNA damage
 - E. There may be activation of oncogenes or inactivation of tumor suppressor genes, leading to uncontrolled cell division and malignancy
32. Examples of gene therapy include all of the following except which one?
- A. Using inhaled DNA carrying the gene encoding a normal chloride transporter to genetically modify airway cells in patients with cystic fibrosis, to improve lung function.
 - B. Removing T-cells from a patient with ADA-SCID, inserting the normal adenosine deaminase gene into the nuclei of T-cells, allowing the T-cells to divide, then injecting the genetically modified T-cells back into the patient
 - C. Inserting a normal tumor suppressor gene into a virus genome, packaging the recombinant DNA into virus particles, and injecting the resulting virus into patients with cancer, allowing delivery of the normal tumor suppressor gene to the patient’s cancer cells.
 - D. Incorporating the DNA encoding the normal phenylalanine hydroxylase gene into a plasmid, injecting the plasmid into liver cells taken from a person with phenylketonuria, then replacing the recombinant liver cells back into the patient’s liver
 - E. Monthly injections of a purified preparation of the enzyme sphingomyelinase, into the blood stream of a patient with Neimann-Pick syndrome, which is due to mutation in the sphingomyelinase gene.
33. For which of the following might pharmacogenomics be most valuable?
- A. To determine whether a particular asthmatic patient might respond favorably or adversely to a beta-2-agonist inhaler.
 - B. To determine whether a person who currently is not known to have asthma might be susceptible to developing asthma later in life.
 - C. To compare the genes of asthmatics with those of non-asthmatics in order to determine which gene variants predispose people to asthma
 - D. To screen SNPs with DNA from patients with Paget’s disease and DNA from non-affected individuals, in order to find genes that are involved in Paget’s disease
 - E. To screen ill babies in neonatal care unit for genetic diseases
34. Which is **not true** about DNA sequence variants in humans?
- A. Single nucleotide polymorphisms (SNPs) are the most common DNA sequent variant
 - B. Deletions of DNA sequences are a type of DNA sequence variant
 - C. Insertions of DNA sequences are a type of DNA sequence variant

- D. Some people have more copies of certain DNA sequences than other people: these are known as copy-number variants
- E. Most people have just a small number of SNPs, less than a few hundred, in their genome.
35. A patient comes to see you, complaining of a painful, red, swollen right first metatarsophalangeal joint (1st MTP joint). He was fine yesterday, but the pain and swelling came on quickly during the night. If you examined fluid from the joint under the microscope, you would expect to see:
- A. Needle-shaped crystals showing no birefringence under polarized light microscopy
- B. Needle-shaped crystals showing negative birefringence under polarized light microscopy
- C. No crystals, but abundant gram negative bacteria
- D. Hexagonal crystals showing no birefringence
- E. Clear fluid with no crystals or bacteria
36. Which of the following is **not** the importance of glutamine?
- A. Synthesis of nucleotides
- B. Synthesis of glucose in liver via gluconeogenesis
- C. Energy production in rapidly proliferating cells
- D. Ammoniogenesis in the kidney
- E. All are important functions
37. Cyanide poisoning can result in total arrest of oxidative phosphorylation by locking electron flow through which complex?
- A. Complex I
- B. Complex II
- C. Complex III
- D. Complex IV
38. Which of the following enzymes is a target for the action of quinolones?
- A. DNA polymerase
- B. DNA gyrase
- C. Primase
- D. Helicase
39. Which of the following is **NOT true** concerning medium chain acyl CoA dehydrogenase deficiency?
- A. Decreased fatty acid oxidation
- B. Hypoglycemia
- C. Primarily affects adults
- D. Autosomal disorder
40. Mitochondrial oxidative phosphorylation is catalyzed by:
- A. Cytochrome oxidase
- B. Cytochrome reductase
- C. ATP synthase complex
- D. Succinate dehydrogenase
41. Which of the following coenzymes transfers the most oxidized form of carbon (CO₂)?
- A. Biocytin
- B. Tetrahydrofolate
- C. PLP
- D. FAD
- E. Thiamine pyrophosphate

42. The enzyme fumarase catalyzes the conversion of fumarate to malate in the citric acid cycle; it is grouped under which of the following enzyme classes?
- A. Oxidoreductases
 - B. Lyases
 - C. Ligases
 - D. Transferases
 - E. Isomerases
43. Which of the following is correct regarding the rate of a chemical reaction?
- A. Initially it is dependent on substrate concentration
 - B. When the concentration of the substrate is equal to the K_m value of the enzyme, it will achieve half maximal velocity
 - C. Maximum velocity could only be achieved with ideally large concentration of substrate
 - D. If adding more substrate doesn't change the rate, the enzyme is said to be "saturated"
 - E. All of these
44. All of the following is true regarding metal ion catalysis, except:
- A. They participate in oxidation-reduction catalysis
 - B. They shield negative charges
 - C. They bind and orient substrates to the active site
 - D. They transiently carry functional groups
45. Which of the following is **incorrect** regarding formation of ES (Enzyme substrate) complex?
- A. Binding energy helps in lowering the activation energy of a chemical reaction
 - B. Binding energy contributes to the specificity of an enzyme for its substrate
 - C. In order to catalyze reactions an enzyme must be complimentary to reaction transition state
 - D. The key and lock enzyme substrate binding model is the most accepted model presently
 - E. Weak interactions at the active site of enzymes are crucial for both substrate specificity and binding
46. Suicide inhibitors
- A. Are one of the reversible inhibitors
 - B. Bind to a different site than the active site and affect the conformation of the enzyme
 - C. Use the mechanism of action of the enzymes to yield product
 - D. Are called mechanism-based inhibitors
 - E. All of them are correct
47. All of the following are true regarding the active site of an enzyme **except**:
- A. It is a small 3-dimensional cleft of an enzyme
 - B. It is the place of substrate binding and catalysis
 - C. It is complementary to its substrate
 - D. It can bind noncompetitive inhibitors
 - E. It is a hydrophobic site with hydrophilic catalytic groups
48. All of them are correct about enzymes, except:
- A. Most are proteins
 - B. They are highly specific
 - C. They are catalytically more powerful than inorganic catalysts
 - D. All of them are regulated depending on the energy need of cells

- E. Their activity can be influenced by extreme pH and temperature
49. Uncompetitive inhibitors
- A. bind to the active site and interfere with the ES complex formation
 - B. raise V_{max} value of the enzyme
 - C. raise K_m value of the enzyme
 - D. enzyme activity will be reduced but not lost
 - E. bind only to the ES complex
50. Which of the following enzyme regulation is irreversible regulation?
- A. Covalent modification
 - B. Feedback inhibition
 - C. Allosteric regulation
 - D. Zymogen activation
 - E. A and D
51. What do we mean by the genetic code is degenerate?
- A. The genetic code is specific for a single amino acid
 - B. One amino acid can be coded by more than one genetic code
 - C. Some of the genetic codes do not represent any amino acid
 - D. The code is similar in almost all organisms
52. Which of the following prevents premature association of small ribosomal subunit with the large one in prokaryotes during protein synthesis?
- A. IF 1
 - B. IF 2
 - C. IF 3
 - D. eIF4E
 - E. eIF4G
53. Which of the following is required for both prokaryotic and eukaryotic protein synthesis?
- A. Binding of the small ribosomal subunit to the Shine-Dalgarno sequence
 - B. Translocation of the peptidyl-tRNA from the A site to the P site
 - C. fMet-tRNA^{fMet}
 - D. Movement of the mRNA out of the nucleus and into the cytoplasm
 - E. Recognition of the 5'-cap by initiation factors
54. All of the following are important in prokaryotic translation, **except**:
- A. mRNA
 - B. DNA
 - C. Large ribosomal unit
 - D. Small ribosomal unit
 - E. GTP
55. Which of the following is **incorrect** about the termination of translation?
- A. Termination of codons will be recognized by the release factors
 - B. The reaction of the peptidyl transferase is with the last charged tRNA
 - C. The large and the small ribosomal subunits will dissociate
 - D. Release factors mimics the structure of tRNA
 - E. All
56. All of the following statements about anaerobic glycolysis are correct EXCEPT:

- A. The control enzyme is PFK-1 which converts Fructose 6-P to Fructose 1,6-bisphosphate.
 - B. The pathway is allosterically activated by Fructose 2,6-bisphosphate
 - C. Glycolysis takes place in the cytosol of all cells
 - D. Glycolysis produces L-lactate and NAD^+
 - E. Glycolysis can begin with Hexokinase or Glucokinase
57. The enzyme that catalyzes the second substrate level phosphorylation of glycolysis:
- A. is Phosphoglucomutase
 - B. produces lactate as a biproduct.
 - C. uses phosphoenolpyruvate as substrate
 - D. is found in the mitochondria
 - E. is a reversible reaction
58. Which statement about the Lactate Dehydrogenase reaction is CORRECT?
- A. The enzyme converts glucose to L-lactate
 - B. The enzyme utilizes NADH and produces NAD^+
 - C. The reaction is irreversible
 - D. It is the last reaction in aerobic glycolysis
 - E. The enzyme is found only in the liver
59. Which of the following is a product of the Pentose Phosphate Pathway?
- A. NADPH
 - B. Ribulose 5-P
 - C. CO_2
 - D. Fructose 6-P
 - E. All of the above
60. The following can be precursors for biosynthesis of glucose EXCEPT:
- A. L-alanine
 - B. L-lactate
 - C. Citrate
 - D. Acetoacetate
 - E. Glycerol
61. Which of the following gluconeogenic enzyme is mitochondrial?
- A. Pyruvate carboxylase
 - B. Phosphoenolpyruvate carboxylase
 - C. Glucose 6-phosphatase
 - D. Fructose 1,6-bisphosphatase
 - E. All of the above
62. Deficiency of two pentose phosphate pathway enzymes leads to which disease?
- A. Fructosuria
 - B. Diabetes mellitus
 - C. Hemolytic anemia
 - D. Galactosemia
 - E. Glycogen storage disease
63. Which of the following is WRONG about the pentose shunt?
- A. It produces sugars of different carbon numbers
 - B. It takes place in the mitochondrion of cells

- C. It produces a sugar useful for nucleotide synthesis
 - D. Its product is useful for reductive biosynthesis
 - E. None of the above
64. Which of the following is CORRECT about glucose 6-P?
- A. It is produced only in RBCs.
 - B. It serves as raw material for pentose shunt.
 - C. It is an intermediate formed in cells undergoing glycolysis
 - D. It can serve as an intermediate in glycogen synthesis
 - E. All except A.
65. The synthesis of glycogen in our cells:
- A. doesn't require a high-energy nucleotide
 - B. requires the availability of primer
 - C. takes place when blood [glucose] is high
 - D. All of the above
 - E. B and C
66. Some infants present with stomach distension, cramp and diarrhea after milk feeding. This is due to:
- A. Inability to absorb and utilize glucose
 - B. Deficiency of the enzyme digesting milk sugar
 - C. Inability of undergoing Glycogenolysis
 - D. Defect due to protein digesting enzymes
 - E. Inability to utilize milk proteins
67. Dietary fructose intolerance is a metabolic disorder of fructose metabolism which is due to:
- A. malabsorption of fructose using GLUT5 by enterocytes
 - B. deficiency of the enzyme fructokinase
 - C. deficiency of the enzyme aldolase-b
 - D. All of the above
 - E. A and B
68. Glycogen metabolism is a process?
- A. that is hormonally regulated
 - B. that is reciprocally controlled by opposing enzymes
 - C. which requires set of enzymes found in the cytosol of all cells.
 - D. which is useful for organisms that are only in a state of starvation
 - E. All of the above
69. Which one of the following pathways is NOT activated by insulin?
- A. Gluconeogenesis
 - B. Glycogenesis
 - C. Glycolysis
 - D. Lipogenesis
 - E. None of the above
70. Defect in Glucose 6 phosphate dehydrogenase:
- A. is a common X-linked genetic disorder
 - B. appears in different forms of gene defects
 - C. makes some cells protective to oxygen free radicals

- D. occurs only in the black community
 - E. All except D.
71. Blood glucose under normal circumstances is well regulated and this is possible due to:
- A. Gluconeogenesis
 - B. Glycogenolysis
 - C. Reciprocal regulation by hormones
 - D. All of the above
 - E. A and B only
72. In a highly exercising muscle:
- A. there is large production of lactate
 - B. the pH drops due to production of acid
 - C. anaerobic glycolysis cannot take place
 - D. pyruvate is converted to glucose
 - E. All of the above
73. The glycogen degrading enzyme glycogen phosphorylase:
- A. cleaves α (1,4) glycosidic bonds
 - B. is activated by phosphorylation
 - C. produces glucose 1-P from glycogen
 - D. is activated by glucagon
 - E. All of the above
74. A 2-year-old boy has been diagnosed with Zellweger syndrome, a disorder caused by malformation of peroxisomes, the oxidation of which of the following fatty acids is impaired in this defect?
- A. Short chain fatty acids
 - B. Long chain fatty acids
 - C. Very long chain fatty acids
 - D. Unsaturated fatty acids
75. Which of the following apoproteins is an activator of lipoprotein lipase?
- A. Apo A
 - B. Apo B 48
 - C. Apo C-II
 - D. Apo E
76. All of the following statements correctly describe ketone bodies, EXCEPT:
- A. May be synthesized during starvation
 - B. They are utilized by the liver during long-term starvation
 - C. They may be synthesized in uncontrolled diabetes mellitus
 - D. They are utilized by the cardiac muscles
77. The extramitochondrial acetyl-CoA for fatty acid synthesis comes from all, EXCEPT:
- A. Glucose metabolism
 - B. β -oxidation of fatty acids
 - C. Ketogenic amino acids
 - D. All of the above
78. Which of the following is NOT involved in provision of NADPH in cytosol for fatty acid synthesis?
- A. Malate dehydrogenase

- B. Malic enzyme
 - C. Cytosolic Isocitrate dehydrogenase
 - D. HMP pathway
79. Which of the following is TRUE regarding fatty acid synthesis?
- A. HMG-CoA reductase is the rate limiting step
 - B. One malonyl-CoA and seven Acetyl-CoA are utilized for palmitic acid synthesis
 - C. Acetyl-CoA carboxylase controls the pace of fatty acid synthesis
 - D. The two units of the dimeric fatty acid synthetase are placed in Head-to-Head manner.
80. The acetyl CoA formed on β -oxidation of all long chain fatty acids is metabolized under normal circumstances to:
- A. CO₂ and water
 - B. Cholesterol
 - C. Fatty acids
 - D. Ketone bodies

Blood and Immunity Final (07 Batch)

1. Which of the following cells carry CD4 marker cells on their surface and are thus important targets of HIV infection?
 - A. Cytotoxic T cells
 - B. Helper T cells
 - C. Suppressor T cells
 - D. NK cells
 - E. T lymphocytes
2. With regard to the *lymphoid organs*, which of the following statements is correct?
 - A. Lymph node receives afferent lymphatic vessels
 - B. Hassall's corpuscles are located in the Peyer's patches
 - C. Spleen has a cortex, a medulla, and a subscapular sinus
 - D. The blood-thymus barrier is located in the medulla of the thymus
 - E. Pharyngeal tonsil is covered by nonkeratinized stratified squamous epithelium
3. Which statement is true regarding the *thymus*?
 - A. It has lymphoid tissues in the cortex
 - B. It is made up of epithelioreticular cells
 - C. It is relatively more developed with increasing age
 - D. It contains more B lymphocytes than the T lymphocytes
 - E. NK cells gain their immunocompetence in the thymus
4. The *spleen*:
 - A. Has no B lymphocytes
 - B. Has smooth myocytes in its capsule
 - C. The entire lymphatic region is found in the red pulp
 - D. Contains periarterial lymphoid sheath in the marginal zone of the red pulp
 - E. Has cords of Billroth in its outer cortex and Malpighian corpuscles in its inner medulla
5. Which of the following is NOT a function of the spleen?
 - A. Destruction of old or damaged red blood cells
 - B. Storage of blood
 - C. Filtration of lymph

- D. Fetal blood cell formation
 - E. Lymphocyte and antibody production
6. Identify an innate immune evasion mechanism employed by microbes
- A. Capsular polysaccharides
 - B. Production of catalase
 - C. Sialic acid expression
 - D. Synthesis of modified LPS
 - E. All of the above
7. If a medical personnel is requested to work in West African countries affected by the current Ebola epidemic as a volunteer is supposed to be vaccinated. However, as there is no vaccine for Ebola, the medical personnel received injection of antibodies pooled from individuals who have recovered from the disease. The pooled antibodies, which should protect the medical personnel, are of which type?
- A. IgA
 - B. IgG
 - C. IgE
 - D. IgD
 - E. None of the above
8. A live attenuated vaccine is acceptable in which one of the following groups
- A. children under 8 years of age who are not immunodeficient
 - B. Patients treated with steroids
 - C. Pregnant mothers
 - D. Patients with leukemia
 - E. Patients treated with radiotherapy
9. Variolation refers to:
- A. The deliberate infection with small pox
 - B. The attenuation of virulent organisms
 - C. Inoculation of scab material into small skin wounds
 - D. The removal of scab material from an individual with smallpox
 - E. A and C only
10. Herd immunity refers to:
- A. proportion immune among individuals in a population
 - B. particular threshold proportion of immune individuals that should lead to a decline in incidence of infection
 - C. a pattern of immunity that should protect a population from invasion of a new infection
 - D. all of the above
 - E. none of the above
11. If someone has to develop an effective vaccine against mycobacterial diseases such as tuberculosis, the most important piece of the immune response to be stimulated is:
- A. A high titer of antibody
 - B. Antibody in the gut lumen
 - C. Cytotoxic T-cells
 - D. Macrophage-activating cell mediated immunity
 - E. Neutrophils

12. A potential disadvantage of immunological protection using passive transfer of immunoglobulins is
- A. Waning of immunity after vaccination
 - B. Development of immune complex mediated inflammatory reactions
 - C. Cost
 - D. All of the above
 - E. A and B only
13. Which of the following is a characteristic of T cell exhaustion?
- A. Rapid proliferation
 - B. Robust polyfunctionality
 - C. an effective recall response to infection
 - D. gradual up-regulation of inhibitory receptors
 - E. Cytotoxicity
14. The success of eradicating infectious disease by vaccination depends on all EXCEPT:
- A. if no latency is associated with the infection
 - B. if the pathogen is not associated with high antigenic variation
 - C. if animal reservoir is not involved in the transmission
 - D. if the immunity induced by vaccination is long lasting
 - E. none of the above
15. Immune evasion strategy of parasites
- A. Sequestration of parasites
 - B. Stage specific change in antigen expression
 - C. Antigen shielding
 - D. continuous variation of major surface antigens
 - E. all of the above
16. The classical activation of macrophages is shown to be important in controlling intracellular infection with parasites like Leishmania. Which one of the following statements is TRUE about the classical activation of macrophages
- A. destroy intracellular parasites
 - B. cytokines like IL-4 and IL-13 are shown to responsible
 - C. the enzyme arginase will be activated
 - D. the enzyme inducible nitric oxide synthase will be activated
 - E. all except B
17. The canonical type 2 immunity
- A. mediates defense against helminthic infections
 - B. results in the production of IgE
 - C. initiates activation of mast cells and basophils
 - D. initiates humoral type 2 response
 - E. all of the above
18. Which of the following is TRUE about amplifiers and innate effector response in helminthic infections?
- A. initiated by IL-25 and IL-33
 - B. the innate effector cells secrete IL-13 that induces type 2 responses
 - C. initiates eosinophilia hallmark of helminthic infections

- D. initiates alternative activation of macrophages
 - E. all of the above
19. Identify the WRONG combination of pathogen recognition receptor (PRR) and pathogen
- A. TLR3: dsDNA
 - B. TLR9: CpG DNA
 - C. NLR: bacterial peptidoglycan
 - D. RLR: viral RNA
 - E. None of the above
20. Which of the following is NOT true about adaptive immune response to extracellular bacteria?
- A. Activation of CD4+ T cells by extracellular proteins of bacteria is not important
 - B. Antibodies produced against the pathogens are important to initiate ADCC
 - C. The adaptive immune response could be responsible for pathology associated with the infection
 - D. The humoral immunity is important to block infection
 - E. The humoral immunity is important to neutralize toxins
21. Which of the following is immune evasion mechanism of bacteria?
- A. Antigenic variation
 - B. Inhibition of phagosome maturation
 - C. Inactivation of reactive oxygen and nitrogen species
 - D. Escape from phagosome
 - E. All of the above
22. Quiescent phase of Mycobacterium tuberculosis infection is characterized by
- A. Granuloma formation
 - B. Giant cell formation
 - C. Positive TST
 - D. Balanced activating and regulatory T cell response
 - E. All of the above
23. Killing of extracellular bacteria (S. aureus) by neutrophils involves all EXCEPT:
- A. Oxidative burst
 - B. Production of antimicrobial peptides
 - C. Sequestration of essential nutrients
 - D. Production of nitric oxide
 - E. None of the above
24. The immunopathology that develops following viral infection is associated with all EXCEPT
- A. Upregulation of CTLA-4 by CTL
 - B. Immune complexes formed by antibodies produced against a virus
 - C. Impaired T reg cells that control aggressive CTL and Th cells
 - D. Secretion of pro-inflammatory molecules
 - E. None of the above
25. The persistence of viral infections is associated with
- A. Increased production of IL-10, TGF- β and IL-35
 - B. Expression of TLR signal proteins
 - C. Up-regulation of inhibitory receptors like PD-1 by effector T cells
 - D. Increased production of chemical mediators like resolvins

E. All of the above

26. Which one of the following is associated with the development of immunity to fungal infections?

- A. Phagocytosis
- B. Neutrophil degradation
- C. Antibody production
- D. Activation of T_H2 type immune response

E. All except D

27. In the development of multiple or mixed antigen vaccine, which one of the following is a major concern?

- A. Relative competition between vaccine responses
- B. Route of vaccine administration
- C. Vaccine production methods
- D. Vaccine target groups

E. All of the above

28. Development of a vaccine that induces higher titer of broadly neutralizing antibodies

- A. Has no value in intracellular infection
- B. Is very important against viral infections like HIV and influenza virus
- C. Is good against pathogens that use antigenic variation as immune evasion mechanism
- D. Is impossible as antibodies are highly specific

E. A and D

29. Identify the WRONG combination of vaccine clinical phase and its aim:

- A. Phase I: to assess safety
- B. Phase II: to obtain immunogenicity data
- C. Phase III: to assess the protective efficacy
- D. Phase IV: to detect rare adverse effects

E. None of the above

30. Identify the WRONG statement about the different type of vaccine development strategies and concerns/limitations associated

- A. Live viral vaccines involving recombinant viruses: production of antigens that stimulate CTL responses that kill the infected host cells
- B. Purified Antigen (Subunit) Vaccines: elicit low-affinity antibody responses and may be poorly immunogenic in infants
- C. Protein vaccines generating by coupling polysaccharides with proteins to form conjugate vaccine: do not generate potent CTL
- D. Synthetic Antigen Vaccines (HPV 6,11,16 & 18): predominance of immunity specific to the viruses included

E. None of the above

31. Which one of the following is the limitation of live vaccines?

- A. Induction of long lasting immunity
- B. Poor stability**
- C. Local and systemic immunity produced
- D. Evoke strong immune response
- E. None of the above

32. Which one of the following is not the disadvantage of inactivated vaccine?

- A. Constituents are clearly defined
 - B. Adjuvants are needed
 - C. Need booster doses
 - D. Local reactions are common
 - E. They do not induce long lasting immunity
33. Which one of the following is the correct order in terms of rejection rates in tissue transplantation in non-immunologically privilege sites
- A. Xenogeneic graft > Allogeneic graft > Syngeneic graft > Autologous graft
 - B. Xenogeneic graft > Syngeneic graft > Allogeneic graft > Autologous graft
 - C. Xenogeneic graft > Allogeneic graft > Autologous graft > Syngeneic graft
 - D. Allogeneic graft > Xenogeneic graft > Syngeneic graft > Autologous graft
 - E. Allogeneic graft > Xenogeneic graft > Syngeneic graft > Autologous graft
34. Alloantigen
- A. Stimulate adaptive immune response
 - B. Are encoded by polymorphic genes
 - C. Are antigens found in same members of species
 - D. Can be recognized as foreign without being processed
 - E. All of the above
35. The direct recognition of alloantigen by alloreactive T cells
- A. Reflects the inherent specificity of TCRs for MHC molecules
 - B. Reflects that the positive selection in thymus promotes survival of T cells with strong reactivity to allogeneic MHC molecules
 - C. Reflects that the negative selection in thymus promotes survival of T cells that bind strongly in allogeneic MHC molecules
 - D. Contributes significantly to rejection
 - E. All of the above
36. T cell response to directly presented allogeneic MHC molecules is very strong compared to that for microbial peptides. Which one of the following can be the reason:
- A. There are higher frequencies of T cells specific for allogeneic MHC molecules
 - B. Every APC expresses thousands of copies of different MHC molecules
 - C. There are fewer T cells specific for microbial peptides
 - D. Many of the T cells that respond to allogeneic MHC molecules are memory T cells
 - E. All of the above
37. Which of the following is TRUE about the mechanism of tissue rejection?
- A. Hypoacute rejection is mediated by complement activation
 - B. Acute cellular rejection is mediated by inflammation caused by cytokines produced by Th cells
 - C. Acute cellular rejection is mediated by CTL-mediated killing of graft parenchymal cells and endothelial cells
 - D. Alloantibodies cause acute rejection
 - E. All of the above
38. Which of the following is applicable in minimizing alloantigenic differences between the donor and recipient as a strategy to reduce graft immunogenicity?
- A. ABO blood group tests

- B. HLA typing
 - C. Screening tissue recipients for the presence of preformed antibodies against allongenic HLA molecules prevalent in the population
 - D. Cross matching test if the patient/recipient has antibodies that react specifically with the donor's cells
 - E. All of the above
39. Which one of the following methods is used to treat or prevent allograft rejection?
- A. Use of anti-IL2R to block proliferation of T cells
 - B. Use of CTLA4-Ig to block delivery of co-stimulatory signal required for T cell activation
 - C. Use of metabolic toxins that kill proliferating T cells
 - D. Use of Antibodies that react with T cell surface structures and deplete or inhibit T cells
 - E. All of the above
40. Which of the following is a characteristic of non-Hodgkin lymphomas?
- A. Constitutional symptoms
 - B. Extra-nodal disease
 - C. Contiguous spread
 - D. Painful lymphadenopathy
 - E. None of the above
41. Which of the following is NOT true concerning Reed-Stenberg (R-S) cells?
- A. Reed-Stenberg cells are derived from B lymphocytes
 - B. R-S cells secrete cytokines that attract reactive inflammatory cells
 - C. Reactive cells produce factors that promote survival of RS cells
 - D. Diagnostic R-S cells are more than five times larger than small lymphocytes
 - E. Lympho-histiocytic variant R-S cells are typical for Nodular sclerosis HL
42. The subtype of Hodgkin's lymphoma with the poorest prognosis is
- A. Nodular sclerosis
 - B. Mixed cellularity
 - C. Lymphocyte rich
 - D. Lymphocyte depletion
 - E. Nodular lymphocyte predominance
43. Which of the following is NOT true regarding MALT (Mucosa-associated lymphoid tissue) lymphomas?
- A. They usually present as extra-nodal tumors
 - B. The tumor may regress if the inciting antigen is removed
 - C. Auto-immune diseases may be a risk factor for the development of MALT lymphomas
 - D. Acquired mucosa-associated lymphoid tissue in auto-immune conditions is initially polyclonal
 - E. None of the above
44. One of the following is NOT a feature of small lymphocytic lymphoma/chronic lymphocytic leukemia?
- A. Asymptomatic disease
 - B. Lymphadenopathy
 - C. Splenomegaly
 - D. Mediastinal mass

- E. Recurrent infections
- 45. Diffuse large B cell lymphoma may develop in the following settings, except:
 - A. Small lymphocytic lymphoma
 - B. Lymphoblastic lymphoma
 - C. Follicular lymphoma
 - D. HIV infection
 - E. *De novo* diffuse large B cell lymphoma
- 46. Over-expression of BCL-2 with the inhibition of apoptosis is characteristic of:
 - A. Lymphoblastic lymphoma
 - B. Small lymphocytic lymphoma
 - C. Follicular lymphoma
 - D. Marginal zone lymphoma
 - E. Mantle cell lymphoma
- 47. The translocation that characterizes Burkitt's lymphoma is
 - A. t(14;18)
 - B. t(8;14)
 - C. t(11;14)
 - D. t(12;21)
 - E. t(9;22)
- 48. NOT true regarding multiple myeloma
 - A. May develop on a background of solitary plasmacytoma
 - B. Neoplastic plasma cells are monoclonal
 - C. Secreted immunoglobulin molecules or fragments are monoclonal
 - D. Neoplastic plasma cells may appear morphologically normal
 - E. None of the above
- 49. In patients with plasma cell neoplasms, hyper viscosity symptoms are more severe if the secreted M component is composed of:
 - A. IgG
 - B. IgA
 - C. IgD
 - D. IgE
 - E. IgM
- 50. The mechanism that best explains sinus histiocytosis in lymph nodes draining tissue harboring a tumor is
 - A. Tumor cells deposited in lymph nodes
 - B. Antigen delivery to lymph nodes
 - C. Infection of lymph nodes
 - D. Increased lymphatic fluid reaching the nodes
 - E. All of the above
- 51. Which of the following is NOT true regarding the pathogenesis of CML?
 - A. BCR-ABL fusion gene is present in all cases
 - B. Majority of BCR-ABL fusion is a result of t(9;22)
 - C. Activation of ABL tyrosine kinase is ligand independent
 - D. Point mutation of JAK2 causes accelerated phase

- E. All of the above
52. In which of the following cases do you expect 'tear-drop RBCs'?
- A. Hereditary spherocytosis
 - B. Chronic myeloid leukemia
 - C. Polycythemia vera
 - D. Essential thrombocytosis
 - E. Primary myelofibrosis
53. Which of the following is a mark of myeloid blasts?
- A. Positive myeloperoxidase reaction
 - B. Positive PAS
 - C. CD 20
 - D. CD 3
 - E. None of the above
54. Works as an anticoagulant only *in vivo*?
- A. Warfarin
 - B. Heparin
 - C. Sodium citrate
 - D. A and C
55. Increases the risk of hemorrhage if taken together with warfarin?
- A. Oral contraceptive pills
 - B. Rifampin
 - C. Vitamin K
 - D. Aspirin
56. Identify fibrin specific fibrinolytic agent
- A. Alteplase
 - B. Streptokinase
 - C. Abciximab
 - D. Clopidogrel
57. Toxicity manifestation of oral iron preparations to a child who consumed them in overdose does not include
- A. Bloody diarrhea
 - B. Necrotizing gastroenteritis with vomiting
 - C. Metabolic alkalosis
 - D. Shock
58. Maturation of RBC is accelerated by intake of Vitamin B₁₂ and folic acid which are important for the synthesis of:
- A. RNA molecule
 - B. DNA molecule
 - C. Endoplasmic reticulum
 - D. Mitochondria
 - E. Lysozymes
59. In what conditions do Rh⁻ mothers produce anti-D antibody?
- A. During sexual contact with Rh⁺ husband
 - B. Leakage of Rh⁺ blood of the fetus during the 1st birth

- C. Leakage of Rh⁺ blood of the fetus during the 1st birth
 - D. During the birth of the 3rd child
 - E. None of the above
60. Which type of WBC doesn't use the mechanism of phagocytosis to attack germs?
- A. Neutrophil
 - B. Basophil
 - C. Monocyte
 - D. Lymphocyte
 - E. Eosinophil
61. If you are ordered to calculate the MCV of the RBC indices, you are dealing with identifying:
- A. the size or volume of individual RBC
 - B. the percent volume of packed cell volume (Hct)
 - C. the concentration of hemoglobin of individual RBC
 - D. RBC-number in mm³ of blood
 - E. All of the above
62. T cell antigen receptor
- A. Recognizes conformational epitopes on the native antigen
 - B. Has Ig light chains
 - C. Is made up of a heavy chain and beta-2 microglobulin
 - D. Are associated with Ig α and Ig β to form a complex
 - E. Recognize epitopes on linear peptides associated with MHC molecules
63. The class I MHC processing pathway primarily
- A. Processes antigens that are present in the cytosol
 - B. Processes antigens from the extracellular environment
 - C. Generates peptides, complexes them with class I MHC molecules for presentation to helper T cells.
 - D. Generates peptides, complexes them with class I MHC molecules for presentation to NK cells
 - E. Is involved in the process of the antibody response
64. Dendritic cells are characterized by
- A. The expression of BCR
 - B. Expression of CD19
 - C. Expression of IgM molecules
 - D. Their ability to release histamine
 - E. Their interface between the innate and adaptive immune systems
65. C3b is
- A. Chemotactic
 - B. An anaphylatoxin
 - C. Opsonizes bacteria
 - D. Directly injures bacteria
 - E. The inactive form of C3
66. B cells are distinguished from T cells by the presence of
- A. CD4
 - B. Surface Ig

- C. CD8
 - D. CD3
 - E. Class I MHC antigen
67. Specific antibodies are readily detectable in the serum following primary contact with an antigen after:
- A. 11 min
 - B. 1 day
 - C. 5-7 days
 - D. 3-5 weeks
 - E. Only following a second contact with antigen
68. Which is NOT correct about heme?
- A. Heme is a component of cytochromes in the electron transport chain
 - B. Hemopexin, a plasma protein, binds to heme and reduces its toxicity
 - C. Heme regulates the expression of genes involved in cell differentiation and proliferation
 - D. Heme is never toxic, even at high concentrations in the blood
 - E. Heme can enter cell membranes easily and cause membrane damage
69. Which of the following diseases is associated mainly with an unconjugated hyperbilirubinemia?
- A. Crigler-Najjar syndrome type 1
 - B. Dubin-Johnson syndrome
 - C. Rotor syndrome
 - D. A gallstone obstructing the common bile duct (choledocholithiasis)
 - E. A tumor of the head of the pancreas obstructing the common bile duct
70. Which is NOT true about neonatal jaundice?
- A. Unconjugated hyperbilirubinemia is found essentially in all newborn babies
 - B. Unconjugated jaundice is worse in premature babies
 - C. Even mild elevations of serum bilirubin can be damaging to a newborn child
 - D. Phototherapy with blue light is useful for treating increasing jaundice in newborns
 - E. Unconjugated jaundice in a newborn child is partly due to hemolysis of excess fetal red blood cells during the neonatal period
71. Not visible
72. Which of the following antibodies are useful for the diagnosis of systemic lupus erythematosus?
- A. Anti-cardiolipin
 - B. Anti-nuclear antibody
 - C. Anti-SS-A(Re), anti-SS B(La)
 - D. Anti-21-hydroxylase
 - E. Anti-glomerular basement membrane
73. Which of the following is not organ-specific autoimmune disease?
- A. Myasthenia gravis
 - B. Multiple Sclerosis
 - C. Hashimoto's thyroiditis
 - D. Rheumatoid arthritis
 - E. Type 1 (Insulin dependent) diabetes mellitus
74. Susceptibility to the development of ankylosing spondylitis is very strongly associated with possession of

- A. HLA-DR4
 - B. HLA-DR3
 - C. HLA-B27
 - D. HLA-DQ
 - E. HLA-DQ8
75. All of the following diseases are examples of type II hypersensitivity expect
- A. Acute rheumatic fever
 - B. Grave's disease
 - C. Myasthenia gravis
 - D. Autoimmune hemolytic anemia
 - E. Poststreptococcal glomerulonephritis
76. A 6-year-old boy was suspected to suffer from pulmonary tuberculosis. After 2 days, intradermal injection of purified protein derivative (PPD); the boy developed reddening and induration of the injection site. This is indicative of which type of hypersensitivity reactions?
- A. Immediate hypersensitivity
 - B. Antibody mediated hypersensitivity
 - C. Immune complex mediated hypersensitivity
 - D. Delayed type hypersensitivity
 - E. Allergic reaction
77. A 13-year-old girl was recently diagnosed to be allergic to egg. What is the effect of repeated intake of eggs in this individual?
- A. The rapid response gets worse
 - B. Antibody production is decreased
 - C. Antibody formation remains constant
 - D. Patient will be desensitized
 - E. T cell activity increases significantly
78. A one year old male with known X-linked agammaglobinuria came with recurrent bacterial infections such as otitis media, bronchitis and pneumonia. This is due to:
- A. antibody mediated cell destruction
 - B. lack of opsonizing antibodies
 - C. drug resistant strains of the bacteria
 - D. autoantibody against self-antigen
 - E. lack or depressed cell mediated immunity
79. A gynecologist while doing cesarean section for an HIV positive female got accidentally pricked by a needle. What is his risk of seroconversion?
- A. 3.0%
 - B. 30%
 - C. 33%
 - D. 13%
 - E. 0.3%
80. Which immune marker is present during window period of HIV infection?
- A. antibodies to gp120
 - B. antibodies to gp41
 - C. p24 antigen

- D. p17 antigen
 - E. antibodies to CD4+
81. Which of the following statement is incorrect about HIV/AIDS?
- A. HIV infected cells evade cytotoxic T cells through down regulation of class I MHC molecule expression
 - B. Persistent generalized lymphadenopathy is seen in middle chronic phase of HIV infection
 - C. Antibody production against HIV is usually seen within 3 to 7 of presumed infection
 - D. HIV is a DNA virus
 - E. HIV can infect monocytes or macrophages and dendritic cells
82. Activation of alternative pathway involves
- A. C1q
 - B. C1s
 - C. C2
 - D. C3
 - E. C4
83. CD 8⁺ T cells
- A. Can be classified into Th1 and Th2 subgroups based on biological functions
 - B. Do not produce IFN- γ
 - C. Can recognize and kill virus infected cells
 - D. Can bind free virus
 - E. Do not require cell to cell contact with their targets for killing
84. B cells do not express
- A. CD2
 - B. MHC-I molecule
 - C. MHC-II molecule
 - D. CR2
 - E. LFA-3
85. The secondary, but not the primary, immune response is based on
- A. Memory
 - B. The bonus effect of multivalency
 - C. Complement activation
 - D. Mast cell degranulation
 - E. Clonal selection
86. The cytokine which is most involved in the class switch to IgE production is
- A. IL-1
 - B. IL-2
 - C. IL-3
 - D. IL-4
 - E. IL-5
87. The C3 convertase in classical pathway of complement activation is
- A. C3bBb
 - B. C4b2b
 - C. C4a2b
 - D. C4b2b3b

- E. C3bBb
88. Prior to class switching, B-cells express
- A. IgA alone
 - B. IgA and IgG
 - C. IgM and IgD
 - D. IgD alone
 - E. No surface Ig
89. Which of the following is produced by CD4 Th1 cells?
- A. IL-2 receptor
 - B. IL-8
 - C. C3
 - D. IL-4
 - E. IL-12
90. The class I MHC processing pathway primarily
- A. Processes antigens that are present in the cytosol
 - B. Processes antigen from the extracellular environment
 - C. Generates peptides, complexes them with class I MHC molecules for presentation to helper T cells
 - D. Generates peptides, complexes them with class I MHC molecules for presentation to NK cells
 - E. Is involved in the process of antibody response
91. A complement component which is strongly chemotactic for neutrophils is
- A. C9
 - B. C5a
 - C. C3
 - D. C3b
 - E. C5b
92. With regard to *blood*, which statement is *false*?
- A. Serum is the yellowish fluid remaining after blood has clotted
 - B. Albumin preserves more oncotic pressure within the vascular system
 - C. Fibrinogen is present in lower concentrations in plasma than in serum
 - D. The buffy coat of centrifuged blood consists of leukocytes and platelets
 - E. Platelets are cell fragments from megakaryocytes within the bone marrow
93. With regard to blood cells, which statement is *true*?
- A. Circulating antibodies are produced by monocytes
 - B. Reticulocytes are the most abundant polymorphonuclear leukocytes
 - C. Basophils have nuclei pinched into several lobes connected by narrow constrictions
 - D. Erythrocytes may undergo cell division after leaving the hematopoietic organ of their origin
 - E. A patient with a parasitic infection is likely to reveal an increase in the circulating eosinophils
94. *Erythrocytes*
- A. Can originate from a mast cell
 - B. Form H₂O₂ during phagocytosis

- C. Have precursors that produce hemoglobin in their SER
 - D. Are removed from circulation by macrophages after approximately 120 days
 - E. Enter the circulation by active amoeboid movement only after becoming fully mature
95. Which statement is TRUE about hematopoiesis?
- A. Myeloblasts possess specific granules
 - B. Metamyelocyte nucleus becomes highly polyploid
 - C. Megakaryocytes are the first recognizable cells in the erythroid series.
 - D. Large numbers of band cells in the blood usually indicate bacterial infection
 - E. The first identifiable progenitor of lymphoid cells is the basophilic myeloblast
96. Which of the following immunoglobulin types crosses the placental barrier?
- A. IgG
 - B. IgA
 - C. IgM
 - D. IgD
 - E. IgE

Blood and immunity Final (06 Batch)

1. Which of the following is false with regards to the blood?
 - A. Serum is the yellowish fluid remaining after blood has clotted
 - B. Albumin preserves osmotic pressure within the vascular system
 - C. Monocytes, the smallest, leukocytes, are classified as polymorphonuclear granulocytes
 - D. Basophils contain heparin and histamine
 - E. Plasma cells are derived from B lymphocytes
2. Erythrocytes
 - A. enter the circulation only after becoming fully mature
 - B. have precursors that produce hemoglobin in their RER
 - C. undergo mitosis in the circulation in response to erythropoietin
 - D. contain mitochondria and are capable of oxidative phosphorylation
 - E. are removed from circulation by macrophages after approximately 120 days
3. The correct order of frequency of the leukocytes is
 - A. Lymphocyte, monocyte, eosinophil, basophil, neutrophil
 - B. Neutrophil, lymphocyte, monocyte, eosinophil, basophil
 - C. Neutrophil, lymphocyte, basophil, monocyte, eosinophil
 - D. Neutrophil, monocyte, eosinophil, lymphocyte, basophil
 - E. Neutrophil, basophil, eosinophil, monocyte, lymphocyte
- ✓ Mnemonics = Never Let Monkeys Eat Bananas (N>L>M>E>B)
4. Concerning the blood cells, which statement is true?
 - A. Erythrocytes and platelets contain rounded central nuclei
 - B. Neutrophils, the largest leukocytes, survive and perform optimally in an anaerobic environment
 - C. Corticosteroids produce a rapid increase in the number of blood eosinophils
 - D. Erythrocytes and blood platelets perform their functions exclusively in the blood stream
 - E. Monocytes are the only type of leukocytes that return from tissue back to the blood, after diapedesis.
5. With regard to the lymphocytes, which statement is false?

- A. T cells constitute 65-75% of blood lymphocytes
 - B. T lymphocytes, which gain their immunocompetence in the thymus, recognize only epitopes
 - C. B cells recognize soluble antigens or antigens present on cell surfaces
 - D. Natural Killer Cells lack the marker molecules characteristic of B and T cells
 - E. None of the above
6. Megakaryocytes
- A. are multinucleated
 - B. are formed by the fusion of many haploid cells
 - C. contain a network of intracellular membranes
 - D. are located primarily in the spleen
 - E. serve as precursors to bone marrow macrophages
7. Which of the following is an encapsulated lymphatic organ?
- A. Lymph follicles
 - B. White pulp
 - C. Peyer's patches
 - D. Lymph node
 - E. Diffuse lymphatic tissue
8. What is the characteristic feature of a secondary nodule?
- A. Lymphocytes
 - B. Germinal center
 - C. Capsule
 - D. Trabeculae
 - E. None of the above
9. Which of the following is NOT a function of the spleen?
- A. Filtration of lymph
 - B. Destruction of old and damaged red blood cells
 - C. Lymphocyte and antibody production
 - D. Storage of blood
 - E. Fetal blood cell formation
10. When looking at a lymph node, what are lymphatic nodules?
- A. Deep cortex
 - B. Outer cortex
 - C. Tertiary cortex
 - D. Juxtamedullary cortex
 - E. Paracortical cortex
11. Concerning the immune and lymphoid organs, select the correct statement.
- A. Pharyngeal tonsil is partly encapsulated and covered by keratinized stratified squamous epithelium
 - B. A lymph node possesses blood-thymus barrier in the medulla
 - C. The lymphatic tissue in the spleen is called white pulp
 - D. The thymus receives afferent lymphatic vessels through its hilum
 - E. The appendix is a thymic dependent secondary lymphoid tissue
12. Regarding the thymus, which statement is true?

- A. Is relatively more developed with increasing age
 - B. Has Hassall's corpuscles located in the cortex, especially in the cords of Billroth
 - C. Has lymphoid nodules with germinal centers in its medulla and the paracortical sinus
 - D. Contains more B lymphocytes than T lymphocytes
 - E. None of the above
13. The *spleen*:
- A. Has smooth myocytes in its capsule
 - B. Filters lymph
 - C. Has no B lymphocytes
 - D. Has an outer cortex and inner medulla
 - E. Contains periarterial lymphoid sheath in its red pulp
14. The specificity of antibody secreted by a daughter B-cell may deviate from clonal parent to some extent due to:
- A. Class switching
 - B. Somatic hypermutation
 - C. Allelic exclusion
 - D. Alternative splicing
 - E. Different heavy: light pairing
15. Immune cell NOT phagocytic in nature?
- A. Neutrophil polymorphonuclear leukocytes
 - B. B lymphocytes
 - C. Microglial cells
 - D. Macrophages
 - E. Kupffer cells
16. Involved in the initial presentation of antigen to T lymphocytes?
- A. Dendritic cells
 - B. Plasma cells
 - C. Neutrophil polymorphonuclear leukocytes
 - D. Eosinophil
 - E. NK cells
17. Produce immunoglobulins of all isotypes
- A. Mast cells
 - B. Naïve B cells
 - C. Basophils
 - D. T lymphocytes
 - E. Plasma cells
18. Helper CD4+ T lymphocytes recognize _____ if the T cell has the correct antigen specificity in the TCR
- A. HLA class I antigen
 - B. HLA class III antigen
 - C. Processed peptides from antigen
 - D. CD8 antigen
 - E. Surface immunoglobulin
19. A chemo-attractant for eosinophils

- A. IL-5
 - B. IL-10
 - C. IFN γ
 - D. IL-2
 - E. IL-7
20. Which one of the following statements about antibody production is true?
- A. B cells cannot produce specific antibody without T-cell help
 - B. Tregs can produce antibodies to regulate autoantibody production
 - C. The BCR of B cells is a member of the immunoglobulin superfamily
 - D. A single B cell can make several different antibodies of different antigen-binding specificities
 - E. All of the above
21. The cells that die by neglect are
- A. Cells with TCR that have a low affinity for self
 - B. Cells with TCR that have a high affinity for self
 - C. Cells with TCR that have intermediate affinity for self
 - D. A and C
 - E. None of the above
22. Positive selection of thymocytes
- A. in the cortex
 - B. in the medulla
 - C. involves Epithelial cells of the thymic cortex (cTECs)
 - D. medullary thymic epithelial cells (mTECs)
 - E. A and C only
23. The cells that die by clonal deletion are
- A. Cells with TCR that have a low affinity for self
 - B. Cells with TCR that have a high affinity for self
 - C. Cells with TCR that have intermediate affinity for self
 - D. A and B
 - E. None of the above
24. Foxp3-expressing CD4⁺ T cells have been well characterized by regulatory T cells (Tregs). These cells suppress immune responses through numerous mechanisms:
- A. Production of anti-inflammatory cytokines
 - B. Direct cell-cell contact
 - C. Modulating the activation state and function of APCs
 - D. All of the above
 - E. A and C only
25. IgD
- A. Is lost up on antigen stimulation of B cells
 - B. Memory B cells lack this class of immunoglobulin
 - C. Co-expressed with IgG on B cells prior to class switching
 - D. Co-expressed with IgM on B cells prior to class switching
 - E. All except C
26. Class switch to IgE production is mediated by

- A. IL-4
 - B. What is described in A plus IL-13
 - C. What is described in A plus IL-5
 - D. What is described in A plus IL-13
 - E. A and B only
27. The classical complement pathway can be activated best by
- A. Endotoxin
 - B. Immunoglobulin IgE
 - C. IgD
 - D. IgM
 - E. None of the above
28. A live attenuated viral vaccine is acceptable to which one of the following groups
- A. Children under 8 years of age who are not immunodeficient
 - B. Patients treated with steroids
 - C. Pregnant mothers
 - D. Patients with leukemia
 - E. Patients treated with radiotherapy
29. A coreceptor for HIV is:
- A. CXCR4
 - B. CD8
 - C. CD4
 - D. CCR5
 - E. A and D only
30. Variolation refers to:
- A. The deliberate infection with small pox
 - B. The attenuation of virulent organisms
 - C. Inoculation of scab material into small skin wounds
 - D. The removal of scab material from an individual with small pox
 - E. A and C only
31. Herd immunity refers to
- A. Proportion immune among individuals in a population
 - B. Particular threshold proportion of immune individuals that should lead to a decline in incidence of infection
 - C. A pattern of immunity that should protect a population from invasion of a new infection
 - D. All of the above
 - E. None of the above
32. If someone has to develop an effective vaccine against mycobacterial diseases such as tuberculosis, the most important piece of the immune response to be stimulated is:
- A. A high titer of antibody
 - B. Antibody in the gut lumen
 - C. Cytotoxic T-cells
 - D. Macrophage-activating cell mediated immunity
 - E. Neutrophils

33. A potential disadvantage of immunological protection using passive transfer of immunoglobulins is
- A. Waning of immunity after vaccination
 - B. Development of immune complex mediated inflammatory reactions
 - C. Cost
 - D. All of the above
 - E. A and B only
34. A hapten is
- A. A carrier
 - B. A paratope
 - C. A small chemical grouping which reacts with preformed antibodies
 - D. An epitope
 - E. An immunogen
35. Antigen to antibody interaction:
- A. Involves covalent bonding
 - B. Is optimized by spatial complementarity
 - C. Is usually unaffected by pH
 - D. Unaffected by the presence or absence of water molecules
 - E. Usually unaffected by molecular rigidity
36. Intermolecular forces that contributes to the interaction between antibody and antigen:
- A. Electrostatic
 - B. Van der Waals
 - C. Hydrophobic
 - D. Hydrogen bonds
 - E. All of the above
37. Affinity is
- A. Measure of strength of the binding of antigen to antibody
 - B. Association constant of the Ag/Ab equilibrium
 - C. Related to the free energy change of the Ag/Ab interaction
 - D. Related to specificity
 - E. All of the above
38. The $\alpha\beta$ T-cell receptor recognizes
- A. MHC alone
 - B. Native antigen
 - C. Native protein antigen plus major histocompatibility complex (MHC) molecule
 - D. Processed peptide antigen plus MHC
 - E. Processed peptide antigen
39. Antigen processing for presentation by MHC class II molecules involves
- A. Invariant chain DM
 - B. TAP
 - C. Endosomal and lysosomal proteases
 - D. Cytosolic proteasome
 - E. A and C
40. TCR recognition of peptide-MHC complex

- A. Involves β and α chain of TCR
 - B. The immunodominant epitopes are only recognized
 - C. Involves CDR-mediated binding
 - D. All of the above**
 - E. A and C only
41. Cross-presentation of exogenous antigen to T cells doesn't require the involvement of:
- A. T cell receptor
 - B. Antigen processing
 - C. MHC class I
 - D. MHC class II**
 - E. An antigen presenting cell
42. NKT cells
- A. express $\alpha\beta$ TCR
 - B. recognize lipid antigens
 - C. Following stimulation secrete gamma interferon but not IL-4
 - D. Are restricted by CD1 but not by classical MHC molecules
 - E. All except C**
43. Super-antigens
- A. Bind to all members of a given V Beta T-cell receptor family
 - B. Bind to MHC class II
 - C. Cross link the T-cell to the MHC class II on an antigen presenting cell
 - D. Do not cause pathology
 - E. A and C**
44. A boy who presented at Tikur Anbessa Specialized Hospital Pediatrics clinic was diagnosed with defective development of the thymus. You would not expect this patient to have an increased incidence of infection with
- A. Viruses
 - B. Blood borne parasites
 - C. Fungi
 - D. Gram negative bacteria**
 - E. Intracellular bacteria
45. RAG-1 and RAG-2 genes trigger
- A. Somatic mutation
 - B. Somatic recombination**
 - C. Memory cell formation
 - D. Alternative Splicing
 - E. Isotype switching
46. Humoral immunity
- A. Is effective against extracellular microbes
 - B. Involves the complement system
 - C. Secreted antibodies are effector molecules
 - D. All of the above**
 - E. None of the above
47. Properties of adaptive immunity

- A. Diversity enables immune system to respond to large variety of antigens
 - B. Contraction and homeostasis allows the immune system to respond to newly encountered antigens
 - C. Memory leads to enhanced response to repeated exposures
 - D. Clonal expansion enables the immune system to cope up with high multiplication rate of microbes
 - E. All of the above
48. Identify the true statement about the stages in the life history of lymphocytes
- A. Memory cells have longer life than effector cells
 - B. The affinity of immunoglobulin on naïve and memory cells is similar
 - C. Naïve B cells have antigen receptor whereas activated once have reduced
 - D. All of the above
 - E. A and C only
49. Identify the wrong statement: Recognition of microbes by the innate immunity
- A. The specificity is for structural detail of microbial molecules
 - B. The specificity is for structures shared by classes of microbes
 - C. Receptors have limited diversity
 - D. Identical receptors are expressed on all cells of the same lineage
 - E. Have the capacity to discriminate self from non-self
50. Which one of the following is not the effector function of activated phagocytes
- A. Killing of microbes by the production of reactive oxygen species
 - B. Killing of microbes by the production of nitric oxide
 - C. Induction of inflammation
 - D. Activation of adaptive immunity
 - E. All of the above
51. Effector function of the complement includes
- A. Induction of inflammation
 - B. Induction of opsonization
 - C. Induction of phagocytosis
 - D. Lysis of microbes
 - E. All of the above
52. Identify an innate immune evasion mechanism employed by microbes
- A. Capsular polysaccharides
 - B. Production of catalase
 - C. Sialic acid expression
 - D. Synthesis of modified LPS
 - E. All of the above
53. Identify the wrong statement about the features of MHC
- A. Bind many different peptides
 - B. Display one peptide at a time
 - C. Bind only peptides
 - D. Display molecules for short period of time
 - E. Stability requires binding of peptides

54. If a medical personnel is requested to work in West African countries affected by the current Ebola epidemic as a volunteer, he/she is supposed to be vaccinated. However, as there is no vaccine for Ebola, the medical personnel received injection of antibodies pooled from individuals who recovered from the disease. The pooled antibodies, which should protect the medical personnel, are of which isotype?
- A. IgA
 - B. IgG**
 - C. IgE
 - D. IgD
 - E. None of the above
55. Which of the following compound is most likely to block ADP receptors and prevent platelet aggregation
- A. Clopidogrel**
 - B. Aspirin
 - C. Prostacyclin
 - D. Abciximab
 - E. Montelukast
56. The primary advantage of enoxaparin over heparin is that it
- A. is unlikely to cause bleeding
 - B. more effectively inhibits the synthesis of clotting factors
 - C. has a more rapid onset
 - D. does not cause thrombocytopenia
 - E. has a longer life**
57. Which of the following statements is true regarding the parenteral administration of Alteplase?
- A. It increases the formation of plasminogen
 - B. It is less effective than streptokinase when given after a myocardial infarction
 - C. It causes a high incidence of thrombocytopenia
 - D. It may cause bleeding reversible by aminocaproic acid**
 - E. It activates free plasminogen
58. A 54-year-old male with a prosthetic aortic valve replacement complained to his family physician of black and tarry stools. Physical examination and vital signs were unremarkable except for subconjunctival hemorrhages and bleeding gums. Stools tested positive for heme, and hematuria was observed. The patient has been receiving oral warfarin since his valve replacement 1 year earlier. Prothrombin time was found to be significantly elevated. Which one of the following therapies would provide the most rapid recovery from the observed bleeding secondary to warfarin treatment?
- A. Intravenous vitamin K
 - B. Transfusion of fresh-frozen plasma**
 - C. Intravenous protamine sulfate
 - D. Immediate withdrawal of warfarin treatment
 - E. Intravenous administration of anti-warfarin antibodies
59. The common enzyme produced by intrinsic and extrinsic clotting mechanisms that converts Prothrombin to thrombin is:
- A. Plasminogen

- B. Factor Xa
 - C. Factor VIIIa
 - D. Factor IX
 - E. None of the above
60. Squeezing out through the capillary bed by WBC's is generally called:
- A. Margination
 - B. Chemotaxis
 - C. Amoeboid motion
 - D. Diapedesis
 - E. None of the above
61. A factor that is not applied to calculate RBC indices is
- A. Hematocrit (%)
 - B. WBC number
 - C. RBC number
 - D. Hemoglobin concentration
 - E. None
62. The major cause of initiation of the extrinsic clotting mechanism involves the release of:
- A. Thromboplastin from traumatized tissues
 - B. Von Willebrand Factor in the blood
 - C. Factor VIII (antihemophilic factor) in damaged tissues
 - D. Histamine
 - E. Nitric oxide to nearby tissues
63. Serum is:
- A. Blood plasma from which globulin is removed
 - B. A blood that contains Fibrinogen and albumin
 - C. Blood plasma from which red and white blood cells are removed
 - D. Hematocrit portion of the blood
 - E. Blood plasma from which fibrinogen is removed
64. The false statement concerning the general function of blood:
- A. It transports CO₂ and O₂ gases
 - B. It transports Urea
 - C. It transports bilirubin
 - D. It transports urine
 - E. It transports heat
65. False anemia could occur in
- A. Fluid retention
 - B. Dehydration
 - C. Hemolytic anemia
 - D. In polycythemia
 - E. None of the above
66. Microcytic hypochromic anemias are caused by disorders of
- A. Globin synthesis
 - B. Red cell membrane
 - C. Hemoglobin synthesis

- D. DNA synthesis
 - E. None of the above
67. Sickle cell disease is caused by point mutation in the
- A. sixth codon of β -globin
 - B. sixth codon of α -globin
 - C. Replacement of a glutamate residue with a Lysine residue
 - D. A and C are correct
 - E. None of the above
68. Which of the following hemolytic anemia is caused by an acquired genetic defect?
- A. Sickle cell anemia
 - B. Immune hemolytic anemia
 - C. Paroxysmal Nocturnal Hemoglobinuria (PNH)
 - D. Pernicious anemia
 - E. None of the above
69. Qualitative defect of the VWF (Von Willebrand Factor) results in
- A. Type 1 von Willebrand disease
 - B. Type 2 von Willebrand disease
 - C. Type 3 von Willebrand disease
 - D. Type 4 von Willebrand disease
 - E. None of the above
70. Specific forms of megaloblastic anemia caused by autoimmune gastritis and an associated failure of intrinsic factor production is known as:
- A. Iron deficiency anemia
 - B. Anemia of chronic disease
 - C. Pernicious anemia
 - D. Sideroblastic anemia
 - E. None of the above
71. In the human body, iron is found as a component of all of the following except which one?
- A. Cytochromes
 - B. Myoglobin
 - C. Ferritin
 - D. Immunoglobulins
 - E. Hemosiderin
72. Which of the following is not associated with hemolytic jaundice?
- A. Sickle cell anemia
 - B. Obstruction of the bile duct due to a tumor in the head of the pancreas
 - C. *Plasmodium falciparum* infection
 - D. Glucose-6-phosphate dehydrogenase (G6PD) deficiency
 - E. Mismatched blood transfusions
73. The formation of biliverdin from heme breakdown releases which of the following?
- A. Carbon monoxide
 - B. Nitric oxide
 - C. Oxygen
 - D. Carbon dioxide

- E. Hydrogen peroxide
74. Which of the following diseases is associated with a largely conjugated hyperbilirubinemia?
- A. Crigler-Najjar syndrome type 1
 - B. Crigler-Najjar syndrome type 2
 - C. Gilbert's syndrome
 - D. Dubin-Johnson syndrome
 - E. Newborn jaundice
75. The enzyme in macrophages that breaks down heme to form biliverdin is:
- A. Heme reductase
 - B. Heme oxygenase
 - C. Heme hydrolase
 - D. Heme synthetase
 - E. Biliverdin reductase
76. Which of the following is not a cause of splenomegaly?
- A. Chronic myeloid leukemia
 - B. Acute myeloid leukemia
 - C. Chronic lymphocytic leukemia
 - D. Non-Hodgkin lymphoma
 - E. None of the above
77. Which of the following is not a feature of myeloid blasts?
- A. Auer rods
 - B. Prominent nucleoli
 - C. Cytoplasmic granules
 - D. Positive myeloperoxidase reaction
 - E. None of the above
78. Which one of the following mechanisms is involved in the development of anemia in CLL?
- A. Immune hemolysis
 - B. Decreased hematopoiesis
 - C. Non-immune hemolysis
 - D. A and B are answers
 - E. All of the above
79. Which of the following is not correctly matched?
- A. Nodular sclerosis Hodgkin lymphoma --- Lacunar variant of R-S cell
 - B. Lymphocyte depletion Hodgkin lymphoma --- L & H cell
 - C. Burkitt lymphoma --- "Starry-sky pattern"
 - D. Mixed cellularity HL --- B symptoms
 - E. None of the above
80. Which of the following is true regarding Burkitt lymphoma?
- A. It is more likely to involve lymph nodes than extra nodal tissue at the beginning of illness
 - B. Epstein-Barr virus genome is seen more frequently in sporadic than endemic Burkitt's
 - C. This tumor shows a low rate mitosis
 - D. Tumor cells undergo high rates of apoptosis
 - E. None of the above
81. Which of the following is associated with a poorer response to therapy in ALL?

- A. Female sex
 - B. Age range between two and ten
 - C. Philadelphia chromosome negative ALL
 - D. Higher total leukocyte count**
 - E. All of the above
82. Which of the following typically results in a diffusely effaced lymph node architecture?
- A. Small lymphocytic lymphoma**
 - B. Follicular lymphoma
 - C. Follicular hyperplasia
 - D. Sinus histiocytosis
 - E. None of the above
83. Which of the following is not a feature of MALT lymphoma?
- A. Increased incidence in patients with H. Pylori infection
 - B. Regression of some early lesions with antibiotic therapy
 - C. Association with autoimmune diseases
 - D. Common occurrence in the gastro-intestinal tract
 - E. None of the above**
84. What is more characteristic of Hodgkin's than Non-Hodgkin lymphoma?
- A. Extra-nodal disease
 - B. Waldeyer's ring involvement
 - C. Peripheral blood involvement (leukemic phase)
 - D. Younger age peak**
 - E. All of the above
85. A 45-year-old man comes to you complaining of an erythematous skin rash on his arms, thighs, and abdomen. Physical exam reveals no evidence of hepatosplenomegaly. The peripheral blood showed numerous lymphocytes with convoluted, highly irregular deep nuclear folding. Immunostaining shows these cells to be CD3+, CD5+ and CD4+. The patient is HTLV-1 negative. With the results of this information you determine that this man most likely has:
- A. T lymphoblastic lymphoma
 - B. Sezary syndrome**
 - C. Burkitt lymphoma
 - D. Mycosis fungoides
 - E. Adult T cell leukemia/lymphoma
86. A 58-year-old high school teacher complains of fatigue which is gradually getting worse over the past one year. Blood count reveals total leukocyte count of 84,000/ μ L with 4% neutrophils. You do a peripheral smear and see that there are frequent smudge cells on the slide and a large number of normal appearing small lymphocytes that make up 90% leukocytes. The most likely diagnosis is:
- A. Burkitt lymphoma**
 - B. Small lymphocytic lymphoma
 - C. Chronic lymphocytic leukemia
 - D. Acute lymphoblastic leukemia
 - E. None of the above
87. Which of the following is less likely to be seen in multiple myeloma?

- A. Vertebral body lytic lesion
 - B. Pathological fracture
 - C. Spinal cord compression
 - D. Renal failure
 - E. Hyperviscosity syndrome
88. Which of the following hypersensitivity reactions involves the formation of antigen-antibody (immune) complexes that get deposited on vessel walls or in extracellular tissue?
- A. Type I
 - B. Type II
 - C. Type III
 - D. Type IV
89. After going on a walk through the forest, a 30-year-old woman notices a slightly raised and tender irregular reddish rash on one forearm. This rash gradually increases in intensity for 2 days and then fades away in two weeks. Which of the following forms of immunologic hypersensitivity is most likely demonstrated in this patient?
- A. Type I
 - B. Type II
 - C. Type III
 - D. Type IV
90. Autoimmunity can result from all of the following hypersensitivities except:
- A. Type I
 - B. Type II
 - C. Type III
 - D. Type IV
91. A 23-year-old woman has had worsening fever, malaise along with a malar (butterfly) skin rash persisting for 3 weeks. Laboratory studies show a positive serologic test for anti-Smith antibody, with a titer of 1:2048. Which of the following is the most likely diagnosis?
- A. Diffuse systemic sclerosis
 - B. Polyarthritis nodosa
 - C. Systemic lupus erythematosus (SLE)
 - D. Rheumatoid Arthritis
92. Individuals with immunodeficiencies are at risk for:
- A. Hypersensitivity reactions
 - B. Fungal infections only
 - C. Autoimmune diseases
 - D. Opportunistic infections
93. The microorganism that causes acquired immunodeficiency syndrome (AIDS) is a:
- A. gram-negative bacterium
 - B. gram-positive bacterium
 - C. retrovirus
 - D. protozoan
94. A 30-year-old male with known HIV infection presented with multiple reddish purple nodules on the skin of both legs. This is typical for
- A. Herpes Zoster

- B. Kaposi's Sarcoma
 - C. Candidiasis
 - D. Toxoplasmosis
95. Acquired immunodeficiency syndrome (AIDS) is defined as CD4⁺ T cell count below
- A. 1200 cells per μL
 - B. 500 cells per μL
 - C. 400 cells per μL
 - D. 200 cells per μL

Neoplasia Test-1 06 Batch

1. Tumors having a mixed pattern EXCEPT
 - A. Teratomas
 - B. Pleomorphic adenoma of salivary gland
 - C. Fibroadenoma of the breast
 - D. Adenocarcinoma of the colon
2. If the epithelial tumor is almost entirely composed of parenchymal cells, it is called:
 - A. Anaplastic carcinoma
 - B. Scirrhus carcinoma
 - C. Medullary carcinoma
 - D. Anaplastic carcinoma
3. Not a paraneoplastic syndrome
 - A. Polycythemia by renal cell carcinoma
 - B. Hypoglycemia by fibrosarcomas
 - C. Hypercalcemia by squamous cell carcinoma of the lung
 - D. Clubbing of the fingers by lung cancers
4. Methotrexate is
 - A. A purine antagonist
 - B. A folic acid antagonist
 - C. An antibiotic
 - D. An alkylating agent
5. The antibiotic for cancer chemotherapy:
 - A. Cytarabine
 - B. Doxorubicin
 - C. Gentamycin
 - D. Etoposide
 - E. Chloroquine
- ✓ Doxorubicin is cardiotoxic – it results cardiomyopathy
6. Fluorouracil belongs to:
 - A. Antibiotics
 - B. Antimetabolites
 - C. Plant alkaloids
 - D. Bone marrow growth factor
 - E. Antifungal
7. The mechanism of action of anticancer drugs belonging to plant alkaloids?
 - A. Inhibition of DNA-dependent RNA synthesis

- B. Mitotic arrest at a metaphase
 - C. Nonselective inhibition of aromatases
 - D. Inhibit cell wall synthesis
 - E. None
8. miRNAs may be better targets than mRNAs for cancer diagnosis because:
- A. Increased transcription coverage
 - B. Fewer sample required for validation
 - C. Detectable in more tissue/biofuels
 - D. All of the above
9. Which of the following are known to cause cancer?
- A. Viruses
 - B. Radiations
 - C. Chemicals
 - D. All of these
10. Genes that normally prevent cell division are:
- A. Tumor suppressors
 - B. Transcription factors
 - C. Proto-oncogenes
 - D. Growth factors
 - E. Oncogenes
11. Eating 5 fruits or vegetables each day can cut your cancer risk by 20%
- A. TRUE
 - B. FALSE
12. The function of microRNA (miRNA) is to:
- A. Serve as part of the structure of the ribosome
 - B. Transfer amino acids from the cellular fluid to the ribosome for protein synthesis
 - C. Transfer genetic code to the ribosome for protein synthesis
 - D. Controls genetic expression by turning some genes on and other off thus controlling genetic expression
13. Genomic scars in cancer is important for:
- A. Most appropriate therapy
 - B. Identification of cancer causing agents
 - C. Identification of cancer cell
 - D. All of the above
14. What is ICGC?
- A. International Cancer Genome Consortium
 - B. International Center for Genome Consortium
 - C. International Cancer Genome Center
 - D. International Cancer Gene Consortium
15. Human papillomavirus can cause cancer in humans and most commonly associated with
- A. Rectal polyps
 - B. Breast cancer
 - C. Prostate cancer
 - D. Anogenital cancer

- E. Mesotheliomas
16. Viruses can cause cancer in humans and animals. A principle of viral carcinogenesis is the
- A. Retroviruses cause most type of human cancers
 - B. Not all infections with a human cancer virus lead to tumor formation
 - C. Short latent periods elapse between time of virus infection and tumor appearance
 - D. Animal models seldom predict cellular mechanisms and human cancer
 - E. Host factors are insignificant in influencing the development of virus induced human cancer.
17. A 38-year-old woman with multiple sex partners is diagnosed with cervical cancer. This cancer has sexual transmission and viral etiology. The causative of this cancer is
- A. Hepatitis C virus
 - B. Hepatitis B virus
 - C. HPV high risk groups (16 & 18)
 - D. Polyomavirus
 - E. Herpes viruses
18. Only those RNA viruses are involved in human cancer who have
- A. Reverse transcriptase
 - B. Zoonotic viruses
 - C. Genome with double stranded RNA
 - D. Genome with single stranded RNA
 - E. Envelope
19. Which of these is recognized as cancer producing bacterium?
- A. Salmonella typhi
 - B. Mycobacterium tuberculosis
 - C. Streptococcus bovis
 - D. Chlamydia pneumonia
 - E. Helicobacter pylori
20. Nonspecific mechanism of carcinogenesis of Helicobacter Pylori seems to be
- A. Toxins production
 - B. Induction of inflammation
 - C. Bacteremia
 - D. Pathologic only to humans
 - E. Actively mobile
21. Which of the aflatoxins is the most important in cancer production?
- A. Aflatoxin B1
 - B. Aflatoxin B2
 - C. Aflatoxin G1
 - D. Aflatoxin G2
 - E. Aflatoxin M1
22. Which one of the following is FALSE about ionizing radiation (IR)?
- A. It is a mixed blessing for human beings
 - B. Low LET ones are more applicable in our countries
 - C. the biologic effect appears prior to chemical effect
 - D. two people may get the same dose of radiation but respond differently

- E. Significant effects are due to the interaction of IR with water
23. Which one of the following is INCORRECT about cancer epidemiology?
- A. Globally, the most prevalent form of cancer is lung cancer
 - B. 40% of cancer are preventable
 - C. Only 10% of cancer is attributed to hereditary factors
 - D. Some sort of cancers is communicable
 - E. Targeting on prevention is invaluable strategy to reduce cancer mortality
24. Which class of drugs binds avidly to tubulin and cause arrest of cells in metaphase?
- A. Vinca alkaloids
 - B. Nitrogen mustards
 - C. Alkylating agents
 - D. Anticoagulants
 - E. Antimetabolites
25. _____ is an alkylating agent which is a pro drug that needs metabolism to have an effect and one of its metabolite causes hemorrhagic cystitis.
- A. Fluorouracil
 - B. Tamoxifen
 - C. Bevacizumab
 - D. Cyclophosphamide
 - E. Hydroxyurea
26. _____ is an antidote for cisplatin induced nephrotoxicity
- A. Deferoxamine
 - B. Amifostine
 - C. Carboplastine
 - D. EDTA
 - E. Flumazenil
27. One of the following is not a mechanism for methotrexate (MTX) resistance?
- A. Decrease drug transport
 - B. Decrease formation of cytosolic MTX polyglutamates
 - C. Increase expression of enzyme that facilitate MTX metabolism
 - D. Synthesis of increased levels of DHFR through gene amplification
 - E. Altered DHFR with reduced affinity for MTX
28. If allopurinol is used in combination with Drug X anticancer agents, the dose of the DRUG X anticancer drug should be decreased. DRUG X is:
- A. Methotrexate
 - B. Cyclophosphamide
 - C. Chlorambucil
 - D. 6-mercaptopurine
 - E. 6-thioguanine
29. One of the following anticancer agents has a potential to cause congestive heart failure:
- A. Doxorubicin
 - B. Cisplatin
 - C. Hydroxyurea
 - D. Fludarabine

- E. None
30. _____ is a recombinant humanized monoclonal antibody that targets all forms of VEGF- α :
- A. Tamoxifen
 - B. Etoposide
 - C. Cytarabine
 - D. Fludarabine
 - E. Procarbazine
- ✓ The Question is incorrect. Bevacizumab is the monoclonal antibody that is VEGF-alpha blocker

Neoplasia Test-2 06 Batch

1. Which one of the following gene is less likely involved in regulating apoptosis?
 - A. p53
 - B. Bcl-2
 - C. Rb
 - D. Bax and Puma
 - E. A and C are answers
2. One of the following genes is NOT concerned about maintenance of genomic stability, which is it?
 - A. P53
 - B. Rb
 - C. Ras
 - D. p21
 - E. None of the above
3. The typical feature of intrinsic apoptotic pathway is:
 - A. The release of cytochrome c from mitochondria
 - B. The involvement of TNF and their ligands
 - C. The involvement of caspases and other enzymes
 - D. The blebbing of the cell membrane
 - E. All of the above
4. Identify the disorder that can happen due to decreased apoptosis:
 - A. Cancer
 - B. Hyperplasia
 - C. Atherosclerosis
 - D. Neurodegeneration
 - E. All of the above
5. The forefront hit tumor suppressor gene in colorectal cancer is:
 - A. APC
 - B. HNPCC
 - C. DCC
 - D. KRAS
 - E. p53
6. DNA damage can occur due to:
 - A. ROS and RNS produced by the cell itself
 - B. Hydrolysis and deamination
 - C. Diagnostic x-ray radiations
 - D. Toxic chemicals in food

- E. All of the above
7. Which types of DNA damage are hard to repair?
- A. DSB
 - B. SSB
 - C. Nucleotide removal
 - D. Both A and B are answers
 - E. All of the above
8. Which one of the following enzyme involved at final step of DNA damage repair?
- A. DNA polymerase
 - B. DNA ligase
 - C. Nucleases
 - D. Proteases
 - E. All
9. Colorectal disease can be linked to:
- A. MSI
 - B. Mutation in MMR
 - C. Mutation in APC
 - D. All of the above
 - E. None
10. Which pathway is incorrect?
- A. ATM --- p53 --- p21 --- cell cycle arrest at G1
 - B. ATR --- CHK1 --- CDC25-G2 arrest
 - C. ATM --- p53 --- Bax --- apoptosis
 - D. ATM --- BRCA1 --- Bak --- apoptosis
 - E. All of the above
11. Which one of the following statement is FALSE about breast cancer?
- A. BRCA 1 mutation is more common in breast cancer than ovarian cancer
 - B. Mutation in both BRCA1 and BRCA2 are important in male breast cancer
 - C. BRCA 1 mutation breast cancers tend not to respond for endocrine therapy
 - D. In ER+ and BRCA 1 mutated breast cancers, tamoxifen alone is a drug of choice
 - E. Only B and D are answers
12. Mutation in which of the following gene may indicate that the man is at higher risk for breast cancer development?
- A. BRCA 2
 - B. BRCA 1
 - C. p53
 - D. Her-2/Neu
 - E. All of the above
13. Kebede is born from two carrier parents for xeroderma pigmentosum, which is an autosomal recessive disorder with repair defects. What is the probability that Tolossa, Bekele's first son, to develop the recessive disorder if Kebede marries a carrier lady?
- A. 25%
 - B. 50%
 - C. 100%

- D. 75%
- E. 12.25%
14. The drawbacks of PSA and tumor biomarkers are that their expression is highly elevated in:
- A. BPH
 - B. Prostatitis
 - C. Infarction
 - D. Prostate CA
 - E. All except D are answers
- ✓ All including D can be answer – the question is not correct.
15. A 40-year-old come to a clinic with a suspected breast cancer. The pathologist has collected biopsy samples from her breast and carried out immunohistochemistry. The results are as follows:

ER	PR	PS2	BRCA 1 mutation	Her-2/Neu
-	-	-	+	+

According to the above result, which one of the following is incorrect?

- A. M will probably benefit from high dose of adjuvant chemotherapy
 - B. Unfortunately, she has high chance to develop ovarian CA
 - C. Tamoxifen can be used in combination with other chemotherapy
 - D. The probability of the tumor to spread into the next breast is high.
 - E. Unfortunately, she will have poor prognosis and short survival
16. EGF promotes cell proliferation through one of the following intracellular signaling cascade:
- A. The RAS/ERK pathway
 - B. PI3 kinase/AKT pathway
 - C. JAK/STAT pathway
 - D. A and B
 - E. All of the above
17. One of the following regulate the activity of Ras protein:
- A. Grb2
 - B. GTPase activating protein
 - C. Guanine nucleotide Exchange Factor
 - D. B and C
 - E. A and B
18. One of the following growth factor uses cell surface receptor with an intracellular serine/threonine kinase domain:
- A. PDGF
 - B. EGF
 - C. VEGF
 - D. TGF beta
 - E. FGF
19. What is the first event which happens to a growth factor/cytokine receptor after binding its cognate signal molecule?
- A. Binding of signal activates the receptor's cytosolic tyrosine kinase domain
 - B. Binding of signal causes the receptor to pick up a tyrosine kinase from cytosol

- C. Binding of signal causes receptors to dimerize
- D. Binding of signal causes tyrosine in the receptor's C-terminal tail to become phosphorylated
20. One of the following is the major mechanism by which proto-oncogene is converted into oncogene:
- A. Point mutation
- B. Chromosomal rearrangements
- C. Gene amplification
- D. Incorporation of new promoter
- E. All of the above
21. Burkitt lymphoma is caused by
- A. Chromosomal translocation
- B. Point mutation
- C. Gene amplification
- D. Incorporation of new promoter
22. One of the following intracellular protein has a tyrosine kinase activity:
- A. SRC
- B. SOS
- C. GAP
- D. RAF
23. The RAS/MAPK pathway
- A. Eventually results in activation of nuclear factors such as c-MYC
- B. In cancer, is only known to be activated by mutation in *RAS*.
- C. Is not involved in cell death regulation
- D. Is often activated via receptor tyrosine kinases
24. In Philadelphia chromosomes (CML), where could be the fused gene?
- A. Chromosome 9
- B. Chromosome 22
- C. Chromosome 8
- D. Chromosome 11
25. Which one of the following EGF receptor families acts as a co-receptor for EGF?
- A. erbB2 and erbB4
- B. erbB3 and erbB4
- C. erbB2 and erbB3
- D. erbB and erbB2
26. What is the outcome of JAK/STAT signaling activated by epidermal growth factor?
- A. Cell proliferation
- B. Migration
- C. Cell survival
- D. All of the above
27. The TGF β acts through its serine/threonine kinase receptor by activating which one of the following intracellular signaling molecules?
- A. SMAD
- B. STAT

- C. JAK
 - D. SRC
28. Which one of the following statements about chromosomal translocation is correct?
- A. Gene disruption occurs but new properties cannot be acquired.
 - B. No loss of genetic material occurs in balanced translocations
 - C. Translocations are carried by fewer than 1 in 10,000 individuals
 - D. Translocations cannot be inherited
 - E. When a translocated chromosome is involved in meiosis, it cannot give rise to a normal gamete.
29. Which ONE of the following is NOT a characteristic of oncogenes?
- A. A significant number encode protein phosphatases
 - B. Their gene products function mainly in growth factor signaling pathways
 - C. They are derived from normal genes
 - D. They are dominant in their mode of action
 - E. They are highly conserved between species
30. Which of the following statements regarding epigenetic inheritance is FALSE?
- A. Epigenetic inheritance can be reset during gametogenesis
 - B. Epigenetic inheritance can temporarily affect an individual
 - C. Epigenetic inheritance doesn't involve a change in DNA sequence
 - D. A and C
 - E. None of the above.

Musculoskeletal Test-1 07 Batch

1. Which one of the following statement is wrong?
 - A. Laterally, lacuna vasorum is lined by lacunar ligament
 - B. Branch of genitofemoral nerve is content of lacuna vasorum
 - C. Lacuna vasorum and lacuna musculorum commonly bounded by iliopectine arch
 - D. Iliopsoas muscle passes through lacuna musculorum
 - E. Lacuna vasorum and lacuna musculorum have common anterior boundary
2. Which of the following is not a content of adductor canal?
 - A. Femoral artery
 - B. Femoral vein
 - C. Great saphenous vein
 - D. Saphenous nerve
 - E. Nerve to vastus medialis
3. Which one is not true about patella?
 - A. It is the largest sesamoid bone in the human body
 - B. Its anterior surface is covered by hyaline cartilage
 - C. It is located in the tendon of quadriceps femoris muscle
 - D. It has large lateral and small medial facets
 - E. It has triangular shape where its apex is located inferiorly
4. Great saphenous vein blood from which of the following veins it drains into femoral vein except:
 - A. Superficial circumflex iliac vein
 - B. Superficial external pudendal nerve
 - C. Superficial epigastric vein

- D. Accessory vein
 - E. None of the above
5. Which bone of the foot does not involve for the attachment of either leg or foot muscles of the human body?
- A. Talus
 - B. Calcaneus
 - C. Metatarsals
 - D. Cuboid
 - E. None of the above
6. Which one does not occur during Pott's fracture?
- A. Fracturing the medial malleolus
 - B. Fracture of lateral malleolus
 - C. Forceful inversion of foot
 - D. Fracture of inferior part of fibula
 - E. None of the above
7. Which one is true about the ligaments of the knee joint and associated structures?
- A. Anterior cruciate ligament prevents hyperextension of knee joint
 - B. Posterior cruciate ligament is weaker than anterior cruciate ligament
 - C. Patellar ligament is an intracapsular ligament of knee joint
 - D. Lateral meniscus is firmly attached to the lateral collateral ligament
 - E. None of the above
8. Which one of the following muscle does not contribute for the flexion of knee joint?
- A. Sartorius
 - B. Short head of biceps femoris
 - C. Soleus
 - D. Plantaris
 - E. Popliteus
9. One of the following statement is not true about the innervation of the foot:
- A. All interossei muscles are innervated by lateral plantar nerve
 - B. First lumbrical and flexor hallucis get innervation from medial plantar nerve
 - C. Dorsum of foot can get cutaneous innervation from deep peroneal nerve
 - D. Oblique head of adductor hallucis is innervated by medial plantar nerve
 - E. Major cutaneous innervation of the sole of the foot is from medial plantar nerve
10. Which muscle raises the hip bone of the suspended leg during walking acting on the supported side of the body?
- A. Gluteus maximus
 - B. Gluteus medius
 - C. Obturator internus
 - D. Obturator externus
 - E. Quadratus femoris
11. During descent of the femoral hernia through the femoral canal, the neck of the sac will lie:
- A. At the saphenous ring
 - B. Medial to the lacunar ligament
 - C. Above and medial to the pubic tubercle

- D. At the mouth of obturator canal
 - E. Below and lateral to the pubic tubercle
12. Which of the following muscle involved in the dorsiflexion of the foot at the ankle joint:
- A. Extensor digitorum brevis
 - B. Tibialis posterior
 - C. Extensor hallucis brevis
 - D. Tibialis anterior
 - E. All except B
13. All of the following structures are passing through greater sciatic foramen except:
- A. Superior gluteal artery
 - B. Obturator internus tendon
 - C. Pudendal nerve
 - D. Inferior gluteal vein
 - E. None of the above
14. Which muscle is wrongly paired with its innervation?
- A. Superior gemellus ---- nerve to obturator internus
 - B. Inferior gemellus --- nerve to quadratus femoris
 - C. Tensor fascia lata --- superior gluteal nerve
 - D. Gluteus minimus --- inferior gluteal nerve
 - E. Sartorius --- femoral nerve
15. Abnormally lateral deviation of great toe is described by:
- A. Hammer toe
 - B. Hallux valgus
 - C. Hallux flexus
 - D. Talipes valgus
 - E. Claw toes
16. Which one of the following muscle helps to unlock the knee joint to start walking?
- A. Popliteus
 - B. Biceps femoris
 - C. Semitendinosus
 - D. Semimembranosus
 - E. Tibialis posterior
17. Foot drop which causes slap walking of the foot is the motor deficit of:
- A. Tibial nerve
 - B. Femoral nerve
 - C. Obturator nerve
 - D. Deep peroneal nerve
 - E. Superficial peroneal nerve
18. Ankle joint gets blood supply from various arteries. Which one of the following artery arises from posterior tibial artery?
- A. Anterior medial malleolar artery
 - B. Anterior lateral malleolar artery
 - C. Medial tarsal artery
 - D. Lateral tarsal artery

- E. None of the above
19. The regions of hyaline cartilage matrix characterized with abundant collagen type II with smaller IX collagen:
- A. Capsular matrix
 - B. Peri-cellular matrix
 - C. Territorial matrix
 - D. Inter-territorial matrix
 - E. None
20. Rows of chondrocytes in lacunae between collagenous fibers and chondrocytes are fewer and smaller and are not seen in isogenous groups is a characteristic of:
- A. Hyaline cartilage
 - B. Elastic cartilage
 - C. Fibrocartilage
 - D. Dens irregular connective tissue
 - E. None of the above
21. The zones of articular cartilage characterized by elongated and flattened chondrocytes surrounded by a parallel collagen II
- A. Tangential zone
 - B. Transitional zone
 - C. Radial zone
 - D. Calcified zone
 - E. Tidemark
22. The collagen molecule that facilitates fibril interaction with the matrix proteoglycan molecules in cartilages:
- A. Type I collagen
 - B. Type II collagen
 - C. Type IX collagen
 - D. Type X collagen
 - E. Type XI collagen
23. A collagen fiber that periosteum with the compact layer of the bone is
- A. Tome's fiber
 - B. Sharpey's fiber
 - C. Volkmann's canal
 - D. Canaliculi
 - E. Trabeculae
24. Which of the following cytoplasmic zone of osteoclast cell houses most of the organelles including multiple nuclei
- A. Clear zone
 - B. Basal zone
 - C. Ruffled border
 - D. Vesicular zone
 - E. None
25. Which of the following is wrong?
- A. Organic matrix of the bone provides resilience and resist twisting force

- B. Enamel is the richest mineralized tissue of your body in hydroxyapatite
 - C. Interstitial growth results from cell division of preexisting chondrocytes
 - D. Chondroblast produce antiangiogenesis factor in cartilage
 - E. Spongy bone lacks osteons
26. Which protein of the smooth muscle cell is related to the troponin C of the skeletal muscle
- A. Caldesmon
 - B. Calmodulin
 - C. α -actin
 - D. Vimentin
 - E. None
27. A skeletal muscle protein thought to lack laminin, which resides in the external lamina of the muscle cell, to actin filaments
- A. Dystrophin
 - B. Cap Z
 - C. Nebulin
 - D. Tropomodulin
 - E. Myomesin
28. A myosin-binding protein that holds thick filaments in register at the M line is:
- A. Myomesin and C proteins
 - B. Cap Z
 - C. Tropomodulin
 - D. Nebulin
 - E. Titin
29. Which of the following permits each myosin II molecule to contact the thin filament, forming a cross-bridge between the two filament types
- A. Heavy meromyosin
 - B. Light meromyosin
 - C. S1 and S2 moieties
 - D. The junction between heavy and light meromyosin
 - E. The junction between S1 and S2 moieties
30. The type of coetaneous gland formed from Simple alveolar glands:
- A. Merocrine sweat glands
 - B. Sebaceous glands
 - C. Ceruminous glands
 - D. Mammary glands
 - E. Apocrine sweat glands
31. In cross-sectional anatomy of the hair, the part that is formed by flat keratinized cells that overlap like shingles with their free edges pointing upward is
- A. Medulla
 - B. Cortex
 - C. Cuticle
 - D. Epithelial root sheath
 - E. Dermal root sheath
32. The layer of epidermis that lack nucleus but have cytoplasm with granules is

- A. Stratum basale
 - B. Stratum spinosum
 - C. Stratum lucidum
 - D. Stratum corneum
 - E. Stratum granulosum**
33. Which of the following **is not** a correct statement regarding the development of the skeletal system?
- A. Develops from mesoderm and neural crest cells
 - B. Somatomes are segmental blocks of the somatic mesoderm that form the skull**
 - C. Somites are segmental blocks formed by paraxial mesoderm
 - D. Sclerotome gives rise to the mesenchymal cells
 - E. Dermatome is formed from the dorsolateral part of somites
34. Which portion of the muscle is formed by the sclerotome cells?
- A. Myofibrils
 - B. Perimysium
 - C. Tendon**
 - D. Fascia
 - E. Muscle fiber
35. Which of the following bones is showing **an incorrect match for** its derivation?
- A. Frontal --- Neural crest
 - B. Occipitalis ---- Somites
 - C. Limbs ---- Somatic mesoderm
 - D. Hyoid ---- Neural crest
 - E. Mandible ---- paraxial mesoderm**
36. Which of the following limb defects are caused by contractive action of amniotic bands?
- A. Syndactyl
 - B. Brachydactyly
 - C. Amputation**
 - D. Polydactyl
 - E. Phocomelia
37. Select the **wrong** statement regarding sutures of a newborn's skull?
- A. Scaphlocephaly is a craniosynostosis of the sagittal suture
 - B. Mastoid and sphenoidal fontanelles close earlier than the lambdoid fontanelle
 - C. Provide molding of the skull during birth
 - D. Cranial bones continue to grow until age 5 to 6
 - E. Anterior fontanelle normally closes at about 3 months after birth**
38. Which of the following fetal skeleton completely ossify before birth?
- A. Vertebrae
 - B. Long bones
 - C. Auditory ossicles**
 - D. Acetabulum
 - E. Skull

47. Which of the following genes are **wrongly** matched with their action:
- A. BMP signaling --- developing edges of neural folds
 - B. Expression WNT1 --- epithelial to mesenchymal transition of neural crest cells
 - C. Midbrain --- neural crest cells in rhombomeres**
 - D. Rhombomeres --- segmental pattern of neural crest cells
48. Select segments of neural crest cells or rhombomeres, which migrate to pharyngeal arches 4-6?
- A. R3 and R5
 - B. R1 and R2
 - C. R4
 - D. R8
 - E. R6 and R7
- ✓ The question is not right – pharyngeal arches 4 and 6 don't have neural crest cells (rhombomeres).
49. Which of the following birth defects is not caused by disruption of neural crest cells?
- A. Occuloauriculovertebral syndrome
 - B. Branchial fistula**
 - C. Mandibulofacial dyastosis
 - D. Robin sequence syndrome
 - E. DiGeorge anomaly
50. Select a **wrong** statement about tongue development?
- A. Mucous of posterior tongue is innervated by lingual nerve**
 - B. Body of the tongue is derived from the first pharyngeal arch
 - C. Second, third and fourth pharyngeal arches form root of the tongue
 - D. Myotomes of occipital somites form tongue muscles
 - E. Terminal sulcus separates anterior and posterior parts of tongue.

Musculoskeletal test-3 07 Batch

1. One of the following statements is **false** regarding risk factors for osteoarthritis:
 - A. Age and presence of osteoarthritis are positively correlated
 - B. An association between osteoarthritis and obesity is not established**
 - C. Bone material density and the presence of osteoarthritis are positively correlated
 - D. A family history of osteoarthritis is common in patients with osteoarthritis
 - E. Osteoarthritis may appear in younger individuals in patients with previous injuries to a joint and a congenital developmental deformity of a joint(s).
2. Brodie's abscess refers to
 - A. A well delineated liver abscess caused by Entamoeba histolytica
 - B. Intra-osseous abscess walled off by reactive bone**
 - C. Ovoid and white opalescent tissue cysts caused by cysticerci
 - D. Subchondral cyst caused by osteoarthritis
 - E. A well circumscribed sclerotic bone lesion on X-Ray
3. Tuberculous osteomyelitis
 - A. Is usually blood borne during the initial stage of primary infection
 - B. Is usually multiple and less destructive than pyogenic infection in healthy individuals**
 - C. Long bones in extremities are the commonest sites of infection
 - D. Usually begins in diaphysis of appendicular skeleton
 - E. Rarely involves joint spaces

4. Which one of the following is not true about primary gout?
- A. It accounts for the majority of idiopathic cases
 - B. It is caused by overproduction of uric acid while uric acid excretion is increased, normal or reduced in the majority of cases
 - C. The main clinical manifestation is that of gouty arthropathy
 - D. May result from chemotherapy of an aggressive tumor or leukemia
 - E. Unknown enzymatic defect underlies the majority of cases
5. Morphologic changes in osteoarthritis **does not** include:
- A. Vertical and horizontal fibrillation and cracking of articular cartilage
 - B. Grossly the surface becomes necrotic and hemorrhagic with abscess and granuloma formation
 - C. Portions of the cartilage are sloughed and dislodged
 - D. Rebuttredding and sclerosis of the underlying cancellous bone
 - E. Matrix water increases while proteoglycans decreases
6. A pathology report from a skin biopsy states the diagnosis as Condyloma accuminata. Which of the following is true?
- A. It is a malignant skin tumor arising from basal cells
 - B. It is a chronic inflammation of the skin due to viral infection
 - C. It is a benign neoplasm caused by viral infection
 - D. It is infection of the skin by tuberculosis
 - E. None
7. A 30-year-old male HIV infected patient developed an itching painful lesion over the right side of the chest following a dermatome, what is the most probable diagnosis?
- A. Herpes simplex infection
 - B. Chicken pox
 - C. Herpes zoster
 - D. Scrofuloderma
 - E. Seborrheic keratosis
8. One of the following is not among the routes of infection of tuberculosis of the skin
- A. Sexually transmitted
 - B. Hematogenous spread
 - C. Scrofuloderma from deeper infected lymph node
 - D. Direct inoculation
 - E. None
9. A 70-year-old man came to the clinic with an ulcer around his nose of 1 year duration, biopsy was taken and histopathology shows nests of basaloid cells with peripheral palisading invading the dermis. Which diagnosis is correct?
- A. Mucocutaneous leishmaniasis
 - B. Squamous cell carcinoma
 - C. Melanoma
 - D. Lymphoma
 - E. Basal cell carcinoma
10. Which statement is incorrect regarding melanomas?
- A. Sun exposure is an important cause

- B. Can be seen in an adult as a congenital lesion with no change over long period of time
- C. Can metastasize to distant organs
- D. Variegated color of the lesion is one of the early signs
- E. None

11. Among the benign bone tumors, the most frequent is:

- A. Osteochondroma
- B. Chondroma
- C. Osteoid osteoma
- D. Osteoma
- E. Chondroblastoma

✓ Note – if the question asks

- i. What is the most common bone tumor?
 - 1. It is metastatic bone tumor (i.e. secondaries are common as a whole)
- ii. What is the most common primary bone tumor?
 - 1. It is Osteochondroma
- iii. What is the most common primary malignant bone tumor?
 - 1. It is osteosarcoma
- iv. What is the most common primary benign bone tumor?
 - 1. It is Osteochondroma (among primary bone tumors – the benign ones are most common)

12. The following two benign but locally aggressive bone neoplasm always arises in the epiphysis of long bones:

- A. Clear cell chondromas and osteosarcomas
- B. Giant cell tumors and chondroblastomas
- C. Chondromyxoid fibromas and chondromas
- D. Osteoid osteomas and osteochondromas
- E. Osteomas and hemangiomas

➤ 13 – 19 are missing

20. The gliding scale of malignancy in soft tissue neoplasms: select the correct level for rhabdomyosarcoma

- A. Benign
- B. Intermediate, locally aggressive
- C. Intermediate, rarely metastasizing
- D. Metastasis in 2 to 10%
- E. Metastasis in up to 100%

21. Mr. JJ came to Black Lion hospital with a complaint of difficulty in breathing (sometimes he hears wheezing sound when he breaths), coughing especially when it is cold and at night. The examining physician wanted to know whether Mr. JJ had some bronchial hyperactivity or not. What is the most likely drug to be given for Mr. JJ for diagnosis?

- A. Methacholine
- B. Pilocarpine
- C. Bethanechol

- D. Oxotremorine
 - E. Varenicline
22. _____ is a muscarinic agonist used for the treatment of dry mouth in Sjogren syndrome?
- A. Varenicline
 - B. Carbachol
 - C. Lobeline
 - D. Cevimeline
 - E. Bethanechol
23. Pharmacologic effect of direct acting cholinomimetic, except:
- A. Decrease in cardiac heart rate and contraction
 - B. Increase peristaltic activity
 - C. Increase uterine contraction
 - D. Contraction of bronchial smooth muscle
 - E. Increase in salivation
24. Organophosphate poisoning is best managed by:
- A. Oximes if before aging + Atropine
 - B. Atropine + Ambenonium
 - C. Depolarizing neuromuscular blocker
 - D. Competitive neuromuscular blocker
 - E. Ipratropium + oximes if before aging
25. _____ is applied for the diagnosis/monitoring effectiveness of treatment of Myasthenia gravis?
- A. Edrophonium
 - B. Pyridostigmine
 - C. Neostigmine
 - D. Ambenonium
 - E. Rivastigmine
26. Antimuscarinics can be used in all of the following cases, EXCEPT:
- A. In motion sickness
 - B. In management of asthma
 - C. In management of Parkinson's disease
 - D. To improve cognitive defect in Alzheimer's disease
 - E. As preoperative medications
27. CNS adverse effect is expected from one of the following carbamate esters
- A. Neostigmine
 - B. Pyridostigmine
 - C. Physostigmine
 - D. Echothiophate
 - E. None
28. Mr. TL, after being injected with a non-depolarization neuromuscular blocker, experienced difficulty in breathing, his face appeared red when his blood pressure was measured it was found to be below the normal range. What was the most likely neuromuscular blocker injected to Mr. TL

- A. Pancuronium
 - B. Cisatracurium
 - C. Pipecuronium
 - D. Mivacurium
 - E. Vecuronium
29. Correct concerning succinylcholine
- A. It is devoid of histamine releasing activity
 - B. Neostigmine reverses phase I response caused by succinylcholine
 - C. Coadministration of halogenated anesthetics risks hyperkalemia
 - D. May result in apnea in individuals with atypical plasma cholinesterase
 - E. It has slow onset and long duration of action
30. One of the following drugs reverses the effect of non-depolarizing neuromuscular blockers?
- A. Cholinesterase inhibitors
 - B. Halogenated hydrocarbon anesthetics
 - C. Aminoglycosides
 - D. Calcium channel blockers
 - E. All
31. Peripherally (at muscle) acting Spasmolytic
- A. Baclofen
 - B. Tubocurarine
 - C. Dantrolene
 - D. Clonidine
32. Indicated for therapy of Glaucoma
- A. Propranolol
 - B. Latanoprost
 - C. Misoprostol
 - D. Alprostadil
33. The preferred local anesthetic in a patient with liver disease:
- A. Lidocaine
 - B. Bupivacaine
 - C. Procaine
 - D. Etidocaine
34. A patient is to undergo cancer chemotherapy with a highly emetogenic drugs. The antiemetic drug most likely to be included in her regimen is:
- A. Bromocriptine
 - B. Cetirizine
 - C. Ketanserin
 - D. Ondansetron
35. The local anesthetic effect of the agents cannot be prolonged by adding:
- A. Epinephrine
 - B. Norepinephrine
 - C. Dopamine
 - D. Phenylephrine
36. Used only for surface or topical anesthesia

- A. Benzocaine
 - B. Tetracaine
 - C. Procaine
 - D. Bupivacaine
37. Caffeine + ergotamine is indicated for Migraine
- A. Potentiation
 - B. Addictive
 - C. Synergism
 - D. Antagonism
38. Causes of methemoglobinemia?
- A. Prilocaine
 - B. Procaine
 - C. Lidocaine
 - D. Ropivacaine
39. Indication of H1 antagonists:
- A. Allergic reactions (rhinitis)
 - B. Motion sickness
 - C. Morning sickness
 - D. All of the above
40. Side effect of first-generation H1 antagonists:
- A. Aplastic anemia
 - B. Vomiting, tinnitus
 - C. Sedation
 - D. Gastric ulcers
41. Epinephrine reversal (depressor effect) is seen when epinephrine is coadministered with:
- A. Beta agonists
 - B. Beta one selective agonists
 - C. Alpha 1 antagonists
 - D. Beta antagonists
 - E. Alpha 1 agonists
42. A Beta 2 agonist of choice for nocturnal asthma
- A. Isoproterenol
 - B. Albuterol
 - C. Metaproterenol
 - D. Salmeterol
 - E. Ritodrine
43. Mr. XY has been taking clonidine for his moderate hypertension for the past one year. But he got tired of taking the medication and stopped taking the drug at once yesterday. He was brought to the emergency room because he got very sick afterwards. What is the most likely cause of his illness?
- A. Rebound tachycardia
 - B. Marked sedation
 - C. Marked dry mouth
 - D. Marked sexual dysfunction

E. Rebound hypertension

44. Patients taking MAOI should take tyramine containing foods because when MAO is inhibited tyramine will cause:

- A. Bradycardia
- B. Postural hypotension

C. Hypertensive crisis

- D. CNS depression
- E. None

45. The following effects are expected if one took amphetamine, EXCEPT:

A. Bronchoconstriction

- B. Peripheral vasoconstriction
- C. Inhibition of gut motility
- D. Increased heart rate
- E. Raised arterial pressure

46. Phenoxybenzamine is preferred over Phentolamine for long term control of inoperable pheochromocytoma because:

- A. Phenoxybenzamine has less incidence of postural hypotension
- B. Phenoxybenzamine has less incidence of reflex tachycardia

C. Phenoxybenzamine is irreversible alpha antagonist

- D. Phenoxybenzamine is reversible alpha antagonist
- E. Both are equally indicated; no preference

47. A beta 1 selective antagonist with a very short duration of action and used in critically ill patients

- A. Metoprolol
- B. Atenolol
- C. Acebutolol

D. Esmolol

- E. Bisoprolol

48. Which group of beta blockers should be avoided in management of glaucoma

- A. Non-selective beta blockers

B. Beta blockers with intrinsic sympathomimetic activity

- C. Beta blockers with membrane stabilization activity
- D. Beta blockers with high lipid solubility
- E. Beta one selective antagonists

49. All of the following beta blockers can be given in asthmatic hypertensive patients with minimal effect on asthma, EXCEPT:

A. Propranolol

- B. Labetalol
- C. Celiprolol
- D. Bucindolol
- E. Pindolol

50. What is the major risk of using beta blockers in patients with diabetes and hypertension comorbidities

- A. Beta blockers decrease insulin secretion

B. Beta blockers mask the symptoms of hypoglycemia

- C. Beta blockers mask the symptom of hyperglycemia
- D. Beta blockers increase gluconeogenesis
- E. Beta blockers increase Glycogenolysis

OTHER QUESTIONS – REMEMBERED BY 07 STUDENTS

❖ **MICRO**

1. What is the common route of transmission of microorganisms to bone?
✓ Hematogenous spread
2. What is the etiologic agent that does not bring chronic arthritis?
✓ Know that these ones bring chronic arthritis → *M. Tb*, *Spirochetes* (*T. pallidum*, *B. burgoderferi*, *Leptospira* spp.) and fungi (*C. immitis*, *S. schenkii*, *candida albicans*). If anything, out of these comes on the choice, choose that one.
3. What is the most commonly affected area of acute osteomyelitis?
✓ Femur
4. Which one is not etiologic agent of reactive arthritis?
✓ Know that these ones bring reactive arthritis → *Yersinia enterocolitica*, *Camylobacter jejuni*, *Neisseria meningitides*, *Hemophilus influenza*, *Salmonella* spp. If anything, out of these comes on the choice, choose that one.
5. What is the common agent of acute osteomyelitis after puncture wounds
✓ *Pseudomonas aeruginosa* (this is the most common cause of nosocomial or hospital acquired infections). Otherwise, it is always *S. aureus*
6. What is the most common cause of acute osteomyelitis?
✓ *Staphylococcus aureus*
7. Which is not true about crusted “Norwegian” scabies?
✓ Hyper infestation of lots of scabies mites
8. Which is not true about pediculosis?
✓ Lice can borrow into the skin
9. What is vagabond’s disease?
✓ Caused by *P. corporis*. It is a prolonged infestation resulting in thickening and pigmentation around waist, groin and upper thigh.
10. What causes poliomyelitis?
✓ A **picorna virus** which is characterized by neurogenic atrophy. Polio virus is grouped under *picornaviridae*.
11. Dermatomyositis is characterized by:
✓ Rash
✓ Muscle pain
✓ Muscle Weakness and atrophy
12. Skeletal muscle tumors (rhabdomyosarcomas) are
✓ almost ALWAYS malignant – 100% of them.
13. Rhabdomyosarcoma is:
✓ a childhood tumor
14. What is necrotizing fasciitis?
✓ It is a rapidly spreading cellulitis with necrosis

15. What are the general characteristics of Staph aureus?

- ✓ Gram positive
- ✓ Catalase positive
- ✓ Coagulase positive

❖ PHYSIO

16. Intercalated discs are characteristics of?

- ✓ Intercalated disc

17. What are the characteristics of single unit smooth muscle?

- ✓ Most common type – found in GIT, many blood vessels, airways, bile ducts, ureters, uterus
- ✓ Their fibers are connected by **gap junctions** (single unit smooth muscle)

18. What are the characteristics of multiunit smooth muscle?

- ✓ Composed of discrete, **separate** smooth muscle fibers
- ✓ Located in large airways, arteries, eyes
- ✓ Their control is mainly exerted by nerve signals

19. Activated calmodulin activates an enzyme called?

- ✓ Myosin light chain kinase

20. Botulinum toxin result death by:

- ✓ paralysis of diaphragm

21. Calmodulin in smooth muscle is equivalent to

- ✓ Troponin C in skeletal muscle

22. What do muscles use as primary source of energy?

- ✓ **Phosphocreatine**. But during resting, note that it is fatty acids!

23. What does the "**Fenn effect**" state?

- ✓ The **greater amount of work** performed by muscle, the **greater amount of ATP** that is cleaved

24. Smooth muscles have small invaginations of cell membrane called

- ✓ **Caveolae**

25. What is Latch state?

- ✓ is a special state that allows smooth muscle to maintain tone (force) with minimal expenditure of ATP.

26. What do you call the ability of muscles to contract with varying degrees of force?

- ✓ **Summation**. There are two types – motor unit and wave summation

27. The combination of neuron and the muscle it innervates refers to?

- ✓ **Motor unit summation**

28. Which type of summation leads to Tetanization?

- ✓ **Wave (frequency) summation**

29. Which of the following are features of slow fibers?

- A. Abundant mitochondria, extensive capillary supply, low myoglobin, and fatigue resistant
- B. Abundant mitochondria, extensive capillary supply, high myoglobin, and easily fatigued
- C. Abundant mitochondria, extensive capillary supply, high myoglobin, and fatigue resistant**
- D. Scarce mitochondria, extensive capillary supply, high myoglobin, and fatigue resistant
- E. Scarce mitochondria, extensive capillary supply, low myoglobin and fatigue resistant

30. What is the feature of fast fibers (white fibers)?
✓ Fewer mitochondria, less extensive blood supply, low myoglobin, easily fatigued.
31. What is **isometric** contraction?
✓ When muscle **does not shorten** during contraction (Load = tension, velocity ↓)
32. What is **isotonic** contraction?
✓ Muscle shortens but the **tension** on muscle remains **constant** (Tension > load; velocity 0)

❖ PATHO

- The most catastrophic or serious effect of rheumatoid arthritis –
✓ Vasculitis of various organs
- What is the pathogenesis/mechanism of idiopathic or primary osteoarthritis?
✓ Chondrocyte injury → Early, chondrocyte proinflammation → chronic inflammation → Subchondral bone change
- Diffusive type of Synovial Giant Cell Tumor
✓ Affects larger joints
- A 17-year-old man came to a clinic with an ulcer around his nose of one year duration. Biopsy was taken and histopathology shows **nests of basaloid cells** with **peripheral palisading** invading the **dermis**. What is the most likely diagnosis?
✓ Basal cell carcinoma
- A 30-year-old HIV patient developed itching painful lesions over the right side of chest following a dermatome. What is the most probable diagnosis?
✓ Herpes Zoster
- What is condyloma accuminata
✓ It is a benign neoplasm caused by viral infection (low risk HPV 6 and 11)
- What is the most common primary bone tumor?
✓ Osteochondroma
✓ Note – if the question asks
 - What is the most common bone tumor?
✓ It is metastatic bone tumor (i.e. secondaries are common as a whole)
 - What is the most common primary bone tumor?
✓ It is Osteochondroma
 - What is the most common primary malignant bone tumor?
✓ It is osteosarcoma
 - What is the most common primary benign bone tumor?
✓ It is Osteochondroma (among primary bone tumors – the benign ones are most common)
- What are the two locally aggressive primary bone tumors?
✓ Chondroblastoma and Giant cell tumor (osteoclastoma)
- Which is **false** about secondary bone tumors (metastatic bone tumors)?
✓ They commonly affect appendicular skeleton
- A benign cartilage forming tumor that affects the soft tissue by forming hemangiomas is called?
✓ Maffucci syndrome
- What is **false** about giant cell tumor of bone?

- ✓ Giant cells are formed by fusion of mononuclear stromal cells
12. A primary malignant round cell tumor of bone which exhibit a neural phenotype with neural differentiation is called
- ✓ PNET
13. What are the two locally aggressive primary bone tumors which affect the epiphysis of long bones?
- ✓ Chondroblastoma and Giant cell tumor (osteoclastoma)
14. Which locally aggressive primary bone tumor affects epiphysis of long bones in older age group (20-40 years)?
- ✓ Giant cell tumor (osteoclastoma)
15. Which locally aggressive primary bone tumor affects epiphysis of long bones in younger age group?
- ✓ Chondroblastoma
16. Which is **not true** about Osteoarthritis?
- ✓ It is not associated with obesity (overweight)
17. Which is **false** about osteoarthritis?
- ✓ It affects trabecular bone
18. Which one is **false** about crystal induced arthritis (gout)?
- ✓ Primary gout results from chemotherapy
19. What is Brodie's abscess?
- ✓ Intra-osseous abscess walled off by reactive bone

Musculoskeletal Final 06 Batch

1. Serves as a controllable channel and found in SR membrane:
- A. Ryanodine receptor
- B. Nicotine receptor
- C. Dihydropyridine receptor
- D. Muscarinic receptor
- E. A and C
2. The ability of muscles to contract with varying degrees of force:
- A. Multiple fiber summation
- B. Frequency summation
- C. Motor unit summation
- D. Wave summation
- E. Summation
3. A special case that allows smooth muscle to maintain tone with minimal expenditure of ATP:
- A. Fenn effect
- B. Oxygen debt
- C. Latch state
- D. Tetanization
- E. None of the above
4. During smooth muscle contraction, activated calmodulin activates an enzyme called
- A. Myosin phosphatase
- B. Myosin light chain kinase
- C. Myosin ATPase

- D. Creatine kinase
 - E. B and C
5. One is a unique feature of cardiac muscle:
- A. Presence of gap junctions
 - B. Abundant mitochondria
 - C. Uninucleated
 - D. Auto rhythmicity
 - E. No summation due to long refractory period
6. Liver glycogen is used in fasting to provide glucose for use by other tissues including the brain. Muscle glycogen is not. What is the explanation for this?
- A. Muscle does not have debranching enzyme
 - B. Muscle cannot degrade glycogen further than glucose-1-phosphate
 - C. Muscle lacks glucose-6-phosphate
 - D. The liver provides all the glucose necessary for metabolism and there is no need for muscle to do the same
7. Which of the following statements about the control of muscle glycogen phosphorylase is correct?
- A. Muscle glycogen phosphorylase is allosterically activated by cAMP
 - B. Muscle glycogen phosphorylase is allosterically activated by ATP.
 - C. Muscle glycogen phosphorylase normally exists in the α form.
 - D. Muscle glycogen phosphorylase is activated by phosphorylation by an active phosphorylase kinase.
8. Which one of the following statements explains the synthesis of glycogen directly from D-glucose?
- A. It does not use glucose-1-P
 - B. It requires debranching enzyme
 - C. It requires a glucosyl transferase
 - D. It requires a UDP-glucose
9. Fast-twitch fibers contain:
- A. A relatively large number of mitochondria and low oxygen
 - B. A relatively small number of mitochondria and more myoglobin
 - C. A relatively small number of mitochondria and low myoglobin
 - D. No mitochondria and no myoglobin
10. Type I muscle fibers have the following characteristics:
- A. White, glycolytic, slow contracting
 - B. White, oxidative, slow contracting
 - C. Red, glycolytic, slow contracting
 - D. Red, oxidative, slow contracting
11. Subcutaneous mycoses
- A. Are caused primarily by primary pathogens
 - B. Portal of entry is the lungs
 - C. Caused by dimorphic fungi
 - D. Causative organisms are introduced by trauma
12. The following are subcutaneous mycoses, except

- A. Mycetoma
 - B. Sporotrichosis
 - C. Black piedra
 - D. Rhinosporidiosis
13. Which of the following chronic infection of the subcutaneous tissue can be caused both by bacteria and fungi?
- A. Chromoblastomycosis
 - B. Sporotrichosis
 - C. Mycetoma
 - D. Lobomycosis
14. *Sporothrix Schenckii*
- A. Is a thermally dimorphic fungus
 - B. If introduced by trauma, it causes a subcutaneous nodule which slowly progresses
 - C. Is a primarily opportunistic pathogen
 - D. A and B
15. *Candida* species can cause
- A. Oral acute pseudo membranous infection
 - B. Angular cheilitis
 - C. Interdigital infection and nail infection
 - D. All of the above
16. Zygomycosis is an opportunistic infection
- A. Causes a self-limiting infection of the stratum corneum
 - B. is a chronic infection and progresses slowly
 - C. primarily involves the skin, hair and nails
 - D. vascular invasion with subsequent necrosis is the hallmark of this infection
17. All of the following are true of cutaneous mycoses except:
- A. Cutaneous mycoses are transmissible from person to person
 - B. It can be caused by several species
 - C. It can be transmitted through using common comb
 - D. They produce lifelong immunity from previous infections
18. The incorrect statement concerning *Clostridium perfringens* is:
- A. Can be found in intestine of healthy individuals
 - B. Their presence in water indicates fecal pollution
 - C. It is the only motile species of the clostridia
 - D. It is the main cause of gas gangrene
19. Select the WRONG statement about *Clostridium perfringens*
- A. *C. perfringens* are highly pathogenic when introduced into healthy tissues
 - B. *C. perfringens* type A and C produce enterotoxin that cause food poisoning
 - C. Patients at risk of developing gas gangrene are those with blood vessel diseases
 - D. None
20. *Mycobacterium leprae*
- A. Has the lowest growing of all species
 - B. Cultivated in selective artificial culture media
 - C. Lepromatous leprosy has good prognosis than tuberculous leprosy

- D. The organism best grows at 37°C or above
21. Which of the following statements is correct?
- A. *M. leprae* is easily distinguishable morphologically by staining from *M. Tuberculosis*
 - B. *M. leprae* is highly resistant to UV light surviving for 30 days
 - C. Skin and peripheral nerves are infiltrated with lymphocytes in lepromatous leprosy
 - D. Incubation period of leprosy may be 2-10 years
22. Which of the following animals are susceptible to *M. leprae*?
- A. Armadillo
 - B. Mice
 - C. Rabbits
 - D. A and B
23. Which of the following is NOT a correct match concerning type of Strep that exhibit hemolysis?
- A. *S. pneumoniae* – Alpha hemolytic
 - B. *S. pyogenes* and *S. agalactiae* – Beta hemolytic
 - C. *Enterococcus* – gamma hemolytic
 - D. *Staphylococcus aureus* – Beta hemolytic
 - E. *Viridians streptococci* – non-hemolytic
24. Osteoblastoma differs from osteoid osteoma in that:
- A. It involves the spine more frequently than other bones
 - B. The pain is sharp and responds promptly to salicylates
 - C. It is associated with a marked bony reaction
 - D. The tumor is less than 2 cm in the greatest dimension
25. Not a characteristic feature of the most common type of osteosarcoma
- A. It arises in the metaphysis of long bones
 - B. It is primary, solitary, and intermedullary
 - C. Poorly differentiated and produces predominantly bone matrix
 - D. None of the above
26. A benign cartilage forming tumor characterized by cartilage capped outgrowth that is attached to the underlying skeleton by a bony stalk is:
- A. Osteochondroma
 - B. Chondroma
 - C. Chondroblastoma
 - D. Chondromyxoid fibroma
27. A subtype of chondrosarcoma histologically exhibiting islands of well differentiated hyaline cartilage by sheets of small round blue malignant tumor cells is:
- A. Clear cell chondrosarcoma
 - B. Mesenchymal chondrosarcoma
 - C. Chondroblastic osteosarcoma
 - D. Dedifferentiated chondrosarcoma
28. NOT true about Ewing sarcoma and Neuroectodermal tumor (PNET)
- A. They exhibit a neural phenotype
 - B. They have the youngest average age at presentation
 - C. There is a t(11:22) (q24;q12) translocation in 85% of cases
 - D. None of the above

29. NOT true about Giant cell tumor of the bone:
- A. It arises from osteoclastic multinucleated giant cells
 - B. It is a benign but locally invasive neoplasm
 - C. In adults, it involves the epiphysis and the metaphysis
 - D. The stromal cells are the proliferating component of the tumor
30. Metastatic skeletal tumors notorious for producing solitary lesions stimulating primary bone cancers are:
- A. Carcinomas of the kidney and thyroid
 - B. Carcinomas of the prostate and the lung
 - C. Carcinomas of the breast and the colon
 - D. Carcinomas of the pancreas and the liver
31. A pathology report from a skin biopsy states the diagnosis as Condyloma accuminata. Which of the following is true?
- A. It is a malignant skin tumor
 - B. It is a suppurative inflammation of the skin
 - C. It is a benign neoplasm caused by viral infection
 - D. It is an infection of the skin by tuberculosis
 - E. None
32. Which one of the following lesions is caused by human papillomavirus (HPV) and has the worst outcome?
- A. Verruca vulgaris
 - B. Condyloma accuminatum
 - C. Verruca plana
 - D. Verruca plantaris
 - E. Squamous cell carcinoma
33. A 30-year-old male HIV infected patient developed itching painful lesions over the right side of the chest following dermatome, what is the most probable diagnosis?
- A. Herpes simplex infection
 - B. Chicken pox
 - C. Herpes zoster
 - D. Scrofuloderma
 - E. Basal cell carcinoma
34. True about folliculitis
- A. Is an inflammatory disease of the epidermis
 - B. Infection around hair follicles
 - C. It is a viral infection of the sweat glands
 - D. It is a warty lesion caused by HPV
 - E. None
35. One of the following is not among the routes of infection of tuberculosis of the skin
- A. Sexually transmitted
 - B. Hematogenous spread
 - C. Scrofuloderma from deeper infected lymph node
 - D. Direct inoculation
 - E. None

36. A 70 years old man came to the clinic with an ulcer around his nose of 1 year duration. Biopsy was taken and histopathology shows nests of basaloid cells with peripheral palisading invading the dermis. Which diagnosis is correct?
- A. Mucocutaneous leishmaniasis
 - B. Squamous cell carcinoma
 - C. Melanoma
 - D. Lymphoma
 - E. None - or Basal cell carcinoma
37. Which statement is incorrect regarding melanomas?
- A. Sun exposure is an important cause
 - B. Can be seen in an adult as a congenital lesion with no change over long period of time
 - C. Can metastasize to distant organs
 - D. Variegated color of the lesion is one of the early signs
 - E. None
38. A 23-year-old student dies due to cardiac failure (dilative cardiomyopathy); - mark the most probable underlying condition:
- A. Poliomyositis
 - B. Dermatomyositis
 - C. Muscular dystrophy BECKER
 - D. Myasthenia gravis
 - E. WERDING-HOFFMANN neurogenic muscular atrophy
39. Which drug has high surface local anesthetic activity and intrinsic vasoconstrictor actions that reduce bleeding in mucous membranes?
- A. Bupivacaine
 - B. Cocaine
 - C. Lidocaine
 - D. Procaine
 - E. Tetracaine
40. A 42-year-old presented with glaucoma and she might be at risk for an acute angle-closure episode, with markedly increased intraocular pressure when administered with one of the following antihistamine drug.
- A. Promethazine
 - B. Meclizine
 - C. Cetirizine
 - D. Doxylamine
 - E. Doxepin
41. Which drug has high surface local anesthetic activity and intrinsic vasoconstrictor actions that reduce bleeding in mucous membranes?
- A. Bupivacaine
 - B. Cocaine
 - C. Lidocaine
 - D. Procaine
 - E. Tetracaine

42. A vasoconstrictor added to a solution of lidocaine for a peripheral nerve block will be useful except
- A. Decrease the risk of a seizure
 - B. Increase the duration of anesthetic action of the local anesthetic
 - C. Less requirement of the drug
 - D. Increase in bioavailability of the drug
 - E. Local anesthetic drug is short acting
43. Which of the following drugs is the most effective drug in relieving angioneurotic edema produced by histamine in smooth muscle?
- A. Cetirizine
 - B. Epinephrine
 - C. Granisetron
 - D. Ranitidine
 - E. Sumatriptan
44. A 35-year-old man was treated with NSAID for his arthritis problem and he subsequently developed gastric ulceration. Which of the following is used to prevent the ulceration?
- A. Paracetamol
 - B. Misoprostol
 - C. Antacid
 - D. Celecoxib
 - E. Ranitidine
45. A 30-year-old office worker suffers sedation after consuming a cough syrup containing one of the following drug:
- A. Cetirizine
 - B. Chlorpheniramine
 - C. Meclizine
 - D. Doxepin
 - E. Cyclizine
46. A patient with a bleeding tendency presents in the hematology clinic. He is apparently taking large amounts of unidentified drug that inhibits platelet activity. Which of the following directly and reversibly inhibits platelet oxygenase?
- A. Alprostadil
 - B. Aspirin
 - C. Ibuprofen
 - D. Misoprostol
 - E. Prostacyclin
47. Prostaglandins involved in inflammatory processes are typically produced from arachidonic acid by which of the following enzymes?
- A. Cyclooxygenase-1
 - B. Cyclooxygenase-2
 - C. Glutathione-S-transferase
 - D. Lipoxygenase

- E. Phospholipase A2
48. An 18-year-old woman complains of severe dysmenorrhea and was prescribed one of the following drug which further caused increase in menstrual bleeding.
- A. Aspirin
 - B. Hydrocortisone
 - C. Ibuprofen
 - D. Indomethacin
 - E. Zileuton
49. A 35-year-old patient presents to the labor ward for delivery and complains that she developed wheezing and shortness of breath. Which of the following prostaglandin should be used with caution to facilitate labor?
- A. PGE 2
 - B. PGI 2
 - C. PGF 2 α
 - D. PGE 1
 - E. PGF 1 α
50. The following are the adverse effects of misoprostol except:
- A. Abdominal discomfort
 - B. Diarrhea
 - C. Dose-dependent bone pain
 - D. Hyperostosis in patients with liver disease
 - E. Increased gastrointestinal bleeding
51. Which of the following is NOT true about parasympathomimetic drugs?
- A. All muscarinic receptor agonist has parasympathomimetic effect
 - B. Have bronchodilation effect
 - C. Cause bradycardia of the heart
 - D. Increase GI motility
52. Naturally occurring acetylcholine receptor that is indicated for the treatment of glaucoma is:
- A. Methacholine
 - B. Pilocarpine
 - C. Carbachol
 - D. Bethanicol
53. One of the following is given for the management of Parkinson like symptoms in Schizophrenic patient who are under treatment of typical antipsychotic?
- A. Atropine
 - B. Trihexyphenidyl
 - C. Pirenzepine
 - D. Gallamine
54. Parasympathomimetic drug should not be given for patient who has (is contraindicated in):
- A. Bronchial asthma
 - B. Peptic ulcer disease
 - C. Glaucoma
 - D. A and B
55. Effect of ganglionic blocker:

- A. Tachycardia
 - B. Increased blood pressure
 - C. Increased GI motility
 - D. Xerostomia
56. Anti-cholinergic drugs can be characterized by:
- A. Bradycardia, myosis, salivation, diaphoresis
 - B. Tachycardia, blurred vision, constipation, mydriasis
 - C. hyperactive bowel, pin pointed pupil, muscle contraction then paralysis
 - D. Bradycardia, constipation, and altered mental status
57. One of the following is not the clinical use of anti-cholinergic drugs
- A. For glaucoma
 - B. for motion sickness
 - C. for intestinal spasm
 - D. for Parkinson's disease
58. Loss of flexion at metacarpophalangeal joints and extension at interphalangeal joints of the 2nd and 3rd digits suggests lesion of the:
- A. Radial nerve
 - B. Musculoskeletal nerve
 - C. Ulnar nerve
 - D. Median nerve
 - E. Axillary nerve
59. One of the following upper limb muscle do not originate from flexor retinaculum?
- A. Opponens pollicis
 - B. Abductor pollicis brevis
 - C. Flexor pollicis brevis
 - D. Flexor digiti minimi brevis
 - E. Abductor digiti minimi
60. Innervation to the hand muscle that enables the index finger independent extension is provided by which of the following?
- A. Posterior interosseous nerve
 - B. Median nerve
 - C. Ulnar nerve
 - D. Anterior interosseous nerve
 - E. Dorsal scapular nerve
61. The nerve that do not supply the muscle depressing the scapula
- A. Median pectoral
 - B. Spinal accessory
 - C. Thoracodorsal
 - D. Long thoracic
 - E. Dorsal scapular
62. The vein which is (are) formed by the anastomoses of superficial veins from the inguinal region with tributaries of the axillary vein:
- A. Thoracoepigastric vein
 - B. Thoracoabdominal vein

- C. Thoracoacromial vein
 - D. Superior epigastric vein
 - E. None
63. One of the following muscles **do not** attach to the common extensor tendon:
- A. Extensor carpi radialis brevis
 - B. Extensor digiti minimi
 - C. Extensor digitorum
 - D. Extensor carpi radialis longus
 - E. None
64. One of the following is **false** regarding tarsal bones:
- A. The talus is the only tarsal bone that has no muscular or tendinous attachments
 - B. All cuneiform articulates with the navicular posteriorly
 - C. Both tubercles of calcaneal tuberosity contacts the ground
 - D. None
65. Which one of the following muscles **does not** act on the ankle joint?
- A. flexor digitorum longus
 - B. fibularis brevis
 - C. popliteus
 - D. soleus
66. Which of the following muscles is **invertor** and **plantar flexor** of the foot?
- A. fibularis brevis
 - B. fibularis longus
 - C. tibialis anterior
 - D. tibialis posterior
67. One of the following is clinically important for muscle graft?
- A. Tibialis anterior
 - B. Gracilis
 - C. Quadriceps femoris
 - D. Psoas anterior
68. Injury to one of the following nerve causes paralysis of primary evertors:
- A. Femoral
 - B. Tibial
 - C. Superficial femoral
 - D. Deep peroneal
69. Select the true statement about blood vessels of lower limb
- A. Dorsalis pedis is a continuation of posterior tibial artery
 - B. Small saphenous vein drains directly to femoral vein
 - C. Lateral circumflex femoral artery has descending branch to supply knee joint
 - D. None
70. All of the following are components of the Facial Skeleton (viscerocranium), **EXCEPT**:
- A. lacrimal bone
 - B. mandibular bone
 - C. zygomatic bone
 - D. maxillary bone

- E. palatine bone
71. All of the following are synovial joints in the head, **EXCEPT**:
- A. the temporomandibular joint
 - B. the zygomaticotemporal joint
 - C. the joint between the malleus and incus
 - D. the joint between the incus and stapes
 - E. None of the above
72. A tumor infiltrating the jugular foramen might result in Jugular Foramen syndrome. The functions of which of the following nerves may be affected as a result of this syndrome?
- A. Spinal accessory nerve
 - B. glossopharyngeal nerve
 - C. vagus nerve
 - D. all of the above
 - E. B and C only
73. All of the following arteries supply the scalp, EXCEPT:
- A. supratrochlear
 - B. superficial temporal
 - C. posterior auricular
 - D. transverse facial
 - E. occipital
74. Clinically, anesthesia (loss of sensation) of the upper lip may be a sign of injury to one of the following nerves:
- A. ophthalmic nerve
 - B. maxillary nerve
 - C. mandibular nerve
 - D. great auricular nerve
 - E. auriculotemporal nerve
75. The muscles of facial expression are clinically important in evaluating the integrity of the facial nerve. One of these muscles that enables us to smile is:
- A. Buccinator muscle
 - B. Orbicularis oculi muscle
 - C. Orbicularis oris muscle
 - D. Frontalis muscle
 - E. Procerus muscle
76. Gray rami communicates to both C5 and C6 nerves arise from the middle cervical ganglion:
- A. True
 - B. False
77. The submental node receives lymph from the tonsillar and peritonsillar regions?
- A. True
 - B. False
78. Which of the following is a correct matching between an endocrine and exocrine gland and its hormone?
- A. Parathyroid gland --- calcitonin
 - B. Thyroid gland --- parathyroid hormone (parathahormone)

- C. Parathyroid gland --- thyroxine
 - D. Thyroid gland --- calcitonin
 - E. NONE
79. The _____ nerve is a sensory branch of the mandibular nerve, which does not pass through the infratemporal fossa.
- A. Inferior alveolar
 - B. Lingual
 - C. Auriculotemporal
 - D. Buccal
 - E. None of the above
80. Which of the following muscles inserts on the medial surface of the ramus of the mandible?
- A. Lateral pterygoid
 - B. Anterior fibers of the temporalis
 - C. Masseter
 - D. Buccinator
 - E. None
81. Which of the following arteries doesn't arise from the first (mandibular) part of maxillary artery?
- A. Middle meningeal
 - B. Buccal
 - C. Inferior alveolar
 - D. Anterior tympani
 - E. Deep auricular
82. One of the following structures doesn't pass through the superior orbital fissure?
- A. Trochlear nerve
 - B. Lacrimal nerve
 - C. Oculomotor nerve
 - D. Optic nerve
 - E. None
83. Which of the following arteries of the eye is a muscular branch?
- A. Short posterior ciliary artery
 - B. Long posterior ciliary artery
 - C. Anterior ciliary artery
 - D. Central artery of the retina
 - E. None
84. One of the following bones **does not** form any wall of the orbit?
- A. Maxilla
 - B. Zygomatic bone
 - C. Palatine bone
 - D. Ethmoid bone
 - E. Nasal bone
85. One of the following **is not** involved in innervations of the auricle?
- A. Glossopharyngeal nerve
 - B. Vagus nerve
 - C. Posterior auricular nerve

- D. Greater auricular nerve
 - E. None
86. Concerning walls of the tympanic cavity, one is a **wrong** relationship:
- A. Medial wall -----labyrinthine promontory
 - B. Roof -----tegumentum
 - C. Floor -----internal jugular vein
 - D. Anterior wall -----common carotid artery**
 - E. None
- ✓ Anterior wall is formed by internal carotid not common carotid. So D is wrong.
87. Which of the following muscles has a different innervation?
- A. Superior oblique**
 - B. Lateral rectus
 - C. Medial rectus
 - D. Inferior oblique
88. Which of the following statements is correct about the cells of cartilage?
- A. Chondrogenic cells can differentiate into chondroblasts as well as osteogenic cells**
 - B. Chondroblasts are found in the lacunae
 - C. Chondrocytes located peripherally are round, while those located deeper are flattened
 - D. Chondrocytes have eosinophilic cytoplasm
 - E. Chondroblasts have processes which communicate by gap junction with each other
89. With regard to the type of collagens present in the matrix of the different types of cartilage:
- A. Fibrous cartilage has type I, hyaline cartilage has type II and elastic cartilage has type III
 - B. Both fibrous and hyaline cartilages have type I, while elastic cartilages have type II collagen
 - C. Fibrous cartilage has type I, while both hyaline and elastic cartilages have type II collagen**
 - D. Fibrous cartilage has type II, hyaline cartilage has type I and elastic cartilage has type III
 - E. Both fibrous and hyaline cartilages have type II, while elastic cartilage has type I collagen
90. Regarding to the bone:
- A. Compact bone has outer and inner circumferential laminae known as the Haversian system.
 - B. Canaliculi contain the cell bodies of the osteoblasts
 - C. Cell bodies of the osteocytes are found lining the surfaces of developing bone.
 - D. Spongy bone contains more Haversian system than the compact bone.
 - E. Volkmann's canal run perpendicular to the longitudinal axis of the bone.**
91. Which of the following statements is correct regarding the bone?
- A. Organic portion of a bone matrix consists primarily type III collagen
 - B. Sharpey's fibers are calcified collagen bundles that run in concentric manner
 - C. Osteoclasts have processes that run in their canaliculi
 - D. Osteoblasts are found as stratified squamous epithelium
 - E. Periosteum is a dense connective tissue that lines the outer surface of a bone**
92. A newly deposited uncalcified bone is known as:
- A. Osteon
 - B. Perosteoid

- C. Osteoid
 - D. Haversian system
 - E. Cancellous bone
93. Which of the following is NOT true regarding the cardiac muscles?
- A. Divides and anastomose, but the cytoplasm remains separated.
 - B. Intercalated discs correspond to some of the Z lines
 - C. Have peripherally placed nuclei
 - D. Sarcoplasmic reticulum makes diad with a T tubule, but not triad
 - E. Cannot regenerate
94. The cellular junction found at the longitudinal part of the intercalated disc of the cardiac muscle is
- A. Zonula adherens
 - B. Gap junction
 - C. Macula adherens
 - D. Zonula occludens
 - E. All of the above
95. Which of the following bones begin to ossify before the others?
- A. Vertebra
 - B. Scapula
 - C. Humerus
 - D. Clavicle
 - E. Femur
96. The costal processes of the sclerotomes give rise to all of the following, except:
- A. The transverse processes
 - B. Foramina transversaria
 - C. Lateral mass or alae of the sacrum
 - D. Spinous process
 - E. None of the above
97. The body of the 4th cervical vertebra is formed by:
- A. The 3rd cervical sclerotome alone
 - B. The 4th cervical sclerotome alone
 - C. Fusion of caudal half of the 2nd cervical with cranial half of the 3rd cervical sclerotomes
 - D. Fusion of caudal half of the 3rd cervical with cranial half of the 4th cervical sclerotomes
 - E. Fusion of caudal half of the 4th cervical with cranial half of the 5th cervical sclerotomes
98. Which one of the following statements is true regarding the hypomeric and epimeric divisions of the myotomes?
- A. Epimers are larger and located dorsally as compared to the hypomers
 - B. Epimers are smaller and located dorsally as compared to the hypomers
 - C. Epimers are innervated by the dorsal branches of the ventral primary rami of the spinal nerves
 - D. Hypomers give rise to the extensors of the back
 - E. None of the above
99. With regard to the development of the skeletal and muscular system:
- A. Apical ectodermal ridge includes differentiation in limb mesenchyme

- B. Development of the upper limb lags behind that of the hind limb by few days
 - C. Skeletal muscle cells are multinucleated because nuclear divisions without cytoplasmic division
 - D. It is defect in the development of the centrum, but not the vertebral arch that causes spina bifida
 - E. All of the above
100. Which of the following is a derivative of the second pharyngeal arch?
- A. The sphenomandibular ligament
 - B. The stapes of the auditory ossicles
 - C. The incus of the auditory ossicles
 - D. The zygomatic bone
 - E. The malleus of the auditory ossicles

Musculoskeletal Test-1 06 Batch

1. Cutaneous nerve supplying the skin covering the radial part of three and half digits is
 - A. Recurrent branch of the median nerve
 - B. Lateral branch of the median nerve
 - C. Median branch of the median nerve
 - D. Palmar cutaneous nerve of the median nerve
 - E. None
2. The surface marking in the scapulae serves for attachment (insertion) of the serratus anterior of the anterior axioappendicular muscles is:
 - A. Outer surface of the medial border of the scapulae
 - B. Superior border of the scapulae
 - C. Lateral border of the scapulae
 - D. Inferior angle of the scapulae
 - E. None
3. The rotator cuff muscles originate from the following surface of the scapulae except:
 - A. Subscapular fossa
 - B. Supraspinatus fossa
 - C. Infraspinatus fossa
 - D. Lateral border of the scapulae
 - E. Inferior angle of the scapulae
4. Innervation to the rotator cuff muscle that laterally rotates the arm is provided by which of the following?
 - A. Suprascapular nerve
 - B. Median nerve
 - C. Ulnar nerve
 - D. Upper and lower subscapular nerves
 - E. Dorsal scapular nerve
5. Which of the following muscle of the upper limb originate from three separate bone of the upper limb?
 - A. Extensor digitorum
 - B. Triceps
 - C. Flexor digitorum superficialis

- D. Flexor digitorum profunda
 - E. Abductor digiti minimi
6. Which of the following is innervated by the superficial branch of the ulnar nerve?
- A. Flexor pollicis longus
 - B. Extensor pollicis longus
 - C. Abductor pollicis brevis
 - D. Deep head of flexor pollicis brevis
 - E. Palmaris brevis
7. Which nerve is probably damaged if a patient cannot abduct the arm beyond 15 degrees?
- A. Axillary nerve
 - B. Radial nerve
 - C. Musculocutaneous nerve
 - D. Median nerve
 - E. Ulnar nerve
8. The posterior relation of the first part of axillary artery is
- A. Long thoracic nerve
 - B. Chords of brachial plexus
 - C. Axillary vein
 - D. Posterior chord
 - E. None
9. The following tendons are inserted into the base of the proximal phalanx of the thumb **except**
- A. Extensor pollicis brevis
 - B. Adductor pollicis longus
 - C. Oblique head of adductor pollicis
 - D. Flexor pollicis brevis
 - E. Transverse head of adductor pollicis
10. Regarding the deltoid, the wrong description includes:
- A. It is the most powerful abductor of the arm
 - B. Its anterior part is a strong flexor and lateral rotator of the humerus
 - C. It is inserted into deltoid tuberosity of the humerus
 - D. It originates from scapulae and clavicle
 - E. Is innervated by anterior terminal branch of the axillary nerve
11. A muscle with a single origin is
- A. Pronator teres
 - B. Flexor digitorum superficialis
 - C. Adductor pollicis
 - D. Flexor carpi ulnaris
 - E. None
12. Which of the following artery does not form arterial anastomoses around the scapula
- A. Anterior circumflex humeral artery
 - B. Profunda brachial artery
 - C. Suprascapular artery
 - D. Thoracoacromial artery

- E. Transverse cervical artery
13. A structure on the anterior aspect of the wrist beneath the flexor retinaculum include all except
- A. Palmaris longus
 - B. Flexor digitorum profundal
 - C. Flexor digitorum superficialis
 - D. Flexor pollicis longus
 - E. Median nerve
14. Loss of supination of flexed forearm suggests a lesion of the
- A. Radial nerve
 - B. Musculocutaneous nerve
 - C. Ulnar nerve
 - D. Median nerve
 - E. Axillary nerve
15. The intrinsic muscle of the hand having dual innervations is:
- A. Flexor pollicis brevis
 - B. Adductor pollicis
 - C. Opponens pollicis
 - D. Abductor digiti minimi
 - E. Opponens digiti minimi
16. The arterial supply of the thumb is derived from
- A. Common palmar digital artery
 - B. Superficial palmar arch
 - C. Deep palmar arch
 - D. Princeps pollicis artery
 - E. Superficial branch of radial artery
17. All **except** one of the following muscles is supplied by two nerves
- A. Brachialis
 - B. Flexor pollicis brevis
 - C. Subscapularis
 - D. Flexor digitorum profunda
 - E. None
18. Which of the following is false regarding to the nerve supply of the upper limb?
- A. Ulnar nerve has no branch in the arm
 - B. Axillary nerve supply the skin over lower half of the deltoid
 - C. Radial nerve supply skin of the medial side of the dorsum of hand
 - D. Median nerve has no branch in the upper arm
 - E. Musculocutaneous nerve is the terminal branch of medial cord
19. Which of the following is **false** regarding the potential spaces located in the upper limb?
- A. The posterior wall of the axilla is formed chiefly by subscapularis muscle
 - B. The medial boundary of auscultation triangle is formed by trapezius
 - C. Brachialis muscle from the superior boundary of the cubital fossa
 - D. Extensor pollicis longus form the posterior boundary of anatomical snuff box
 - E. Radial nerve leaves the axilla through quadrangular space

20. One of the following branch of median nerve arises before the nerve crosses the flexor retinaculum:
- A. Recurrent branch of median nerve
 - B. Lateral branch of median nerve
 - C. Median branch of median nerve
 - D. Palmar cutaneous branch of median nerve**
 - E. None
21. One of the following muscles do not originate from the lateral border of scapulae?
- A. Teres major
 - B. Teres minor
 - C. Latissimus dorsi
 - D. Triceps brachii
 - E. None of the above**
22. The triangle of auscultation is bordered by all of the following except:
- A. The superior horizontal border of latissimus dorsi
 - B. The medial border of the scapula
 - C. The inferolateral border of the trapezius
 - D. The lateral border of the scapula**
 - E. None
23. The lymph from the upper lateral quadrant of the breast drains mainly into the
- A. Lateral axillary nodes**
 - B. Internal thoracic nodes
 - C. Posterior axillary nodes
 - D. Anterior axillary nodes
 - E. Deltopectoral group of nodes
24. The sensory innervation of the nail bed of the index finger is the
- A. Median nerve**
 - B. Radial nerve
 - C. Dorsal cutaneous branch of the ulnar nerve
 - D. Superficial branch of the ulnar nerve
 - E. Palmar cutaneous branch of the ulnar nerve
25. The radial nerve gives off the following branches in the posterior compartment of the arm except which?
- A. Lateral head of the triceps
 - B. Lower lateral cutaneous nerve of the arm
 - C. Medial head of the triceps
 - D. Brachioradialis**
 - E. Anconeus
26. Select the true statement regarding pelvic bone:
- A. Pelvic bones are separated by cartilage before puberty**
 - B. Obturator foramen is bounded by part of ilium, ischium and pubis
 - C. In a sitting position, iliac crest is involved in weight transfer to the ground/chair
 - D. All
27. One of the following is false regarding femur except:

- A. Greater trochanter provides attachment for abductors
 - B. The shaft is convex anteriorly
 - C. Quadrate tubercle is a round elevation on the intertrochanteric line
 - D. Linea aspera is a rough line that forms posterior border of the shaft
28. Choose the weak area of tibia which is prone for fracture commonly:
- A. Between proximal end and upper 1/3 of shaft
 - B. Between upper 1/3 and middle 1/3 of shaft
 - C. Between middle 1/3 and distal 1/3 of shaft
 - D. Between lower 1/3 of the shaft and distal end
29. One of the following is true about fibula?
- A. It does not bear weight
 - B. Serves as muscle attachment
 - C. It is not part of the knee joint
 - D. All
30. Select the wrong statement about calcaneus
- A. Its distal end articulates with medial cuneiform
 - B. Its rough posterior surface provides attachment site for tendocalcaneus
 - C. Largest and strongest bone in the foot
 - D. None
31. Which one of the following bone has head, neck and body?
- A. Talus
 - B. Calcaneus
 - C. Cuneiform
 - D. Cuboid
32. Select the true statement regarding deep fascia of the lower limb
- A. On the lateral side of the thigh fascia lata is thickened to form iliotibial tract
 - B. Flexor retinaculum of the leg is found b/n the lateral malleolus and lateral side of calcaneus
 - C. Adductor hiatus is a gap in the deep fascia between which lies about 4 cm below and lateral to the pubic tubercle
 - D. All
33. Injury to which of the following nerve causes paralysis of primary hip abductors and leads to waddling gait?
- A. Sciatic
 - B. Superior gluteal
 - C. Inferior gluteal
 - D. Inferior clunial
34. The following structures pass through greater sciatic foramen inferior to piriformis, **except**:
- A. Obturator
 - B. Sciatic nerve
 - C. Posterior cutaneous nerve of thigh
 - D. Pudendal nerve
35. Which of the following muscle act on the hip joint?
- A. Vastus medialis

- B. Vastus intermidus
 - C. Vastus lateralis
 - D. Rectus femoris
36. One of the following is a strong hip flexor:
- A. Sartorius
 - B. Quadriceps
 - C. Iliopsoas
 - D. None
37. Select the correct arrangement of the following structures in the femoral triangle from medial to lateral
- A. Femoral nerve, femoral artery, femoral vein
 - B. Femoral vein, femoral artery, and femoral nerve (VAN)
 - C. Femoral nerve, femoral nerve, femoral artery
 - D. Femoral artery, femoral vein, and femoral nerve
38. One of the following nerves is the only branch of femoral nerve which supplies lower limb distal to knee joint
- A. Sural
 - B. Saphenous
 - C. Genitofemoral
 - D. All
39. From muscles of adductor compartment, one has dual innervation
- A. Adductor brevis
 - B. Adductor longus
 - C. Adductor magnus
 - D. Gracilis
40. Hamstring muscles are originated from -----
- A. Ischial spine
 - B. Ischial tuberosity
 - C. Ischial spine
 - D. Ischial tubercle
41. Identify the wrong statement
- A. From nerves, arteries and vein, nerve is the most superficial structure in the popliteal fossa
 - B. Medially and superiorly, popliteal fossa is bounded by semimembranosus
 - C. Saphenous nerve is a cutaneous nerve that passes through popliteal fossa
 - D. None
42. Which of the following muscles does not act on the ankle joint?
- A. Flexor digitorum longus
 - B. Fibularis brevis
 - C. Polpiteus
 - D. Soleus
43. Cutaneous innervation to the medial side of the leg is supplied by ----- nerve
- A. Sural
 - B. Superficial fibular

- C. Deep fibular
 - D. Saphenous
44. One of the following is **false**
- A. All posterior compartment of leg is supplied by tibial nerve
 - B. Tibial division of sciatic nerve supplies all posterior thigh muscles
 - C. Commonly the sciatic nerve passes inferior to piriformis muscle.
 - D. None
45. Choose the muscles that act antagonistically on inversion of foot and synergistically on dorsiflexion of ankle
- A. Tibialis anterior and tibialis posterior
 - B. Tibialis anterior and Fibularis tertius
 - C. Tibialis anterior and Fibularis longus
 - D. Tibialis posterior and Fibularis longus
46. Quadratus plantae is an example of ----- muscle layer of the foot
- A. 1st
 - B. 2nd
 - C. 3rd
 - D. 4th
47. One of the following is invertor and plantar flexor of the foot:
- A. Fibularis brevis
 - B. Fibularis longus
 - C. Tibialis anterior
 - D. Tibialis posterior
48. One of the following muscles is most important to unlock knee joint
- A. Plantaris
 - B. Soleus
 - C. Popliteus
 - D. Gastrocnemius
49. One of the following muscles is important to plantarflex the foot when the knee is flexed.
- A. Gastrocnemius
 - B. Soleus
 - C. Tibialis anterior
 - D. Poplitus
50. Fracture of the head of fibula results the following **except**:
- A. Loss of subcutaneous innervations between 1st and 2nd toes
 - B. Loss of eversion of foot
 - C. Loss of cutaneous innervations on the plantar surfaces of foot
 - D. Foot drop
51. With regard to the type of collagens present in the matrix of the different types of cartilage:
- A. Fibrous cartilage has type I, hyaline cartilage has type II, and elastic cartilage has type III.
 - B. Both fibrous and hyaline cartilages have type I, while elastic cartilage has type II collagen
 - C. Fibrous cartilage has type I while both hyaline and elastic cartilages have type II collagen
 - D. Fibrous cartilage has type II, hyaline cartilage has type I and elastic cartilage has type III

- E. Both the fibrous and hyaline cartilages have type II, while elastic cartilage has type I collagen.
52. Endochondral bone ossification includes all of the following except:
- A. A cartilage model
 - B. Proliferation and enlargement of cartilage cells
 - C. Chondrocytes changing to osteocytes
 - D. Calcification of cartilage matrix
 - E. Interruption of blood supply to the area of ossification
53. Which type of bone cells are involved in bone ossification?
- A. Osteoclasts
 - B. Osteoblasts
 - C. Osteocytes
 - D. a & b
 - E. All
54. A newly deposited uncalcified bone matrix is known as:
- A. Osteon
 - B. Perosteoid
 - C. Osteoid
 - D. Haversian system
 - E. Cancellous bone
55. Which one of the following is not true regarding the cardiac muscles?
- A. Divides and anastomose but the cytoplasm remains separated.
 - B. Intercalated discs correspond to the zone of the Z lines
 - C. Have peripherally located nuclei.
 - D. Sarcoplasmic reticulum makes diad with T-tubule but not triad
 - E. Cannot regenerate
56. Region of cross-linking between myosin molecules in a sarcomere is represented by:
- A. A-band
 - B. I band
 - C. M line
 - D. H band
 - E. Z line
57. The body of **first lumbar vertebrae** is formed by:
- A. The 12th thoracic sclerotome alone
 - B. The cranial half of the 1st thoracic sclerotome alone
 - C. Fusion of caudal half of the 12th thoracic with the cranial half of the 1st lumbar sclerotomes
 - D. Fusion of caudal half of the 1st lumbar with the caudal half of the 2nd lumbar sclerotomes
 - E. The 1st lumbar sclerotome alone
58. The ribs develop from:
- A. somatic layer of lateral mesoderm
 - B. Visceral layer of lateral mesoderm
 - C. Paraxial mesoderm
 - D. Intermediate mesoderm

- E. Neural crest
59. Which one of the following is true regarding hypomeric and epimeric divisions of the myotomes?
- A. Epimers are larger and located dorsally as compared to the hypomers
 - B. Epimers are smaller and located dorsally as compared to the hypomers.
 - C. Epimers are innervated by the dorsal branches of the ventral primary rami of spinal nerves
 - D. Hypomers give rise to the extensors of the back
 - E. None of the above.
60. The ventral mesodermal condensations of the limb bud give rise to which type of muscles?
- A. Extensors and pronators
 - B. Extensors and abductors
 - C. Extensors and adductors
 - D. Flexors and pronators
 - E. Flexors and abductors

Musculoskeletal Test-2 06 Batch

Questions 1-5 are missing

6. Nasopharynx has all of the following features except:
- a. Fold of levatorveli palatine
 - b. Fold of salpingopharyngeus
 - c. Pharyngeal recess
 - d. Pharyngeal tonsils
 - e. None
7. Which of the following is not located in the gap between superior and middle pharyngeal constrictors?
- a. Vagus nerve
 - b. Glossopharyngeal nerve
 - c. Stylopharyngeus
 - d. Stylohyoid ligament
 - e. None
8. Which pair of arterial supply to the paranasal sinus is wrong?
- a. Frontal sinus --- ophthalmic artery
 - b. Ethmoidal air cells --- ophthalmic artery
 - c. Sphenoidal sinus --- maxillary artery
 - d. Maxillary sinus --- maxillary artery
 - e. None
9. Which association between nasal meatus and their communication is wrong?
- a. Nasolacrimal duct --- inferior nasal meatus
 - b. Frontonasal duct --- middle nasal meatus
 - c. Maxillary sinus --- inferior nasal meatus
 - d. Anterior ethmoidal air cells --- middle nasal meatus
 - e. None
10. The nasal septum receives arterial supply from all of the following except

- a. Ethmoidal
 - b. Sphenopalatine
 - c. Superior labial
 - d. Infraorbital
 - e. None
11. Kisselbach's plexus of the nose could be formed by all of the following arteries except:
- a. Anterior ethmoidal
 - b. Posterior ethmoidal
 - c. Sphenopalatine
 - d. Facial
 - e. None
12. Walls and external parts of the nasal cavities receive innervation from all of the following except:
- a. Nasociliary
 - b. Anterior ethmoidal
 - c. Posterior ethmoidal
 - d. Nasopalatine
 - e. Greater palatine
13. All of the following intrinsic muscles of the larynx have the same action except:
- a. Lateral cricothyroid
 - b. Transverse arythenoid
 - c. Posterior cricoarythenoid
 - d. Oblique arythenoid
 - e. None
- ✓ Lateral cricoarythenoid has different action.
14. Anterior branch of inferior laryngeal nerve innervates all of the following muscles of the larynx except:
- a. Thyroarythenoid
 - b. Cricothyroid
 - c. Lateral cricoarytehnoid
 - d. Vocalis
 - e. None
15. Which of the following is wrong about larynx?
- A. Laryngeal vestibule is drained by superior deep cervical lymph nodes
 - B. Infraglottic cavity is drained by inferior deep cervical lymph nodes
 - C. Internal laryngeal nerve innervates laryngeal mucosa
 - D. Venous drainage from inferior part of larynx ultimately reaches brachiocephalic vein
- ✓ The question is not correct. All are true statements.
16. Keratinocytes show mitotic activity in which layer of epidermis?
- A. Stratum lucidum and stratum spinosum
 - B. Stratum basale and stratum granulosum
 - C. Stratum basale, stratum granulosum and stratum basale
 - D. Stratum basale and stratum spinosum
 - E. All layers, except stratum corneum

17. An eosinophilic layer of dead cells that contain densely packed filaments embedded in an electron dense matrix and may be absent in thin skin is:
- A. **Stratum lucidum**
 - B. Stratum spinosum
 - C. Stratum corneum
 - D. Stratum basale
 - E. Stratum granulosum
18. Which one of the following is **not true** about melanocytes?
- A. **Are less in number in white skinned than dark skinned races.**
 - B. Produce melanin pigments which are dark brown pigments
 - C. Have capacity of mitotic activity
 - D. Appear as cells with elongated nuclei surrounded by a clear cytoplasm under light microscopy following H & E staining.
 - E. Have long irregular and branched processes.
19. The most inner layer of the inner epithelial root sheath of hair follicle that anchors the hair in its root is:
- A. Cortex
 - B. Huxley's layer
 - C. **Cuticle**
 - D. Henle's layer
 - E. Medulla
20. Outer coat of hair follicle is:
- A. **Made by connective tissue continuous with the dermis**
 - B. Equivalent to stratum corneum
 - C. Equivalent to the most inner layer of epidermis
 - D. Absent from fine hairs
 - E. None of the above
21. Body of the nail or hair plate is equivalent to:
- A. Epidermis, dermis, and subcutaneous tissue of the skin
 - B. Epidermis and dermis of the skin
 - C. All layers of epidermis with no dermis
 - D. **Stratum corneum of epidermis**
 - E. Stratum germinativum of epidermis
22. Which one of the following is derived from the second pharyngeal arch?
- A. The sphenomandibular ligament
 - B. **The stapes of auditory ossicles**
 - C. The incus of auditory ossicles
 - D. The zygomatic bone
 - E. The malleus of auditory ossicles
23. Which one of the following is not derived from the mandibular pharyngeal arch?
- A. Mylohyoid
 - B. Tensor palatine
 - C. muscles of mastication
 - D. Tensor tympani

- E. **None of the above**
24. Which one of the following **doesn't** arise from pharyngeal pouches?
- A. Stroma of palatine tonsil
 - B. External auditory canal
 - C. **Middle ear cavity**
 - D. Parathyroid gland and thymus
 - E. Eustachian tube
25. Which prominences form the cheek and lateral portions of upper limb
- A. **Maxillary**
 - B. Mandibular
 - C. Lateral nasal
 - D. Median nose
 - E. All of the above
26. Which prominence forms the bridge of the nose?
- A. Lateral nasal
 - B. Medial nasal
 - C. **Frontonasal**
 - D. Maxillary
 - E. Mandibular
27. Which one of the following is caused by failure of the maxillary prominence to merge with its corresponding lateral prominence?
- A. Isolated cleft palate
 - B. Bilateral cleft lip
 - C. **Oblique facial cleft**
 - D. Cleft upper jaw
 - E. None of the above
28. Structure that is not developed in the third brachial arch?
- A. **Lesser cornu of hyoid bone**
 - B. Glossopharyngeal nerve
 - C. Common carotid artery
 - D. Stylopharyngeal artery
 - E. None of the above
29. Which of the following gives rise to most of the musculature of the tongue?
- A. Lateral mesoderm
 - B. Local splanchnic mesenchyme
 - C. **Myotomes of the occipital somites**
 - D. All of the above
 - E. None of the above
30. Structures derived from endoderm of the 4th pharyngeal pouch includes:
- A. Thymus gland
 - B. Palatine tonsil
 - C. Thyroid gland
 - D. **Para follicular cells of thyroids**
 - E. None of the above

Questions from 31-34 are skipped purposely because they are True or False questions.

35. The _____ cervical ganglion forms plexus around the internal and external carotid arteries
- A. Inferior
 - B. Superior
 - C. Middle
 - D. Cervicothoracic
 - E. All of the above
36. Which of the following joins the thoracic duct near its junction with venous system?
- A. Right jugular trunk
 - B. Left subclavian trunk
 - C. Left bronchomediastinum trunk
 - D. "a" and "b" only
 - E. "b" and "c" only
37. Which of the following muscle is not a suprahyoid muscle?
- A. Omohyoid
 - B. Stylohyoid
 - C. Mylohyoid
 - D. Geniohyoid
 - E. Digastric, anterior belly
38. The esophagus is a muscular tube that extends between the pharynx and stomach. Which portion of the esophagus is completely formed by skeletal muscle?
- A. Lower and middle one third
 - B. Cervical part
 - C. Upper and middle one third
 - D. "A" and "B" only
 - E. All of the above
39. Correct statement about the innervation of the esophagus?
- A. Somatic innervation arises from branches of recurrent laryngeal nerves to the lower half of esophagus
 - B. Parasympathetic innervation is provided by the cervical ganglion nerves
 - C. Vasomotor sympathetic innervation to cervical esophagus reaches through the plexus of inferior thyroid arteries
 - D. Vasomotor parasympathetic nerves reach the esophagus through the plexus of superior thyroid arteries
 - E. All of the above
40. Which nerve carries postganglionic parasympathetic fibers to the parotid gland?
- A. Chorda tympani
 - B. Greater petrosal
 - C. Auriculotemporal
 - D. Lingual
 - E. Inferior alveolar
41. Which of the following forms the lateral border of the infratemporal fossa?
- A. Zygomatic arch
 - B. Pterygoid plate

- C. Mastoid process
 - D. Posterior margin of maxilla
 - E. Ramus of mandible
42. Which of the following lymph nodes receive lymph from the tonsillar and peritonsillar regions?
- A. Submental nodes
 - B. Jugulodigastric nodes
 - C. Juguloomohyoid node
 - D. Occipital node
 - E. None of the above
43. Which of the following gland have short ducts?
- A. Sublingual
 - B. Submandibular
 - C. Parotid
 - D. "a" and "b" only
 - E. All of the above
44. Which of the following gland is supplied by the postganglionic parasympathetic fibers of the submandibular ganglion?
- A. Lacrimal
 - B. Sublingual
 - C. Parotid
 - D. Palatine
 - E. All of the above
45. The _____ artery is the first branch of the subclavian artery?
- A. Thyrocervical
 - B. Costocervical
 - C. Vertebral
 - D. Internal thoracic
 - E. Occipital
- ✓ Use the mnemonic VIT C (Vertebral, Internal thoracic, Thyrocervical trunk, Costocervical)
46. A tumor infiltrating the foramen ovale might result in which of the following:
- A. paralysis of tensor palitini muscle
 - B. sensory loss to the skin of the forehead
 - C. sensory loss to the skin of tip of the nose
 - D. paralysis of masseter muscle
 - E. sensory loss to the hard palate
47. In surgical removal of malignant parotid tumor, the facial nerve may be cut resulting in paralysis of one of the following muscles:
- A. tensor palatine
 - B. masseter
 - C. stylopharyngeus
 - D. orbicularis oris
 - E. anterior belly of digastric
48. All of the following are branches of the cervical plexus, EXCEPT:
- A. Supraclavicular nerve

- B. Transverse cervical nerve
 - C. Great auricular nerve
 - D. Greater occipital nerve
 - E. Lesser occipital nerve
49. Which nerve supplies sensory innervation to fascial skin over the parotid gland?
- A. Trigeminal
 - B. Great auricular
 - C. Lesser petrosal
 - D. Facial
 - E. Auriculotemporal
50. Clinically, four muscles are important in evaluating the integrity of the facial nerve. One of these muscles that raises the eyebrows in surprise or horror is:
- A. Buccinator muscle ... (this helps to whistle or suck)
 - B. Orbicularis Oculi muscle... (this helps to close eyelids)
 - C. Orbicularis oris... (this helps to smile)
 - D. Frontalis muscle
 - E. Procerus muscle
51. A patient comes to the Out-Patient Department (OPD) with a left sternocleidomastoid muscle paralysis. Which one of the following symptoms might she also have?
- A. Weakness in turning her head to the left
 - B. Weakness in turning her head to the right
 - C. Inability to protrude her tongue
 - D. Inability to elevate the shoulder
 - E. Inability to close the eyelid
52. Which of the following statements concerning the scalp is NOT true?
- A. Sweat glands and hair follicles lie in the skin layer
 - B. Superficial vessels and nerves lie in the loose connective tissue layer
 - C. The aponeurotic layer contains the frontal and occipital bellies of the occipitofrontalis muscle
 - D. Scalp is supplied by branches of both internal and external carotid arteries
53. The pterion is an important landmark to indicate position of:
- A. Anterior branch of superficial temporal artery.
 - B. Posterior branch of superficial temporal artery
 - C. Anterior branch of middle meningeal artery.
 - D. Posterior branch of middle meningeal artery
54. The structures that pass through the superior orbital fissure include all of the following, EXCEPT
- A. Trochlear nerve
 - B. Ophthalmic artery
 - C. Ophthalmic veins
 - D. Oculomotor nerve
 - E. a and c
55. After exiting the stylomastoid foramen, the facial nerve is correctly described by each of the following EXCEPT:
- A. It innervates the posterior belly of the digastric muscle

- B. It gives rise to posterior auricular nerve
 - C. It innervates a muscle that closes the eye
 - D. It innervates the parotid gland
 - E. It innervates the muscles that elevate the upper lip
56. Each of the following is found within the carotid sheath in the carotid triangle of the neck, EXCEPT the:
- A. Vagus nerve
 - B. Common carotid artery
 - C. Internal jugular vein
 - D. Cervical sympathetic trunk
 - E. None of the above
57. The carotid triangle is separated from the submandibular triangle by which muscle?
- A. Posterior belly of digastric
 - B. Anterior belly of digastric
 - C. Superior belly of omohyoid
 - D. Sternohyoid
 - E. Sternothyroid
58. The following muscles are innervated by Ansa Cervicalis, EXCEPT:
- A. Stylohyoid
 - B. Omohyoid
 - C. Sternohyoid
 - D. Thyrohyoid
 - E. Sternothyroid
59. Which of the following is a direct branch of the external carotid artery?
- A. Inferior alveolar artery
 - B. Superior alveolar artery
 - C. Infraorbital artery
 - D. Middle meningeal artery
 - E. Superior thyroid artery
60. All of the following concerning the cervical vertebrae are true EXCEPT:
- A. Transverse foramina except the transverse foramina of C7, transmit the vertebral artery
 - B. C1 vertebra is atypical because it lacks a vertebral body
 - C. One structure most likely to be compressed by a cervical rib is the brachial plexus
 - D. C7 vertebra is atypical because it lacks a spinous process

PC-2 Exams (currently unavailable)

Mock and Exit Exams

Mock 06 Batch

1. A 25-year-old patient is admitted with fracture of the humerus and wrist drop. Which part of the bone is involved in this accident?
 - a. Mid-shaft
 - b. Greater tubercle

- c. Deltoid tuberosity
 - d. Surgical neck
 - e. Anatomical neck
- 2. Paralysis of the serratus anterior muscle is often associated with winging of the scapula. Which nerve of the brachial plexus is injured?
 - a. Lateral pectoral
 - b. Subscapular
 - c. Suprascapular
 - d. Dorsal scapular
 - e. Long thoracic
- 3. Mesothelium lines the:
 - a. Respiratory tract
 - b. Gastrointestinal tract
 - c. Nasal cavity
 - d. Peritoneal cavity
 - e. Urinary tract
- 4. Which of the following is **not** lined by epithelial tissue?
 - a. Surface of skin
 - b. Corneal surface of the eye
 - c. Luminal surface of the intestinal tract
 - d. Luminal surface of the respiratory tract
 - e. Surface of bones
- 5. With regard to fertilization, which statement is **not** correct?
 - a. It usually occurs in the ampulla of the uterine tube
 - b. It causes metabolic activation of the ovum and initiates cleavage of the zygote
 - c. It follows capacitation of the sperm cells in the female genital tract
 - d. It determines genetic, but not phenotypic sex of the embryo
 - e. It follows shedding of the zona pellucida
- 6. During the third week of development, which statement is correct?
 - a. Somites are formed
 - b. The primitive streak appears
 - c. The notochord forms the primitive axis of the embryo
 - d. Mesoderm exists between the ectoderm and endoderm everywhere
 - e. Neural tube is formed
- 7. In humans, body water content of about 45% is found in:
 - a. Thin infant
 - b. Average female
 - c. Thin male
 - d. Fat male
 - e. Fat female
- 8. All of the following are responsible for electrolyte deficit **except**:
 - a. Sweating
 - b. Excess water drinking
 - c. Reduced osmolality

- d. Watery diarrhea
 - e. Nephritis
9. The branched chain amino acids are:
- a. Leucine, Isoleucine and valine
 - b. Tyrosine, phenylalanine, and tryptophan
 - c. Serine, threonine, and tyrosine
 - d. Methionine, cysteine and serine
 - e. Lysine, Arginine, and Histidine
10. Which one of the following is an essential fatty acid?
- a. Palmitic acid
 - b. Stearic acid
 - c. Butyric acid
 - d. Linoleic acid
 - e. Oleic acid
11. Which of the following vitamin is a component of acyl carrier protein?
- a. Biotin
 - b. Lipoic acid
 - c. Pantothenic acid
 - d. Folic acid
 - e. Thiamine pyrophosphate
12. Which of the following is required as a reductant in fatty acid synthesis?
- a. NADH
 - b. NADPH
 - c. FADH₂
 - d. FMNH₂
 - e. cAMP
13. A newborn becomes lethargic and drowsy 24 hours after birth. Blood analysis shows Hyperammonemia, coupled with orotic aciduria. The baby has an enzyme deficiency that leads to an inability to directly produce which of the following?
- a. Carbamoyl phosphate
 - b. Ornithine
 - c. Citrulline
 - d. Argininosuccinate
 - e. Arginine
14. Abebe had been prescribed a drug to treat his depression. One of the effects of the drug is to maintain elevated levels of a particular neurotransmitter that has been derived from which of the following amino acids?
- a. Tryptophan
 - b. Tyrosine
 - c. Glutamate
 - d. Histidine
 - e. Glycine
15. Which of the following bases is analog to 5-bromouracil that can react with deoxyribonucleic acid to produce a polymer with increased susceptibility to mutation?

- a. Thymine
 - b. Guanine
 - c. Cytosine
 - d. Uracil
 - e. Adenine
16. Which of the following most accurately describes allosterism?
- a. Inhibitor binds to an enzyme at the active site and competes with substrate
 - b. Activator binds to a site on an enzyme that is distant from the active site and reduces the affinity of the enzyme for its substrate
 - c. Feedback inhibition is a common mechanism for controlling the rates of metabolic pathways
 - d. Regulation occurs only with non-enzyme proteins such as hemoglobin
 - e. Refers to the control of gene transcription by proteins that bind to DNA
17. In patients with systemic inflammatory response syndrome, the signs and symptoms are mainly the result of:
- a. Prostaglandins
 - b. C3a
 - c. Serotonin
 - d. Clotting factors
 - e. Tumor necrosis factor – alpha
18. Which of the following statements is correct about wound healing?
- a. Keloids are scars that occur after surgical intervention
 - b. Infection impairs a wound healing by reducing the inflammatory phase
 - c. A wound is made smaller by the action of myofibroblasts
 - d. A well-approximated sutured surgical wound results in excessive scar formation
 - e. Generally, from injury to resolution, wounds go through two phases
19. A 3-year-old child presented with vomiting and fever. The best dosage form to administer paracetamol is:
- a. Tablet
 - b. Suppository
 - c. Capsule
 - d. Pill
 - e. Inhaler
20. Which of the following is wrong regarding drug interactions?
- a. Fatty food increases the absorption of efavirenz
 - b. Diuretics and vasodilators have synergistic hypotensive effect
 - c. Grape fruit juice is Cyp3A4 enzyme inducer
 - d. Antacids decrease absorption of ketoconazole
 - e. Concomitant use of monoamine oxidase inhibitors and wine may result in hypertensive crisis
21. Which of the following is not true regarding lymphoma?
- a. HIV infection increases one's risk of Non-Hodgkin's lymphoma
 - b. The majority of Non-Hodgkin's lymphoma are B-lymphocytic origin
 - c. The neoplastic cells in Hodgkin's lymphoma are T cell origin

- d. Extra-nodal lymphoma is less common than lymph node involvement
 - e. Survival for untreated cases is better for low grade than high grade lymphoma
22. Which of the following is not an example of paraneoplastic syndrome?
- a. Hypoglycemia induced by hepatoma
 - b. SIADH in renal cell carcinoma
 - c. Hypercalcemia due to skeletal metastasis
 - d. Acanthosis Nigricans by lung carcinoma
 - e. Cushing syndrome by small cell carcinoma
23. Which of the following immune cells is most effective against tumors?
- a. B-lymphocytes
 - b. Cytotoxic T cells
 - c. Macrophages
 - d. Helper T cells
 - e. Regulatory T cells
24. A patient is having trouble with keeping food out of the vestibule of the oral cavity. Which muscle of facial expression is affected?
- a. Mentalis
 - b. Temporalis
 - c. Orbicularis oris
 - d. Buccinator
 - e. Orbicularis oculi
25. Immediately after death, muscles undergo rigor mortis because:
- a. Most of the calcium is pumped back into the sarcoplasm reticulum
 - b. Myosin with its hydrolyzed products is detached from actin
 - c. The muscle fibers are autolyzed by lysosomal enzymes
 - d. Myosin remains attached with thin filaments
 - e. Newly synthesized ATP molecules attach to myosin heads
26. The most common cancer in bone is:
- a. Osteosarcoma
 - b. Ewing sarcoma
 - c. Chondrosarcoma
 - d. Metastatic cancer
 - e. Primitive neuroectodermal tumor
27. A 20-year-old man complains of localized bone pain and x-ray shows the presence of a cortical radiolucent lesion surrounded by sclerotic bone. The most likely diagnosis is:
- a. Enchondroma
 - b. Osteoid osteoma
 - c. Osteosarcoma
 - d. Giant cell tumor
 - e. Ewing sarcoma
28. During rib fractures, the broken end of the ribs may injure internal organs. Which of the following rib fracture is **not correctly** matched with the injured organs?
- a. 4th-6th ribs/lung
 - b. 1st rib/subclavian artery

- c. 2nd rib/brachial plexus
 - d. Left 9th-11th rib/spleen
 - e. Right 11th-12th ribs/liver
29. Which one of the following **does not** have direct effect on the normal development of the lungs?
- a. Congenital diaphragmatic hernia
 - b. Eventration of the diaphragm
 - c. Impeded amniotic fluid circulation in the lungs
 - d. Umbilical herniation
 - e. Lack of breathing movements during intrauterine life
30. Which of the following is correct about compliance of the lungs?
- a. Directly related to intrapleural pressure
 - b. Independent of age
 - c. Not influenced by elastic recoil of the lung
 - d. Mainly affected by intrathoracic pressure
 - e. Is inversely related to the surface tension of fluid lining the alveoli
31. Regarding the control of respiration, which one is correct?
- a. The central chemoreceptors detect changes in arterial pO₂, pCO₂ and pH
 - b. The peripheral chemoreceptors are highly sensitive to slight change in arterial pO₂
 - c. The central chemoreceptors play a major role in regulating change in pCO₂
 - d. The dorsal respiratory group of neurons are inactive during quiet breathing
 - e. The respiratory centers in Pons are responsible for setting the normal breathing pattern
32. The Reid index (the ratio of mucus gland thickness to bronchial wall thickness) is increased in:
- a. Chronic obstructive bronchitis
 - b. Emphysema
 - c. Bronchiectasis
 - d. Obstructive bronchial asthma
 - e. Small air way disease
33. Which of the following conditions is least likely to be caused by adenovirus infection in immunocompetent adults?
- a. Keratoconjunctivitis
 - b. Pneumonia
 - c. Pharyngitis
 - d. Meningitis
 - e. Gastroenteritis
34. A phosphodiesterase inhibitor used in advanced heart failure by increasing myocardial force of contraction is:
- a. Aminophylline
 - b. Digitoxin
 - c. Milrinone
 - d. Dobutamine
 - e. Amiodarone
35. A 50-year-old woman presented with blood pressure of 180/110mmHg and coma. The drug of choice to treat this patient is:

- a. Intravenous captopril
 - b. Intravenous nitroprusside**
 - c. Inhalational propranolol
 - d. Intramuscular hydralazine
 - e. Intravenous nifedipine
36. Which of the following conditions would most likely cause obstructive jaundice?
- a. Aneurysm of the splenic vein
 - b. Perforated ulcer of the stomach
 - c. Cancer of head of pancreas**
 - d. Damage to the pancreas tail
 - e. Cancer of body of pancreas
37. Almaz goes to a restaurant and ate 'Doro wot' following the meal her GIT secretion increased tremendously. This phase of secretion is:
- a. Intestinal phase of secretion
 - b. Gastric phase of gastric juice secretion**
 - c. Cephalic phase of gastric juice secretion
 - d. Cephalic phase of pancreatic juice secretion
 - e. Gastric phase of salivary juice secretion
38. Which of the following viruses are most commonly associated with gastroenteritis of all age groups?
- a. Norwalk like viruses**
 - b. ECHO viruses
 - c. Coxsackie virus
 - d. Reoviruses
 - e. Rotavirus
39. Which of the following best describes the delta hepatitis virus?
- a. An incomplete hepatitis B virus
 - b. Related to hepatitis A virus
 - c. Mutant hepatitis B virus
 - d. An incomplete RNA virus**
 - e. Analogous to Hepatitis C virus
40. The importance of autoregulation of renal blood flow is:
- a. To maintain normal blood pH by excreting H^+ ion with the urine
 - b. Maintain stable solute and fluid level in conditions of pressure fluctuations**
 - c. Inhibit the activity of Angiotensin II that causes constriction of afferent arteriole
 - d. Increase the efficient absorptive capacity of water by Antidiuretic hormone
 - e. To stimulate Macula Densa cells to act on Na^+ ions in the filtrate
41. The antidote for reversing organophosphate poisoning:
- a. Atropine + vitamin K
 - b. Atropine + oximes**
 - c. Atropine + naloxone
 - d. Atropine + protamine sulphate
 - e. Atropine + deferoxamine
42. Anticholinergic toxidrome is characterized by:

- a. Tachycardia, mydriasis, constipation
 - b. Bradycardia, miosis, salivation
 - c. H, muscle fasciculation followed by paralysis
 - d. Diaphoresis, tachycardia, blurred vision
 - e. Hyperactive bowel, pinpointed pupil, bradycardia
43. The following factors are important in the pathogenesis of cystic echinococcosis **except**:
- a. Size of cysts and their sites within the affected organs
 - b. Interaction between the expanding cysts and adjacent organ structures
 - c. Presence of previous immunological complications/disorders
 - d. Initial route of infection transmission
 - e. Complications caused by rupture of cysts and larval materials
44. A 32 years old HIV positive pregnant woman with a CD4 count of $150/\text{mm}^3$ is diagnosed with tuberculosis. Which of the following drugs could be considered for treatment?
- a. Nevirapine + rifampicin
 - b. Efavirenz + rifampicin
 - c. Nevirapine + rifabutin
 - d. Efavirenz + rifabutin
 - e. Efavirenz + rifapentine
45. Regarding the male reproductive system, which of the following **is not** correct?
- a. Bulbourethral glands in males are homologous to the Bartholin's glands in females.
 - b. Prostate gland is made of compound tubuloalveolar gland
 - c. Secretion of seminal vesicle is particularly rich in fructose
 - d. Lumina of ductuli efferentes appear serrated in cross section
 - e. Blood-testis barrier is located between primary and secondary spermatocytes
46. Which of the following effects on skeletal muscle is inhibited by insulin?
- a. Entry of glucose
 - b. Synthesis of glycogen
 - c. Uptake of amino acids
 - d. Catabolism of protein
 - e. Synthesis of protein
47. In a 51-year-old woman whose regular sexual cycle is becoming irregular for the following four months and then totally stopped. Possibility of pregnancy worried her and she visited her family physician. The pregnancy test turned out to be negative. Which of the following cocktails of lab results confirm that the woman is in her menopausal years?
- a. $\downarrow$ LH, $\downarrow$ FSH, $\uparrow$ Estradiol
 - b. $\downarrow$ LH, $\uparrow$ FSH, $\downarrow$ Estradiol
 - c. $\uparrow$ LH, $\downarrow$ FSH, $\downarrow$ Estradiol
 - d. $\uparrow$ LH, $\uparrow$ FSH, $\downarrow$ Estradiol
 - e. $\uparrow$ LH, $\uparrow$ FSH, $\uparrow$ Estradiol
48. A 30-year-old lady came with a chief complaint of vaginal bleeding. She gave history of spontaneous abortion six months back. Endometrial aspiration biopsy revealed biphasic proliferation of Cytotrophoblast and syncytiotrophoblast against a hemorrhagic and necrotic background. Chorionic villi were not seen. The best diagnosis is:
- a. Invasive mole

- b. Yolk sac tumor
 - c. Choriocarcinoma
 - d. Placental trophoblastic disease
 - e. None of the above
49. Which statement does not accurately describe tertiary syphilis?
- a. It occurs many years after primary syphilis
 - b. It can be diagnosed by VDRL
 - c. It is characterized by the absence of spirochetes in the lesions
 - d. The patient is highly infectious at this stage
 - e. Condyloma lata do not occur
50. In a neurological examination of a patient with hemiparesis and aphasia, which of the following arteries may have been occluded or thrombosed?
- a. Posterior cerebral
 - b. Middle cerebral
 - c. Anterior cerebral
 - d. Anterior cerebellar
 - e. Posterior cerebellar

Exit 06 Batch

1. If a throat swab culture result shows gram negative, beta-hemolytic and bacitracin-sensitive cocci, which of the following are the most likely bacteria?
 - a. Staphylococcus aureus
 - b. Streptococcus pyogenes
 - c. Streptococcus pneumoniae
 - d. Enterococcus faecalis
 - e. Streptococcus agalactiae
2. Streptococcus pneumoniae evades the immune response in the lung mainly by:
 - a. Producing catalase
 - b. Production of exotoxin
 - c. Producing capsule
 - d. Release of IgA protease
 - e. Release of exotoxin
3. The gold standard laboratory investigation for a patient suspected to have sepsis is:
 - a. Urine culture
 - b. C-reactive protein
 - c. Blood culture
 - d. Urine analysis
 - e. Serum electrolyte
4. Which of the following drugs belongs to class IB antiarrhythmic agents?
 - a. Quinidine
 - b. Lidocaine
 - c. Flecainide
 - d. Amiodarone
 - e. Procainamide

5. In a patient with severe asthma, which of the following is most likely to have adverse effects when used daily over long periods?
- Albuterol by aerosol
 - Beclomethasone by aerosol
 - Cromolyn inhaler
 - Prednisolone by mouth**
 - Theophylline is long-acting oral form
6. The drug of choice for the management of resistant nocturnal asthma is:
- Albuterol
 - Metaproterenol
 - Salmeterol**
 - Terbutaline
 - Ritodrine
7. A 45-year-old man with a previous laparotomy developed a direct inguinal hernia. Which nerve could have been injured?
- Femoral
 - Genitofemoral
 - Lateral femoral
 - Subcostal
 - Ilioinguinal**
8. A 60-year-old male developed perforation of duodenal ulcer. Which artery is most likely to be affected?
- Common hepatic
 - Left gastric
 - Proper hepatic
 - Superior mesenteric
 - Gastroduodenal**
9. A surgeon needs to construct a bypass between the veins of the portal and caval systems to circumvent insufficient drainage through the natural portocaval anastomoses. Which option is likely to be successful?
- Coronary vein to right gastroepiploic vein
 - Inferior mesenteric vein to splenic vein
 - Left colic vein to middle colic vein
 - Splenic vein to left renal vein**
 - Superior mesenteric vein to splenic vein
10. Which of the following is correct about the regulation of GIT functions?
- Excitation of Meissner's plexus increases motility
 - Autonomic nervous system can increase but not decrease the GI functions
 - Effective peristalsis does not require active myenteric plexus
 - Distension of stomach causes enterogastric reflexes
 - Gastrin stimulates gastric secretion and growth of gastric mucosa**
11. A 35-year-old pregnant woman has been brought to the maternal child care health clinic complaining of fatigue. The examination reveals that her eyes, tongue and skin are pale. Her daily

diets are completely cereal based with no animal source. She is suffering from what type of nutritional deficiency disease?

- a. Protein energy malnutrition
 - b. Iron deficiency anemia
 - c. Zinc deficiency
 - d. Iodine deficiency
 - e. Copper deficiency
12. A 2-month-old child has been brought to the pediatric clinic with spina bifida. The cause would be associated with a deficiency of:
- a. B12
 - b. Biotin
 - c. Folate
 - d. Pantothenic acid
 - e. Coenzyme A
13. Choose the correct statement?
- a. Hepatitis A virus infection is common cause of chronic hepatitis in children
 - b. Fatty change is pathognomonic of Hepatitis C virus infection
 - c. The liver biopsy in acute hepatitis due to hepatitis B virus is likely to show ground glass hepatocytes
 - d. Steatorrhea & hyperglycemia can be possible complications of chronic pancreatitis
 - e. In acute pancreatitis, there is a marked elevation of serum transaminase in the first 24 hours.
14. A 30-year-old male patient has had years of intermittent diarrhea and abdominal pain, but has never sought medical advice. Eventually, he begins to pass fecal material in the right anterior abdominal wall and he seeks medical attention. Which one of the following diseases is most likely to cause his complication?
- a. Colorectal adenocarcinoma
 - b. Crohn's disease
 - c. Diverticulitis
 - d. Ulcerative colitis
 - e. Whipple disease
15. Which of the following is **not true** regarding typhoidal salmonellosis?
- a. Serology is the gold standard for diagnosis
 - b. It causes systemic infection with multiple organ involvement
 - c. Fluoroquinolones are drug of choice for treatment
 - d. Mode of transmission is via contaminated foods and drinks
 - e. It usually involves the distal portion of ileum
16. A stool culture from a child with bloody diarrhea reveals non-motile, non-lactose fermenter gram negative rod bacteria that do not produce H₂S. The most likely etiologic agent is:
- a. Campylobacter jejuni
 - b. Enterohemorrhagic Escherichia coli
 - c. Salmonella typhimurium
 - d. Shigella dysenteriae
 - e. Clostridium difficile

17. Which of the following drugs exhibit a maximum gastric acid suppression effect?
- a. Proton pump inhibitors
 - b. H₂ receptor blockers
 - c. Anticholinergics
 - d. Antacids
 - e. Prostaglandin analogues
18. Which of the following stimulant laxative has highest risk of atonic colon on prolonged use?
- a. Methylcellulose
 - b. Magnesium citrate
 - c. Sorbitol
 - d. Bisacodyl
 - e. Docusate sodium
19. Blood from an injured kidney will seep through the perineal fat until it contacts the internal surface of the renal fascia. Without perforating the fascia, it will continue to pass towards:
- a. Pelvic cavity
 - b. Abdominal wall
 - c. Contralateral kidney
 - d. Thoracic diaphragm
 - e. Visceral peritoneum
20. Assuming renal creatinine clearance indicated a normal kidney function, what would be the renal clearance of a substance whose plasma level is 150mg/dL, urine concentration 100mg/dL, and showed urine formation rate of 1mL/min?
- a. 1 ml/min
 - b. 1.5 ml/min
 - c. 125 ml/min
 - d. 0.6 ml/min
 - e. 650 ml/min
21. If a slide shows diffusely packed lymphoid cells organized into cortex and medulla with no lymphoid nodules, which type of lymphoid tissue is it?
- a. Lymph nodes
 - b. Thymus
 - c. Tonsils
 - d. Spleen
 - e. Peyer's patches
22. Which of the following statement is **not correct** about plasma proteins?
- a. Are responsible for the oncotic pressure
 - b. Are filtered at the renal glomerulus
 - c. Are mostly synthesized in the liver
 - d. Transport some hormones in the blood
 - e. Consist of mainly albumin
23. If the blood group of a husband and wife is B, then which statement is **not true**?
- a. All their children can be either blood group-B or O
 - b. The wife's mother might have been blood group O
 - c. The children can be blood group-AB

- d. The wife's mother might have been blood group-B
 - e. The husband's father might have been blood group-O
24. Which of the following is **not true** about albumin?
- a. It is synthesized primarily in the bone marrow
 - b. It is involved in calcium transport in the blood stream
 - c. It functions as a transporter of fatty acids
 - d. It is involved in regulation of the oncotic pressure of the blood
 - e. Its levels in serum are usually low in patients with severe hepatic cirrhosis
25. Which of the following conditions is **not** associated with hemolytic jaundice?
- a. Sick cell anemia
 - b. Obstruction of bile duct
 - c. Plasmodium falciparum infection
 - d. Glucose-6-phosphate dehydrogenase deficiency
 - e. A mismatched blood transfusion
26. Which of the following statement is **not correct** regarding HIV/AIDS?
- a. HIV begins its life cycle when viral envelope gp 120 interacts with CD4 molecule and co-receptors
 - b. In primary infection, patients can experience acute retroviral syndrome
 - c. An HIV positive person can be considered as an AIDS case when CD4 count is < 200
 - d. In AIDS patients, Kaposi sarcoma is caused by Epstein Barr Virus.
 - e. Smear negative tuberculosis is more common in advanced HIV infection.
27. Which of the following is **not correct** concerning the complement system?
- a. Complement component proteins are synthesized in the liver
 - b. Alternative pathway requires antibodies to initiate the pathway
 - c. Classical pathway requires IgM and IgG for activation
 - d. The end results of classical and alternative pathways are similar
 - e. The role of the complement components is opsonization
28. The purpose of lymphocyte recirculation is to:
- a. Remain self-tolerant
 - b. Activate epithelial cells
 - c. Increase exposure to antigen
 - d. Receive activation signals
 - e. Increase the number of cells
29. A 26-year-old woman presented with headache, lightheadedness, rapid heart rate and easy fatigability. Laboratory workup reveals low hemoglobin and microcytic red cells. Which of the following is the most suitable therapy?
- a. Ferrous sulfate
 - b. Folic acid
 - c. Iron dextran
 - d. Pyridoxine
 - e. Vitamin B12
30. A patient has received warfarin for 2 weeks. As a result of this therapy, the patient will mainly exhibit:
- a. Reduced plasma prothrombin (factor II) activity

- b. Reduced plasma factor VIII activity
 - c. Reduced plasma plasminogen activity
 - d. Increased tissue plasminogen activator activity
 - e. Increased platelet adenosine store
31. Which one of the following is present in the Philadelphia chromosomes?
- a. ACK
 - b. AKT
 - c. AML
 - d. APC
 - e. ABL
32. Which of the following molecules induce cell proliferation?
- a. Growth factors
 - b. Growth factor receptors
 - c. Cyclins
 - d. Nuclear lamins
 - e. Checkpoints
33. When grading a malignant tumor, a pathologist considers:
- a. The degree of differentiation of the tumor cells
 - b. Tumor stage
 - c. Lymph node involvement
 - d. Vascular involvement
 - e. The extent of spread of the tumor
34. A 23-year-old woman was presented for a lump in her breast that she palpated on self-breast examination. Her clinical history reveals that her mother and her aunt both had breast and ovarian cancer. Given this presentation, you suspect the patient may have a mutation in which of the following genes involved in DNA repair?
- a. BRCA-1
 - b. Ras
 - c. Bcl-2
 - d. p53
 - e. Rb
35. Which type of tumor is caused by human herpes virus-8?
- a. Kaposi's sarcoma
 - b. Nasopharyngeal carcinoma
 - c. Cervical cancer
 - d. Skin warts
 - e. Burkitt's lymphoma
36. Antineoplastic drugs are not characterized by:
- a. Less selectivity to the cancer cells
 - b. Depression of bone marrow cells
 - c. Immunosuppression
 - d. Increased appetite
 - e. Infertility
37. Which of the following is a target of cyclophosphamide?

- a. DNA
 - b. Microtubules
 - c. Dihydrofolate reductase
 - d. Topoisomerase
 - e. Cell membrane
38. A 40-year-old male patient with an injury around the neck is asked to protrude his tongue. The tongue was deviated to the right, on protrusion and right side of the tongue comes higher than the left on retraction. The patient has injury of:
- a. Right glossopharyngeal nerve
 - b. Right hypoglossal nerve
 - c. Left hypoglossal nerve
 - d. Left glossopharyngeal nerve
 - e. Right vagus nerve
39. A lesion of the lingual nerve immediately after it receives the chorda tympani nerve could result in loss of each of the following **except**:
- a. Sublingual gland secretion
 - b. Submandibular gland secretion
 - c. Sensation from the lower teeth
 - d. Special sensation of taste from anterior two-thirds of the tongue
 - e. General sensation from anterior two-thirds of the tongue
40. One of the following statements **does not** describe micturition reflex?
- a. The reflex is integrated in the sacral segment of the spinal cord
 - b. Parasympathetic stimulation increases voiding
 - c. Destruction of sensory nerves supplying the bladder causes overflow dribbling of urine
 - d. Infants cannot control voiding for their immature brain
 - e. The external urinary sphincter is under autonomic nervous system control
41. What physiological effect **does not** occur during water overload?
- a. Urine osmolality decreases as low as 50 mOsm/L
 - b. Antidiuretic hormone synthesis is inhibited
 - c. Water reabsorption decreases in the late distal and collecting duct
 - d. Body osmolality remains at about 300 mOsm/L
 - e. Atrial natriuretic peptide blocks Na⁺ reabsorption in the distal tubule
42. The microscopic abnormality typically seen in nephritic syndrome:
- a. Thickening of the glomerular basement membrane
 - b. Effacement of podocyte foot processes
 - c. No abnormality on light microscope
 - d. Enlarged, hypercellular glomeruli
 - e. Sub-endothelial immune-complex deposits
43. Which of the following is not correctly matched?
- a. Post-infectious glomerulonephritis --- diffuse proliferative glomerulonephritis
 - b. Post-infectious glomerulonephritis --- crescentic glomerulonephritis
 - c. Good pasture syndrome --- crescentic glomerulonephritis
 - d. C3 glomerulopathies --- minimal change disease
 - e. IgA nephropathy --- mesangioproliferative glomerulonephritis

44. Which of the following statements is wrong about a toxicant?
- a. Agents that fasten intestinal transit reduce absorption of a toxicant
 - b. Bio-transformation of a toxicant may lead to production of more toxic metabolites
 - c. Gastric lavage is recommended after ingestion of a corrosive substance
 - d. Toxicants exposed by oral route are subject to first pass effect
 - e. The use of emetics should be avoided in the management of poisoning due to corrosives
45. Correct match of toxic agent with its antidote:
- a. Acetaminophen Physostigmine
 - b. Benzodiazepines Pyridoxine
 - c. Isoniazid Naloxone
 - d. Methanol Fomepizole
 - e. Opioid flumazenil
46. The most common organ affected by hydatid cyst is?
- a. Liver
 - b. Brain
 - c. Spleen
 - d. Terminal ileum
47. Intestinal amoebiasis in large bowel is characterized by:
- a. Atrophy of villi
 - b. Obstruction of intestine
 - c. Constipation
 - d. Flask-like ulceration
 - e. Mucosal atrophy
48. The most important feature to distinguish *Bacillus anthracis* from *Bacillus cereus* is:
- a. Motility test – Positive
 - b. Aerobic growth
 - c. Capsule test – Positive
 - d. Gram stain – Positive rod shaped
 - e. Coagulase test – Negative
49. Which of the following statements is true about coccidioidomycosis?
- a. Transmitted from person to person by air-borne spherules
 - b. Mostly inapparent and self-limited infection in endemic areas
 - c. Best treated with Griseofulvin
 - d. Most prevalent in the southeastern regions of the United States
 - e. Diagnosed by demonstrating arthroconidia formed in vivo
50. Which of the following statements is **not true** about urinary tract infection?
- a. It affects more men than women
 - b. Hospital procedures are major risk factors for nosocomial infections
 - c. Interference in urine flow increases the risk of urinary tract infections
 - d. Asymptomatic bacteriuria is usually associated with renal abnormalities
 - e. Infecting organisms of urinary tract infections are usually of fecal origin
51. One of the followings is not true of blood/tissue flagellate infections in man:
- a. *Trypanosoma cruzi* infections are not found among people living in Sub-Saharan Africa
 - b. *Leishmania donovani* infections can be acquired from zoonotic sources

- c. Both Chagas disease and human African trypanosomiasis are treatable
 - d. Both male and female tse-tse flies can transmit African trypanosomiasis to humans
 - e. Amastigote stages are found in CSF in patients with human African trypanosomiasis
52. The repeated bouts of fever in louse borne relapsing fever is due to:
- a. Repeated re-infection
 - b. Antigenic variation
 - c. Release of bacteria from reticuloendothelial cells
 - d. Periodic release of endotoxin
 - e. --- Reaction
53. Which of the following penicillin is gastric acid stable and beta lactamase resistant?
- a. Ampicillin
 - b. Cloxacillin
 - c. Piperacillin
 - d. Penicillin V
 - e. Penicillin G
54. Intestinal and Extraintestinal amoebiasis can be effectively treated with:
- a. Metronidazole
 - b. Mebendazole
 - c. Diloxanide furoate
 - d. Tetracycline
 - e. Chloroquine
55. Which of the following should be avoided in treating an infection in a pregnant woman?
- a. Ampicillin
 - b. Cephalexin
 - c. Tetracycline
 - d. Erythromycin
 - e. Amoxicillin
56. A 65-year-old man came to the emergency room with distension of the urinary bladder due to compression of the urethra. Which part of the prostate gland could be the most possible cause of obstruction in this patient?
- a. Anterior lobe
 - b. Posterior lobe
 - c. Median lobe
 - d. Right lateral lobe
 - e. Left lateral lobe
57. An elderly woman has a fracture of the ischial tuberosity disrupting the nerve that passes through the lesser sciatic foramen. In this accident, which structure will not be affected?
- a. Labia minora
 - b. Urethral sphincter
 - c. Posterior fornix of vagina
 - d. Ischiocavernosus muscle
 - e. Skin of the clitoris
58. A 45-year-old mother of five children came with prolapse of uterus. In this condition, the most likely affected structure is:

- a. Pelvic brim
 - b. Pelvic fascia
 - c. Pelvic diaphragm
 - d. Urogenital diaphragm
 - e. Urogenital triangle
59. A 45-year-old patient has difficulty of digital flexion and extension due to muscle damage in the hand. The surgeon decides to transplant gracilis muscle with its blood vessels and nerves to restore the normal function of the hand. What does this imply about gracilis muscle?
- a. Strong flexor
 - b. Weak flexor
 - c. Strong adductor
 - d. Weak adductor
 - e. Weak abductor
60. You are to perform a venipuncture of the median cubital vein. Which of following structures is most suitably located to protect against your needle accidentally entering the brachial artery?
- a. Tendon of biceps
 - b. Bicipital aponeurosis
 - c. Tendon of brachialis
 - d. Brachioradialis
 - e. Tendon of coracobrachialis
61. A medical technologist received a biopsy tissue sample and was asked to prepare a tissue section. Which one of the following procedures should he/she follow before taking the tissue for microtomy?
- a. Dehydration, fixation, clearing, infiltration, & embedding
 - b. Dehydration, clearing, infiltration, fixation & embedding
 - c. Fixation, dehydration, clearing, infiltration, & embedding
 - d. Fixation, dehydration, infiltration, clearing & embedding
 - e. Clearing, dehydration, fixation, infiltration, & embedding
- ✓ Mnemonic: **F**ind **D-C**! In the **E**ast
62. With regard to the epithelial tissue, which of the following statements is correct?
- a. The major constituent of epithelia is abundant extracellular matrix
 - b. Most of the cells have high turnover rate
 - c. It has abundant nerve supply
 - d. Goblet cells produce mucus and represent multicellular glands
 - e. Endothelium is another name for stratified columnar epithelium
63. Concerning gametogenesis, which statement is true?
- a. All oogonia become primary oocytes during ovulation
 - b. Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation
 - c. Secondary oocyte completes its second meiotic division to become a mature ovum during fertilization
 - d. The differences in sex chromosome complement of oocytes from the basis of primary sex determination
 - e. Primary spermatocytes begin the first meiotic division before birth, but completion of prophase does not occur until puberty

64. Which of the following statements is not correct regarding pre-embryonic period?
- Blastocyst is formed about four days after fertilization
 - Morula is formed by 12-16 blastomers
 - Cleavage results in successively smaller blastomers
 - Implantation begins, but is not completed
 - Bilaminar germ disc appears
65. Combined passive and active movements of substances across the cell membrane may be exemplified by:
- Transport of glucose and amino in RBC
 - $\text{Na}^+ - \text{K}^+$ ATPase mechanism
 - $\text{Cl} - \text{HCO}_3$ exchanger of RBC
 - Symport of Na^+ and glucose in the gastrointestinal tract
 - Transport of Ca^{2+} and thyroid hormone in the gut
66. A 20-year-old man stabbed in the abdomen lost about 2.5 liters of blood. Subsequently, his mean arterial blood pressure (ABP) fell down to 60mmHg. Corrective measures with blood transfusion caused a rise in mean arterial blood pressure to 90mmHg. This man on recovery may have:
- New ABP set point of 90mmHg
 - An error signal of ABP of -30mmHg
 - A cardiac output of more than 5 liters per minute
 - A progressive decrease in the total peripheral resistance
 - An elevated blood pressure than normal
67. Regarding synaptic transmission in the autonomic nervous system:
- Parasympathetic cholinergic activity relaxes smooth muscles
 - Vago-vagal syncope is due to increased somatic motor activity
 - Blood vessels of some skeletal muscles are muscarinic cholinergic
 - Increased vagal stimulation to the heart is the cause of resting tachycardia in athletes
 - Beta adrenergic effects on bronchial muscles aggravate asthmatic attack
68. One of the following amino sugars is preferentially involved in facilitating influenza infection in humans
- N-acetylglucosamine
 - N-acetylgalactosamine
 - N-acetylneuraminic acid
 - N-acetylmannosamine
 - N-fructosamine
69. Which of the following glycosaminoglycans serves as natural anticoagulant?
- Chondroitin sulfate
 - Keratan sulfate
 - Hyaluronic acid
 - Heparin
 - Heparan sulfate
70. Acute hemolytic episode after administration of antimalarial, primaquine, is due to deficiency of:
- 6-phosphogluconate dehydrogenase
 - Glucose-6-phosphate dehydrogenase
 - Epimerase

- d. Transketolase
 - e. Transaldolase
71. Von Gierke's disease is caused by which defective enzyme?
- a. Amylo α -1,6 glucosidase
 - b. Branching enzyme (α -1,4 $\rightarrow$ α -1,6)
 - c. Glucose-6-phosphatase
 - d. α -1,4 glucosidase
 - e. Phosphorylase
72. A patient complains of stomach cramp and diarrhea every time he eats ice-cream. He probably has deficiency of which enzyme?
- a. Invertase
 - b. Lactase
 - c. Isomerase
 - d. Maltase
 - e. Amylase
73. Which of the following enzymes is a sensor of blood glucose level in pancreatic β -cells?
- a. Glucokinase
 - b. Hexokinase
 - c. Phosphofructokinase
 - d. Fructokinase
 - e. Glycerol kinase
74. Which of the following inhibits the F_o component of ATP synthase?
- a. Cyanide
 - b. Oligomycin
 - c. Amytal
 - d. Antimycin A
 - e. Penicillin
75. In electron transport chain, 2,4-Dinitrophenol:
- a. Inhibits normal function of the electron transport chain without production of ATP
 - b. Allows normal function of the electron transport chain without production of ATP
 - c. Is a lipophilic molecule and couples electron transport with ATP production
 - d. Increases ATP production
 - e. Cannot freely diffuse across inner mitochondrial membrane
76. Cyanide poisoning results in total arrest of oxidative phosphorylation by inhibition of which of the following?
- a. NADH-dehydrogenase
 - b. Succinate dehydrogenase
 - c. Cytochrome reductase
 - d. Cytochrome oxidase
 - e. $F_o F_1$ ATPase
77. A drug 'X' binds reversibly to an enzyme 'E' which obeys Michaelis-Menten kinetics for its substrate, 'S'. The drug causes **no change** in maximum velocity (V_{max}) of the enzyme for its substrate 'S', but the Michaelis constant (K_m) for S is **increased**. The drug is likely to be:
- a. A non-competitive inhibitor of the enzyme

- b. A competitive inhibitor of the enzyme
 - c. An uncompetitive inhibitor of the enzyme
 - d. A “suicide” inhibitor of the enzyme
 - e. An activator of the enzyme
78. Oxidative stress can be described as:
- a. Disturbance in the balance between the production of reactive oxygen species and antioxidant defenses
 - b. Consumption of more oxygen by mitochondria
 - c. Imbalance between production of reactive oxygen species and antioxidant defenses
 - d. Consumption of more oxygen and releasing less CO₂
 - e. Abnormal folding of protein in Endoplasmic reticulum
79. Fat soluble antioxidants usually protect
- a. Lipid oxidation
 - b. Protein oxidation
 - c. DNA oxidation
 - d. Carbohydrate oxidation
 - e. RNA oxidation
80. Which of the following statements is correct about wound healing?
- a. Keloids are scars that occur after surgical intervention
 - b. Infection impairs a wound healing by reducing the inflammatory phase
 - c. A wound is made smaller by the action of myofibroblasts
 - d. A well-approximated sutured surgical wound results in more intense inflammation and excessive scar
 - e. Generally, from injury to resolution, wounds go through two phases
81. In a patient with severe burn, which of the following substances does not increase capillary permeability?
- a. Bradykinin
 - b. Histamine
 - c. Serotonin
 - d. –
 - e. –
82. Which of the following statements is not correct about bacterial endospores?
- a. Spores are disseminated by air
 - b. Spores are multilayered structures
 - c. Clostridium tetani exhibits terminal spore formation
 - d. Spores can be destroyed by common disinfectants
 - e. Endospores are heat resistant due to high content of calcium dipicolinate
83. Bacterial DNA is transferred from one bacterium to another by a virus through:
- a. Transformation
 - b. Transduction
 - c. Conjugation
 - d. Transposition
 - e. Replication
84. Most disease-causing bacteria are:

- a. Heterotrophs
 - b. Chemotrophs
 - c. Autotrophs
 - d. Chemoheterotrophs
 - e. Phototrophs
85. Which of the following **best** describes interferon's mode of actions in resistance to viral infections?
- a. Stimulates cell-mediated immunity
 - b. Stimulates humoral immunity
 - c. Suppresses rRNA formation
 - d. Inhibits viral mRNA translation
 - e. Alters cell membrane permeability
86. One of the following does not determine the route of administration of a drug:
- a. Mechanism of action of the drug
 - b. Dosage form of the drug
 - c. Onset of action of the drug
 - d. Patient's condition
 - e. Duration of action of the drug
87. Which of the following is not true about drug response?
- a. Drug response may be different in the same patient at different times
 - b. Drug response may be different in different individual at the same dose
 - c. Different concentration may be required to produce the same therapeutic effect in different individual
 - d. Quantal dose response doesn't show variability in drug response among the population
 - e. There is an individual variation for drugs response
88. A blood film slide shows many large sized granulocytes with twisted and horse-shoe shaped nuclei, but not lobulated, and fine granules in the cytoplasm. What is the most likely interpretation?
- a. Mature monocytes with azurophilic granules
 - b. Increased production of eosinophils following parasitic infection
 - c. Increased production of neutrophils following bacterial infection
 - d. Megakaryocyte fragmentations of the metachromatic cytoplasm
 - e. Large sized lymphocytes with endomitosis showing repeated DNA replication
89. Which one of the following is not a correct statement regarding the endocrine glands?
- a. Cells are architecturally arranged as cords, clusters or follicles.
 - b. Have more blood capillaries than exocrine glands
 - c. In the adrenal gland, cells with dark electron dense cytoplasm and pyknotic nuclei are characteristics of zona reticularis
 - d. Folliculostellate cells are nonsecretory cells of adenohypophysis that have long processes making gap junctions
 - e. The cells of zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained with potassium bichromate
90. During the development of the reproductive system, which statement is true?
- a. Hypospadias is constantly associated with exstrophy of urinary bladder

- b. The lower portion of the vagina is derived from sinovaginal bulb
 - c. In the male, failure of the urethral folds to fuse completely results in cryptorchidism
 - d. The cranial end of mesonephric duct in males may persist as paradidymis
 - e. Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester
91. Removal of the adrenal glands generally has all of the following consequences **except**:
- a. Tendency of hyperglycemia with decreased insulin sensitivity
 - b. Poor mobilization and utilization of triacylglycerol
 - c. Poor water excretion by the kidneys and sodium loss in the urine
 - d. Poor resistance to infection and shock
 - e. Psychic changes such as depression or decreased alertness
92. Hypothalamic releasing hormones:
- a. Do not affect secretion of thyroid hormones
 - b. Are transported by hypothalamo-hypophyseal tract to the anterior pituitary
 - c. Have receptors in the anterior pituitary
 - d. Secretion will be impaired by damage of hypothalamus-hypophyseal portal system
 - e. Regulate secretion of calcitonin by negative feedback mechanism
93. Patients with chronic renal disease frequently develop secondary hyperparathyroidism, which is characterized by:
- a. Hypercalcemia
 - b. hyperphosphatemia
 - c. Decreased production of vitamin D3
 - d. Increased production of 1,25-dihydroxycholecalciferol
 - e. Normal plasma level of phosphate
94. Successful fertilization is most likely to occur when the oocyte:
- a. has entered the second meiotic division
 - b. has completed the first meiotic division
 - c. is at metaphase of mitosis
 - d. is at anaphase of mitosis
 - e. is at telophase of mitosis
95. Which of the following best describes hormone response element (HRE)?
- a. HRE is a specific DNA sequence that binds to steroid hormone
 - b. HRE is a receptor that binds to specific DNA sequence
 - c. HRE is a specific DNA sequence that binds only to a hormone-receptor complex
 - d. HRE cannot increase transcriptional activity
 - e. HRE or the receptor can bind to DNA
96. Concerning glucagon receptors, which of the following is correct?
- a. Glucagon receptors are not present on liver cell membrane
 - b. Glucagon receptors act through cAMP
 - c. Glucagon receptors inhibit cAMP production
 - d. Glucagon receptors fail to stimulate the breakdown of glycogen
 - e. Glucagon receptors inhibit lipase in fat cells
97. Which of the following breast lesions has the highest risk of carcinoma?
- a. Atypical ductal hyperplasia

- b. Fibrocystic change
 - c. Duct ectasia
 - d. Large duct papilloma
 - e. Fibroadenoma
98. Which of the following thyroid tumors has the worst prognosis?
- a. Malt lymphoma
 - b. Papillary carcinoma
 - c. Anaplastic carcinoma
 - d. Medullary carcinoma
 - e. Poorly differentiated thyroid carcinoma
99. Which of the following is the least typical consequence or complication of long standing diabetes?
- a. Glomerulosclerosis
 - b. Mucor mycosis of ethmoid sinus
 - c. Beta cell hyperplasia
 - d. Neuropathy
 - e. Myocardial infarction
100. Which of the following statements is true about Neisseria gonorrhoeae?
- a. It is gram positive diplococci bacteria
 - b. The best specimen to isolate the bacteria is urine sample
 - c. It causes infection manifested by urethritis, cervicitis and salpingitis
 - d. It causes ulcerative lesions of the genitalia
 - e. It causes a self-limiting disease with no complications
101. A 31-year-old woman desires a combined oral contraceptive to prevent pregnancy. Which of the following patient factors would lead a health professional to recommend an alternative form of contraception?
- a. Concomitant use of metformin
 - b. History of gastroesophageal reflux disease
 - c. History of pelvic inflammatory disease
 - d. History of migraine headache
 - e. Plan to become pregnant after one year
102. Which of the following agents would be used to facilitate labor and delivery?
- a. Dopamine
 - b. Leuprolide
 - c. Oxytocin
 - d. Prolactin
 - e. Vasopressin
103. In a patient showing analgesia and thermal anesthesia of the face on the left side, and anesthesia of the left side of the body, where is the lesion?
- a. Genu of the left internal capsule
 - b. Trigeminothalamic tract
 - c. Lateral spinothalamic tract
 - d. Medial lemniscus
 - e. Genu of the right internal capsule
104. Which of the following is **not seen** in a patient with Brown-Sequard syndrome?

- a. Ipsilateral paralysis of muscles
 - b. Contralateral loss of discriminative touch
 - c. Contralateral loss of pain sensation
 - d. Ipsilateral dermatomal anesthesia
 - e. Ipsilateral loss of proprioceptive sensation
105. In an infant with obstructive hydrocephalus, the blockage in the ventricular system between the diencephalon and rhombencephalon is located at:
- a. Central canal
 - b. Aqueduct of Sylvius
 - c. Interventricular foramen
 - d. Foramen of Luschka
 - e. Foramen of Magendie
106. With regard to the eye and ear, which of the following statements is true?
- a. The cornea is attached to the ciliary body by zonules
 - b. Aqueous humor is actively secreted by cells of the stria vascularis
 - c. Otoliths are normally found in the saccules and utricles
 - d. The bony labyrinth is filled with lymph
 - e. Reissner's membrane separates the scala media from oval windows
107. Concerning the development of the central nervous system, which one is false?
- a. The basal plate region of the neural tube develops into sensory areas
 - b. Failure of closure of cranial neuropore causes anencephaly
 - c. Caudal neuropore closes before cranial neuropore
 - d. Pontine flexure appears before both cervical and mesencephalic flexures
 - e. Glial cells and neurons develop from the same stem cells
108. Which of the following is true about cerebrospinal fluid (CSF) and Blood Brain Barrier (BBB)?
- a. CSF has more proteins than the plasma
 - b. CSF is reabsorbed actively
 - c. All areas of the brain have BBB
 - d. Hydrocephalus is the result of hypocerebral fluid
 - e. Reduction of CSF causes headache
109. Which of the following is true about the synaptic transmission?
- a. Drugs affect the electrical synapse than chemical
 - b. Cytoplasmic continuity is the feature of chemical synapse
 - c. Synaptic delay is the feature of chemical synapse
 - d. Most chemical synaptic transmissions are bidirectional
 - e. The gap between two cells in electrical synapse is greater than in chemical
110. If you feel pain starting from the last lower ribs around the back pelvic areas, where do you think is the origin of the injury?
- a. Heart
 - b. Ureter
 - c. Stomach
 - d. Urethra
 - e. Liver
111. Bitemporal hemianopia visual defects are associated with lesion of the:

- a. Pyramidal tract
- b. Medial lemniscus
- c. Occipital lobe
- d. Optic nerve
- e. Optic chiasma

112. Air conduction deafness:

- a. Is the decreased ability to hear high frequency sound with age
- b. Results from impaired sound transmission in the external or middle ear
- c. Results from damage to the hair cells or neuronal pathways
- d. Can be observed with prolonged exposure to loud sound
- e. Is present if the sound is louder on the normal ear on the Weber test

113. Catecholamines are synthesized from which of the following amino acids?

- a. Tyrosine
- b. Tryptophan
- c. Glycine
- d. Glutamate
- e. Lysine

114. Acetylcholine binds to which of the following receptors?

- a. α -adrenergic receptors
- b. β -adrenergic receptors
- c. Nicotinic receptors
- d. Dopamine-1-receptors
- e. Nuclear receptors

115. Which of the following is the site of CSF secretion?

- a. Aqueduct
- b. Spinal cord (central canal)
- c. Choroid plexus
- d. Arachnoid villi
- e. Auerbach plexus

116. The brain cell which is mainly involved in forming the Blood Brain Barrier is:

- a. Bipolar neurons
- b. Astrocyte
- c. Microglia
- d. Ependymal cell
- e. Oligodendrocyte

117. Which of the following viral congenital infections **does not** result in birth defects?

- a. Herpes Simplex Viruses
- b. Cytomegalovirus
- c. Rubella virus
- d. HIV
- e. Varicella Zoster virus

✓ The question is **not correct**. All can result in birth defects.

118. A patient taking medications for a psychiatric disorder develops tremor, thyroid enlargement, edema and acneiform eruptions on the face. The drug he is taking is most likely to be:

- a. Carbamazepine
 - b. Haloperidol
 - c. Lamotrigine
 - d. Lithium
 - e. Sertraline
119. The duration of which of the following stages of general anesthesia should be minimized?
- a. Stage 1
 - b. Stage 2
 - c. Stage 3- plane 1
 - d. Stage 3- plane 2
 - e. Stage 4
120. When the femur is fractured, the broken distal end often turns posteriorly to enter the popliteal fossa due to muscle traction. Because of its position deepest in the fossa, which structure is most vulnerable to laceration?
- a. Common fibular nerve
 - b. Great saphenous vein
 - c. Popliteal artery
 - d. Popliteal vein
 - e. Tibial nerve
121. Upon examining a sick child, you notice pus draining from the middle meatus of the nose. Which of the following is the most likely site of origin of the infection?
- a. Nasolacrimal duct
 - b. Sphenoid sinus
 - c. Maxillary sinus
 - d. Mastoid sinus
 - e. Ethmoidal sinus
122. Regarding the tissues of the skeletal system, which of the following is not correct?
- a. Fibrous cartilage is the strongest type of cartilage because it contains type I collagen
 - b. The first stage of intracartilagenous bone ossification is bone collar formation through intramembranous ossification
 - c. Hyaline cartilage has abundant blood supply and could easily heal if damaged
 - d. Osteoclasts are involved in bone decalcification
 - e. Bone forming osteoprogenitor cells are located in both periosteum and endosteum
123. Excitation-contraction coupling does not involve one of the following:
- a. Generation of end plate potential
 - b. Release of calcium from troponin
 - c. Formation of linkage between actin and myosin
 - d. Depolarization along T tubule
 - e. Hydrolysis of ATP to ADP
124. Which of the following blocks calcium-mediated acetylcholine release and death due to paralysis of the diaphragm?
- a. Botulinum toxin
 - b. Neostigmine
 - c. Methacholine

- d. Curare
 - e. Organophosphate
125. Which of the following is not true regarding electromyography (EMG)?
- a. Test electrical activity of nerves
 - b. Record action potential of the innervated muscle
 - c. Is an indicator of muscle tension or performance
 - d. If the weight is held at arm's length, signal remains the same
 - e. Signal is sustained longer if all motor units are recruited
126. When a muscle is at rest, the main source of energy is from:
- a. Serum fatty acid
 - b. Blood glucose
 - c. Muscle glycogen
 - d. Liver glycogen
 - e. Blood free amino acids
127. Muscle glycogen synthesis in resting muscle is increased by action of which of the following hormones?
- a. Epinephrine
 - b. Norepinephrine
 - c. Glucagon
 - d. Insulin
 - e. Fasting and starvation
128. A malignant mesenchymal bone tumor in which cancerous cells produce bone matrix
- a. Ewing sarcoma
 - b. Chondrosarcoma
 - c. Osteosarcoma
 - d. Primitive neuroectodermal tumor
 - e. Leiomyosarcoma
129. A Brodie's abscess stands for abscess collection in:
- a. Joint space
 - b. Anterior cruciate ligament
 - c. Bone
 - d. Skeletal muscle
 - e. Articular cartilage
130. Which of the following is not true regarding tuberculous arthritis?
- a. It affects a single joint in most cases
 - b. It may be a manifestation of progressive primary tuberculosis
 - c. It is always accompanied by active pulmonary tuberculosis
 - d. It may be complicated by fibrous ankylosis
 - e. The knee joints are one of the usual sites of involvement
131. Concerning Mycobacterium species, which of the following is correct?
- a. Acid fast stain is used to distinguish M. leprae from M. tuberculosis
 - b. M. leprae is highly sensitive to ultraviolet light
 - c. The skin in lepromatous leprosy is infiltrated with lymphocytes
 - d. Incubation period of leprosy is measured in weeks

- e. *M. leprae* is genetically transmitted
132. Which of the following is not a superficial mycosis?
- a. **Sporotrichosis**
 - b. Tinea nigra
 - c. White piedra
 - d. Tinea capitis
 - e. Pityriasis versicolor
133. The effect of competitive neuromuscular blockers e.g. pancuronium may be antagonized by:
- a. Pyridostigmine
 - b. Verapamil
 - c. Gentamicin
 - d. Halothane
 - e. **Sugammadex**
134. A patient presented with sharp chest pain that worsens when the patient takes deep breath. Irritation of which structure would most likely produce this type of pain?
- a. Visceral pericardium
 - b. Visceral peritoneum
 - c. **Parietal pleura**
 - d. Visceral pleura
 - e. Parietal pericardium
135. During physical examination, murmur is heard at the left 5th intercostal space, lateral to mid clavicular line. Which valve is affected?
- a. Tricuspid valve
 - b. **Mitral valve**
 - c. Pulmonary valve
 - d. Aortic valve
 - e. Semilunar valves
136. The chest x-ray of 30-year-old patient showed right sided pleural effusion at which intercostal space should the needle be inserted to safely remove excess fluid at mid-axillary line?
- a. 2nd
 - b. 4th
 - c. 6th
 - d. **9th**
 - e. 12th
137. A blood vessel has tunica media showing 40 layers of smooth muscle cells and tunica intima with well-developed internal elastic lamina. What type of blood vessel is this?
- a. Large artery
 - b. **Medium artery**
 - c. Small artery
 - d. Large vein
 - e. Medium vein
138. Select the correct statement pertaining to the morphology of the kidneys?
- a. Macula densa cells are located in the urinary pole of the nephrons
 - b. The distal convoluted tubules exhibit brush borders and apical canaliculi

- c. The renal pyramid and its associated cortex is referred to as the renal lobe
 - d. Interstitial cells in the renal medulla secrete erythropoietin
 - e. Renin is secreted by the podocytes from the parietal layer of Bowman's capsule
139. A person has been hypertensive for 10 years, with resting BP of 180/120. Compared to normotensive person with blood pressure of 120/80, if the blood pressure increased 10mmHg above the mean of both patients, the hypertensive patient would demonstrate:
- a. Increased parasympathetic outflow
 - b. Increased sympathetic outflow
 - c. Decreased parasympathetic outflow
 - d. Decreased sympathetic outflow
 - e. No difference in autonomic outflow
140. The physiological function of the relatively slow conduction through the AV node is to allow sufficient time for:
- a. Repolarization of the ventricles
 - b. Contraction of the ventricles
 - c. Venous return to the atria
 - d. Filling of the ventricles
 - e. Run-off of blood from the aorta to the arterioles
141. Increased pressure within the carotid sinus causes all of the following **except**:
- a. Atrial tachycardia
 - b. A decrease in aortic pressure
 - c. Reflex bradycardia
 - d. Vasodilation of arterioles
 - e. A decrease in sympathetic tone to arterioles
142. Which of the following does not fit the clinical picture of heart failure from aortic atherosclerosis?
- a. Ventricular hypertrophy
 - b. Decreased pulse pressure
 - c. Decreased stroke volume
 - d. Pulmonary edema
 - e. Congestive heart failure
143. All of the following help maintain circulation during state of hypovolemic shock **except**:
- a. An increased heart rate
 - b. Transcapillary filtration of plasma into the interstitial space
 - c. Rapid respiratory effort to promote venous return
 - d. Vasoconstrictive contribution from increases in circulating epinephrine
 - e. Autotransfusion of interstitial fluid into capillaries
144. Which one of the following is a reason for infant respiratory distress syndrome?
- a. High atmospheric pressure
 - b. Absence of autonomic nerve supply
 - c. Absence of surfactant
 - d. Pneumothorax
 - e. Increased airway resistance
145. Which of the following statements is **not true**?
- a. Prostacyclins are effective vasodilators

- b. Thromboxanes are mainly vasoconstrictors
 - c. Leukotrienes are involved in asthmatic and allergic reactions
 - d. Aspirin acts as irreversible inhibitor of cyclooxygenase
 - e. Glutathione peroxidase is a selenium containing metalloenzyme
- ✓ The above question has a mistake in it. All statements are true.
146. Which amino acid is used for the biosynthesis of nitric oxide?
- a. Glutamate
 - b. Aspartate
 - c. Arginine
 - d. Serine
 - e. Threonine
147. The critical initial stage in the formation of atherosclerotic plaque is:
- a. An increased LDL concentration and its oxidation by reactive oxygen species
 - b. Formation of foam cells
 - c. Formation of fatty streak
 - d. Migration of pro-inflammatory agents
 - e. Production of adhesion molecules
148. Which of the following is not a feature of Mycobacterium avium intracellulare infections occurring in the immunodeficient host?
- a. Organisms multiply freely within macrophages
 - b. The bacteria disseminate widely in the body
 - c. The lesions consist of foamy macrophages that contain large number of mycobacteria
 - d. Infections occur in the absence of effective cell mediated immunity
 - e. The lesions consist of well-defined granulomas with caseous necrosis
149. Mark the condition which has the least risk to be the cause of hypertension
- a. Dilative cardiomyopathy
 - b. Pheochromocytoma
 - c. Coarctation of the aorta
 - d. Chronic glomerulosclerosis
 - e. Adenoma of the adrenal cortex (Cushing Syndrome)
150. Mark the typical histologic features of the involved mitral valve rheumatic acute verrucous endocarditis.
- a. Fibrinopurulent exudates with streptococci
 - b. Sterile granulomatous inflammation
 - c. Pure thrombus covering the endothelial defect
 - d. Pure fibrous thickening of the leaflet
 - e. Ulcer/perforation of the leaflet accompanied by staphylococci

Mock 07 Batch

1. Injury or ligation of which of the following arteries will result in distal gangrene of the injured upper limb?
 - a. Brachial
 - b. Radial
 - c. Ulnar
 - d. Palmar

- e. Scapular
- 2. A 60-year-old woman diagnosed with cancer in supero-lateral quadrant of the breast developed lymphatic metastasis. Which of the following group of axillary lymph nodes is most likely to be enlarged?
 - a. Posterior
 - b. Apical
 - c. Central
 - d. Lateral
 - e. Anterior
- ✓ Most lymph goes to anterior
- 3. Which of the following cells are responsible for synthesis of ground substance of connective tissue?
 - a. Macrophages
 - b. Mast cells
 - c. Fibroblasts
 - d. Langerhans cells
 - e. Histocytes
- 4. Which of the following statements is correct regarding nervous tissue?
 - a. Astrocytes are phagocytic cells in the central nervous system
 - b. Pseudounipolar neurons appear angular under microscope
 - c. Microfilaments and microtubules are found in Nissl bodies
 - d. One Schwann cell makes myelin sheath for many axons
 - e. Oligodendrocytes are found in both gray and white matters
- 5. When monozygotic twins are formed by split at morula stage, the twins will have:
 - a. Same chorionic sac and placenta, but separate amniotic cavity
 - b. Same chorionic sac, but separate placenta and amniotic cavity
 - c. Same chorionic sac, placenta and amniotic cavity
 - d. Separate chorionic sac, placenta and amniotic cavity
 - e. Separate chorionic sac and placenta, but same amniotic cavity
- 6. During which period of development does most birth defects occur?
 - a. First 2 weeks
 - b. 3-8th weeks
 - c. 9-16th weeks
 - d. 17-28th weeks
 - e. 30-36th weeks
- ✓ Teratogens are most likely going to kill the fetus if exposed during first 2 weeks than producing birth defects – so the correct answer is B.
- 7. All of the following are examples of positive feedback mechanisms except:
 - a. progress of labor
 - b. Release of luteinizing hormone
 - c. Reversible shock
 - d. Nerve impulse
 - e. Duplication of viruses

- ✓ Reversible shock is an example of negative feedback...rather progressive shock is an example of negative feedback.
8. Regarding synaptic transmission in the autonomic nervous system:
- The acetylcholine receptors in both parasympathetic and sympathetic ganglia are predominantly muscarinic.
 - Norepinephrine secreted by sympathetic postganglionic fibers acts preferentially on nicotinic receptors
 - The adrenal medulla secretes epinephrine in response to activation of muscarinic sympathetic postganglionic fibers in splanchnic nerves
 - Acetylcholine secreted by parasympathetic postganglionic fibers acts on muscarinic receptors.
 - Beta adrenergic stimulation on bronchial muscle aggravates asthmatic attack
9. In which of the following is the combined effect of renin, angiotensin, and aldosterone not required?
- Arterial blood pressure
 - Blood volume
 - Fluid osmolality
 - H⁺ and K⁺ secretion
 - Ca⁺⁺ serum level
10. Which is **not** an amphipathic molecule?
- Phosphatidylcholine
 - Cholesterol
 - Triacylglycerol
 - Phosphatidylserine
 - Phosphatidic acid
11. What is the role of topoisomerases in eukaryotic gene transcription?
- Topoisomerase enzyme cut, uncoil, and reseal the double stranded DNA
 - Topoisomerase enzymes bind to the origin of replication sites within double stranded DNA
 - Topoisomerase enzymes open up the double stranded DNA at the replication fork
 - Topoisomerase enzymes join the Okazaki fragments together with phosphodiesterase bonds
 - Topoisomerase enzymes open up the double stranded DNA at the termination point
- ✓ Through the above MOA described in A – they help in removing supercoils in the helix.
12. A newborn infant had trouble breathing. The infant was 3 months premature. The physicians treated the infant with a solution, which was directly injected to the lungs. Within seconds, the infant responded with much improved breathing. Which of the following is a major component of this preparation?
- Dipalmitoyl phosphatidylcholine
 - Palmitate containing ceramide
 - Sphingosine
 - Sphingomyelin
 - Diacylglycerol

13. Why are statins **not effective** in lowering cholesterol levels in individuals with homozygous familial hypercholesterolemia?
- a. HMG-CoA reductase is resistant to statins
 - b. Statins cannot enter the liver cells
 - c. **LDL receptors are nonfunctional**
 - d. Reverse cholesterol transport is inoperative in these patient
 - e. LCAT is resistant to statin action
- ✓ Remember LDL receptors are defective in **familial hypercholesterolemia**.
14. Which of the following **best** describes chemiosmotic theory of oxidative phosphorylation?
- a. Inner mitochondrial membrane is permeable to protons and hydroxyl ions
 - b. Protons are pumped into the mitochondrial matrix to outer membrane
 - c. **Protons are pumped from mitochondrial matrix to inner membrane**
 - d. Free energy formation is blocked as proton pass down the concentration gradient through the ATP synthase
 - e. ATP is synthesized in mitochondrial intermembrane space
15. Which of the following is **not true** of Xeroderma pigmentosum?
- a. Is a genetic disease in which cells are unable to repair DNA damage
 - b. when cells fail to repair DNA damage caused by ultraviolet light
 - c. **Patients die in their 40s from premature aging**
 - d. Encounter increased risk of skin cancers and blindness
 - e. Involve in a defect of nucleotide excision repair
16. Which enzyme removes an abnormal base from a mutated DNA molecule by breaking the bond between the abnormal base and deoxyribose sugar during base-excision repair?
- a. DNA exonuclease
 - b. DNA helicase
 - c. DNA topoisomerase
 - d. **DNA glycosylase**
 - e. DNA polymerase
17. Which of the following is **true** about allopurinol?
- a. Should be given immediately for an attack of acute gouty arthritis
 - b. Prevents gout by inhibiting phosphoribosyl pyrophosphate (PRPP) synthetase
 - c. Is an intermediate in purine catabolism
 - d. **Is converted to oxopurinol, which, in addition to allopurinol itself, inhibits xanthine oxidase**
 - e. Works by disrupting macrophage microtubules, so reducing inflammation
18. Myocardial infarction results in elevation of all of the following serum enzymes **except**:
- a. **Alkaline phosphatase**
 - b. Lactate dehydrogenase
 - c. Aspartate transaminase
 - d. Creatine kinase
 - e. Alanine transaminase
19. Free radical mediated cellular damage is prevented by:
- a. **Calorie restriction**
 - b. Over exposure to sunlight

- c. Consume foods containing trans fatty acids
 - d. Consumption of less antioxidant
 - e. Reactive oxygen species
20. Hypoxic death of cells within the central nervous system often evokes:
- a. Gangrenous necrosis
 - b. Liquefactive necrosis
 - c. Coagulative necrosis
 - d. Fat necrosis
 - e. Caseous necrosis
21. Which of the following statements is **not correct** about endotoxin?
- a. Stimulates leukocytes to release endogenous pyrogens
 - b. Causes leukopenia
 - c. Activates alternate complement pathway
 - d. Promotes disseminated intravascular coagulation
 - e. Decreases vascular permeability
22. Which of the following sterilizing methods is appropriate for critical medical devices?
- a. Autoclaving
 - b. Boiling
 - c. Steaming
 - d. Chemicals
 - e. Filtration
23. A patient with acute severe pain needs a drug with:
- a. slow onset of action
 - b. Short duration of action
 - c. short plasma half life
 - d. fast onset of action
 - e. long duration of action
24. Which of the following is **true**?
- a. A competitive antagonist has no intrinsic activity
 - b. A partial agonist has less affinity to receptor than full agonist
 - c. Partial antagonist cannot reverse the effect of full agonist
 - d. Antagonist has no affinity to receptors
 - e. Antagonist and partial agonist have similar intrinsic activity
25. Which of the following statements is **not true** with regard to early development of the blood and immune system?
- a. Leukocytes appear before the appearance of erythrocytes
 - b. The normal histological architecture of the thymus may be affected by the migration of neural crest cells
 - c. If you see a lobulated spleen while dissecting a cadaver in the DR it is nothing unusual, for it reflects its development
 - d. The replacement of thymic tissue by fat tissue with age indicates highly reduced arrival of T lymphocytes during early period of life, but no abnormality nor degeneration
 - e. Liver is the major source of erythrocytes during early development before any other tissues in the embryo

- ✓ Yolk sac precedes liver. Remember the sequence by the mnemonic = Young Liver Synthesizes Blood (Yolk sac → Liver → Spleen → Bone marrow)
26. An individual produces normal erythropoietin levels in response to hypoxia, but he still suffers from anemia. Which statement does not satisfy the above condition?
- a. Deficiency of iron intake in his diet
 - b. Increased folic acid intake in his diet
 - c. Deficiency of Vit. B₁₂ intake in his diet
 - d. Decreased transferrin production in the liver
 - e. Decreased intrinsic factor production in the stomach
27. The formation of biliverdin from heme breakdown releases which of the following?
- a. Carbon monoxide
 - b. Nitric oxide
 - c. Oxygen
 - d. Carbon dioxide
 - e. Hydrogen peroxide
- ✓ Remember heme break down releases CO₂
28. Hypoxic death of cells within the central nervous system often evokes:
- a. Gangrenous necrosis
 - b. Liquefactive necrosis
 - c. Coagulative necrosis
 - d. Fat necrosis
 - e. Caseous necrosis
29. One of the following is not a clinical feature of extravascular hemolysis?
- a. Anemia
 - b. Splenomegaly
 - c. Jaundice
 - d. Hemoglobinemia
 - e. None of the above
30. A 28-year-old pregnant woman who is otherwise healthy developed deep venous thrombosis two days ago. Which of the following risk factors is relevant in this case?
- a. Excess estrogen levels
 - b. Prolonged bed rest
 - c. Excess progesterone
 - d. Thrombophlebitis
 - e. Obesity
31. Which of the following is not secondary lymphoid tissue?
- a. Tonsils
 - b. Bone marrow
 - c. Spleen
 - d. Lymph nodes
 - e. Mucosa associated lymphoid tissue
32. The drug of choice to treat deep vein thrombosis in a 34-year-old pregnant woman is:
- a. Aspirin
 - b. Clopidogrel

- c. Enoxaparin
 - d. Lepirudin
 - e. Warfarin
33. Which of the following **best** defines an oncogene?
- a. An oncogene codes for a cell cycle control protein
 - b. An oncogene codes for a mutated form of a protein that forms part of a signal transduction pathway
 - c. An oncogene codes for a protein that prevents the cell from undergoing apoptosis
 - d. An oncogene is a dominantly expressed mutated gene that gives a cell a growth or survival advantage
 - e. An oncogene expresses a protein that regulates receptors
34. A 34-year-old woman was presented for a lump in her breast that she palpated on self-breast examination. Her clinical history reveals that her mother and her aunt both had breast and ovarian cancer. Given this presentation, you suspect the patient may have a mutation in which of the following genes involved in DNA repair?
- a. BRCA-1
 - b. Ras
 - c. Bcl-2
 - d. p53
 - e. Rb
35. When grading a malignant tumor, a pathologist considers:
- a. The degree of differentiation of the tumor cells
 - b. Tumor stage
 - c. Lymph node involvement
 - d. Vascular involvement
 - e. The extent or spread of the tumor
36. Which of the following viral family is not oncogenic?
- a. Retroviridae
 - b. Herpesviridae
 - c. Hepadnaviridae
 - d. Papillomaviridae
 - e. Flaviviridae
- ✓ The question has an error -
37. Which of the following anticancer drugs has a potential to cause dilated cardiomyopathy?
- a. Doxorubicin
 - b. Cisplatin
 - c. Hydroxyurea
 - d. Fludarabine
 - e. Vincristine
38. A soldier is injured by a bullet and a major artery is lacerated in the posterior leg and the sole of his foot is cold and pale. The dorsum of the foot is warm and colored normal. Which of the following arteries could possibly be injured?
- a. Posterior tibial artery
 - b. Anterior tibial artery

- c. Fibular artery
 - d. Popliteal artery
 - e. Plantar artery
39. A quick way to check the function of one of the nerves of the leg is to pinch the skin between the first and second toes. Which nerve are you checking?
- a. Superficial peroneal nerve
 - b. Deep peroneal nerve
 - c. Lateral plantar nerve
 - d. Sural nerve
 - e. Saphenous nerve
40. Which one of the following structures may not be affected during defective development of the 1st arch?
- a. Upper jaw
 - b. Lower jaw
 - c. Masseter muscle
 - d. Internal carotid artery
 - e. Mandibular nerve
41. Which of the following is associated with myasthenia gravis?
- a. Antibody production against calcium channel in the presynaptic membrane
 - b. Continuous contraction of muscle
 - c. Destruction of myelin sheath in the central nervous system
 - d. An autoimmune disease characterized by muscle rigidity
 - e. Destruction of nicotinic acetylcholinesterase receptors at post-junctional membrane
42. Among the following which one best improves performance of an athlete competing in short distance event (e.g. 100 meters sprinting)
- a. Exogenous erythropoietin
 - b. Glycerol solution
 - c. Creatinin supplementation
 - d. High altitude training
 - e. Analogous blood transfusion
43. The degradation of muscle glycogen primarily produces which of the following metabolites?
- a. More glucose than glucose-6-P
 - b. More glucose-6-P than glucose
 - c. Equal amounts of glucose-6-P and glucose
 - d. Neither glucose nor glucose-6-P
 - e. Glucose-6-P
44. A common joint disease characterized by primary abnormalities in articular cartilage is
- a. Osteoarthritis
 - b. Rheumatoid arthritis
 - c. Gouty arthritis
 - d. Infectious arthritis
 - e. Rheumatic fever
45. A 53-year-old woman presents with weakness of both extremities and skin lesions on the face. Mark the probable underlying condition:

- a. Brain infarction (Left motor region)
 - b. Myotonic dystrophy
 - c. BECKER type muscular dystrophy
 - d. DUCHENE muscular dystrophy
 - e. Dermatomyositis
46. The commonest etiologic agent for acute osteomyelitis is
- a. Pseudomonas aeruginosa
 - b. Borrelia Burgdorferi
 - c. Staphylococcus epidermidis
 - d. Mycobacterium tuberculosis
 - e. Staphylococcus aureus
47. The drug of choice for the treatment of early benign prostatic hyperplasia?
- a. Terazosin
 - b. Phenoxybenzamine
 - c. Phentolamine
 - d. Xylometazoline
 - e. Atenolol
48. A cardiac patient is experiencing anterior left ventricular myocardial weakness with poor conduction through her ventricles. You assume that the coronary artery supplies the Bundle of His and the anterior wall of the left ventricle is occluded. The occluded artery is most likely the:
- a. Right marginal
 - b. Circumflex
 - c. Right coronary
 - d. AV nodal artery
 - e. Anterior interventricular
49. A section of an airway with less than 1mm width diameter has no cartilage in its wall and is lined by simple ciliated cuboidal epithelium throughout its luminal surface. Which part of respiratory tract is it?
- a. Segmental bronchus
 - b. Primary bronchiole
 - c. Terminal bronchiole
 - d. Respiratory bronchiole
 - e. Alveolar duct
50. Which of the following **is not** associated with anterior displacement of the trans-coronal septum leading to Tetralogy of Fallot?
- a. Pulmonary stenosis
 - b. Interatrial septal defect
 - c. Interventricular septal defect
 - d. Right ventricular hypertrophy
 - e. Overriding aorta
- ✓ Use the mnemonics PROVe (=Pulmonary stenosis, RRight Ventricular Hypertrophy, OOverriding of aorta, and VVentricular septal defect)
51. Cardiac glycosides, such as digitalis, enhances the contractile performance of cardiac muscle fibers by:

- a. Stimulating the intracellular production of cAMP
 - b. Decreasing the amount of Ca^{2+} available in myofibrils
 - c. Stimulating cardiac beta-adrenergic receptors
 - d. Blocking cardiac beta-adrenergic receptor
 - e. Inhibiting the $\text{Na}^+\text{-K}^+$ ATPase of cell membranes
52. The second heart sound **is not** associated in time with which of the following?
- a. Aortic incisura
 - b. Onset of isovolumetric relaxation
 - c. End of ventricular ejection
 - d. Onset of atrial contraction
 - e. Closed state of all valves
53. Regarding respiratory acidosis, which one is **correct**?
- a. Is primarily caused by elevated plasma level of bicarbonate
 - b. Is associated with increased concentration of lactic acid
 - c. Occurs as a result of hyperventilation
 - d. Occurs in conditions that impair gas exchange at the respiratory membrane
 - e. Is compensated by excretion of bicarbonate in the urine
54. Which of the following are **not** correctly matched?
- a. Exercise at altitude: hyperventilation mediated by peripheral chemoreceptors
 - b. Arterial hypoxemia at altitude: more alkaline arterial blood
 - c. Pulmonary hypertension: left ventricular hypertrophy
 - d. 2,3 – phosphoglycerate: maladaptive at very high altitude
 - e. Acute mountain sickness: treated by carbonic anhydrase inhibitors
55. The reason that nitroglycerin is prescribed to reduce pain of angina pectoris is, that it:
- a. Directly causes the relaxation of coronary artery muscle
 - b. Is metabolized to nitrate that causes relaxation of coronary artery muscle
 - c. Is metabolized to nitrite that causes relaxation of coronary heart muscle
 - d. Is metabolized to nitric oxide that causes the relaxation of coronary artery muscle
 - e. Is metabolized to peroxynitrite that causes the relaxation of coronary artery muscle
56. All of the following are used in the diagnosis of myocardial infarction **except**:
- a. Troponin I
 - b. Myoglobin
 - c. Hemoglobin
 - d. Aspartate transaminase
 - e. Lactate dehydrogenase
57. Primary carcinoma of the lung arising in small bronchioles or alveoli probably from the Clara cells or type II pneumocytes
- a. Small cell undifferentiated carcinoma
 - b. Large cell undifferentiated carcinoma
 - c. Carcinoid tumor
 - d. Squamous cell carcinoma
 - e. Bronchioalveolar carcinoma
58. Thrombosis; Mark the most common site of thrombosis in a bed ridden patient:
- a. Basilar artery (Circle of Willis)

- b. Deep calve vein
 - c. Thrombus in the left atrium (Auricle)
 - d. Abdominal aorta
 - e. Pulmonary artery
59. A 5-year-old boy presented with maculopapular rash which appeared first on his face then later spread to the trunk and limbs. Further examination revealed Koplik's spot. The most likely diagnosis is:
- a. Varicella Zoster virus
 - b. Measles virus
 - c. Rubella virus
 - d. Erythrovirus-B19
 - e. Mumps virus
60. The best way to prevent the occurrence of rheumatic heart disease is:
- a. Treatment of rheumatic fever
 - b. Treatment of acute throat infections by group A streptococcus
 - c. Treatment of throat infections by Staphylococcus aureus
 - d. Surgical treatment of the damaged heart valves
 - e. Immunotherapy
61. Which of the following drugs is preferred for the treatment of acute attack of angina pectoris?
- a. Atenolol
 - b. Verapamil
 - c. Propranolol
 - d. Diltiazem
 - e. Nitroglycerin
62. The drug of choice in the treatment of ascites secondary to liver cirrhosis is:
- a. Hydrochlorothiazide
 - b. Amiloride
 - c. Acetazolamide
 - d. Spironolactone
 - e. Furosemide
63. A photomicrograph of the GI tract shows all several mucosal projections made by a core of lamina propria covered by simple columnar epithelium, and well-formed simple tubular glands in the submucosa. What part of the GI tract could this be?
- a. Fundus
 - b. Pylorus
 - c. Duodenum
 - d. Ileum
 - e. Large intestine
64. Concerning the development of the GI tract and associated structures, which statement is true?
- a. Stomach rotates 270° counterclockwise, similar to midgut
 - b. The midgut gives rise to both ascending and descending colons
 - c. The hepatocytes of liver are of endodermal origin
 - d. Meckel's diverticulum is a congenital remnant of jejunum
 - e. The exocrine portion of pancreas develops from endoderm

65. A 61-year-old obese woman who is a mother of five notes intermittent abdominal pain, primarily in the right upper quadrant. The pain is most commonly experienced by eating a fatty meal. The hormone most likely associated with the increase in her symptoms is:
- a. Gastrin
 - b. Cholecystokinin
 - c. Insulin
 - d. Secretin
 - e. Vasoactive intestinal peptide
66. Which disease is caused by complete absence of Hypoxanthine guanine phosphoribosyl transferase?
- a. Primary gout
 - b. Calcium phosphate
 - c. Immunodeficiency
 - d. Lesch-Nyhan syndrome
 - e. Dwarfism
67. A young man with severe burn injury developed abdominal pain, nausea and vomiting. Endoscopic biopsy of gastric mucosa shows mucosal erosion with edema and scattered neutrophils. What is the diagnosis?
- a. Zollinger Ellison syndrome
 - b. Chronic gastritis
 - c. Peptic ulcer
 - d. Acute gastritis
 - e. Achalasia
68. In a chronic carrier of hepatitis B virus infection, which positive test is most indicative of high infectivity?
- a. Hepatitis B Surface Antigen (HBsAg)
 - b. Hepatitis B Core Antigen (HBcAg)
 - c. Hepatitis B e Antigen (HBeAg)
 - d. Anti-HBsAg antibody
 - e. Anti-HBeAg antibody
69. Which of the E. coli strains produce shiga like toxin?
- a. Enterotoxigenic E. coli
 - b. Enterohemorrhagic E. coli
 - c. Enteroinvasive E. coli
 - d. Enteropathogenic E. coli
 - e. Enteroaggregative E. coli
70. Which of the following antiemetic drugs has additional appetite stimulant activity?
- a. 5-HT₃ receptor antagonists
 - b. Anticholinergics
 - c. Antihistamines
 - d. Dopamine antagonists
 - e. Cannabinoids
71. Concerning the development of the urinary system, which statement is true?
- a. Urine production begins during the 3rd trimester

- b. Accessory renal arteries are anatomical end arteries
 - c. In adults in both sexes, the urachus is represented by the ligamentum teres hepatis
 - d. The urinary bladder becomes a pelvic organ during the 9th week of development
 - e. The kidneys attain their definitive adult position toward the end of the fetal period
72. Which of the following is correct regarding renal blood flow?
- a. Reduced when perfusion pressure increases from 100-to-150 mmHg
 - b. Greater in the medulla than the cortex part of the kidney
 - c. Reduced during emotional stress and fear
 - d. Greater in the glomerular capillaries than peritubular capillaries
 - e. Increased when afferent arteriolar resistance increases
73. Acid base regulatory function of the kidney involves the following except:
- a. Secretion of H⁺ in the proximal tubules
 - b. Reabsorption of HCO₃⁻ in the proximal tubules
 - c. Reabsorption of H⁺ in the distal tubules
 - d. Bulk secretion of H⁺ in the distal tubules
 - e. Reabsorption of some of HCO₃⁻ in the distal tubule
74. Which of the following is not a feature of diabetic nephropathy?
- a. Kimmelstiel-Wilson lesion
 - b. Diffuse proliferative glomerulonephritis
 - c. Glomerular basement membrane thickening
 - d. Hyaline arteriosclerosis
 - e. Papillary necrosis
75. A heavy metal that interferes with calcium-dependent release of neurotransmitters:
- a. Arsenic
 - b. Cadmium
 - c. Mercury
 - d. Lead
 - e. Copper
76. Hemoperfusion:
- a. requires semi-permeable membrane for removal of toxins
 - b. is more effective than dialysis for correcting acid-base & electrolyte abnormalities
 - c. is limited by plasma protein binding capacity
 - d. is preferred for removal of toxins with high volume of distribution
 - e. requires anticoagulant therapy
77. A disease characterized by high infectivity but low symptomatology:
- a. Amoebiasis
 - b. Toxoplasmosis
 - c. Hydatid disease
 - d. Giardiasis
 - e. Intestinal obstruction
78. Which of the following sentence is false about Brucella species?
- a. It causes undulant fever
 - b. It is gram-negative coccobacilli in morphology
 - c. The bacteria is acquired by consuming contaminated milk products

- d. It is obligate intracellular bacteria
 - e. There is an effective human vaccine
79. If a person gets malarial infection through blood transfusion, which of the following stages will be absent?
- a. Trophozoite
 - b. Hypnozoite
 - c. Schizont
 - d. Gametocyte
 - e. Merozoite
80. Which of Schistosoma species has well-formed lateral spine?
- a. Schistosoma japonicum
 - b. Schistosoma mansoni
 - c. Schistosoma hematobium
 - d. Schistosoma makongi
 - e. Schistosoma intercalatum
81. Antimalarial agent effective in prophylaxis and radical cure of malaria caused by Plasmodium vivax is:
- a. Chloroquine
 - b. Quinine
 - c. Primaquine
 - d. Mefloquine
 - e. Artesunate
82. An anti TB drug which is least considered for treatment of TB during early childhood is:
- a. Rifampicin
 - b. Isoniazid
 - c. Ethambutol
 - d. Pyrazinamide
 - e. Streptomycin
83. A 60-year-old woman has been diagnosed with carcinoma of the cervix. Investigations revealed that the carcinoma has spread to the surrounding lymph nodes. Which lymph nodes are the first to be involved?
- a. External iliac
 - b. Internal iliac
 - c. Common iliac
 - d. Superficial inguinal
 - e. Deep inguinal
84. Which of the following is absent from placental barrier during the 2nd half of pregnancy?
- a. Syncytiotrophoblast
 - b. Cytotrophoblast
 - c. Basal lamina of trophoblast
 - d. Fenestrated capillary of endothelium
 - e. Capillaries of the chorionic villi
85. With regard to the development of the endocrine system, which of the following statements is false?

- a. The pituitary gland is purely ectodermal in origin
 - b. The thyroid gland is the first endocrine gland to develop in the embryo
 - c. The pineal gland develops as a median diverticulum of the caudal part of the roof of diencephalon
 - d. C cells of the thyroid gland differentiate from neural crest cells that migrate from pharyngeal arches into the 4th pair of pharyngeal pouches
 - e. The oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal alcohol metabolism
86. Which of the following **is not** correctly associated?
- a. Adipose tissue—leptin to regulate food intake and energy metabolism
 - b. Thymus: use of iodine to make thyroxine for regulation of body growth
 - c. Pineal gland: production of melatonin and involvement in seasonal affective disorder
 - d. Pancreas: glucagon for regulation of blood glucose levels
 - e. Posterior pituitary gland: oxytocin for regulation of milk ejection
87. When puberty commences, which of the following hormonal profiles of a boy changes?
- a. Onset of nocturnal peaks of testosterone
 - b. Establishment of 90 min pulses luteinizing hormone during sleep hours
 - c. Pulsatile secretion of thyroid stimulating hormone begins
 - d. Gonadotrophin releasing hormone level falls
 - e. Levels of leptin and adrenal androgens will remain at the basal level
88. The normal testicular function depends on:
- a. Pulsatile secretion of GnRH, LH, and FSH
 - b. Stimulation of Sertoli cells by PRL
 - c. Regulation of LH secretion by inhibin
 - d. The aromatization of testosterone to DHT
 - e. The interaction of lactotrophs and somatotrophs
89. Concerning phospholipase C, which of the following is correct?
- a. It releases calcium from endoplasmic reticulum
 - b. It hydrolyzes phosphatidyl inositol 4,5 bisphosphates to diacylglycerol and IP₃
 - c. It is a membrane phospholipid
 - d. It is short lived and rapidly degraded
 - e. Phospholipase C is present freely in cytosol
90. Majority of squamous cell carcinomas and pre-invasive insitu lesions in the cervix arise from:
- a. Squamocolumnar junction
 - b. Transformation zone
 - c. Ectocervix
 - d. Endocervix
 - e. Endocervical polyp
91. Which of the following thyroid carcinomas is derived from parafollicular cells?
- a. Undifferentiated carcinoma
 - b. Poorly differentiated carcinoma
 - c. Medullary carcinoma
 - d. Papillary carcinoma
 - e. Follicular carcinoma

92. Which of the following diagnostic tests would be most useful to predict progression of HIV/AIDS?
- a. HIV p7 antigen
 - b. HIV antibody test
 - c. HIV RT PCR
 - d. Virus isolation
 - e. HIV p24 antigen
93. A combination of metformin and ethanol increases the risk of:
- a. A disulfiram-like reaction
 - b. Excessive weight gain
 - c. Hypoglycemia
 - d. Lactic acidosis
 - e. Serious hepatotoxicity
94. In a patient with deviation of the tongue to the left side when protruded, ipsilateral loss of pain and thermal sensations from the face, and contralateral loss of pain & thermal sensation of the body, the lesion is possibly in the:
- a. Spinal cord
 - b. Medulla oblongata
 - c. Pons
 - d. Midbrain
 - e. Thalamus
95. Which of the following is **NOT correct** in the central nervous system?
- a. Both motor neurons and interneurons are Pseudounipolar
 - b. A ganglion is a collection of neuron cell bodies outside the central nervous system
 - c. The conglomerations of gray matter deep within the cerebrum and cerebellum are called nuclei
 - d. Basket cells are found in the cerebellum
 - e. The vast majority of neurons are interneurons
96. Regarding the development of ear and eye, which of the following statements is **not correct**?
- a. The tympanic membrane of the ear has its origin from all the three embryonic germ layers
 - b. The membranous labyrinth of the inner ear develops from surface ectoderm
 - c. Lens develops from mesodermal tissue located at the brim of the optic cup
 - d. Failure of proper fusion of the inner and outer layers of the optic cup leads to detached retina causing degeneration of retina because of lack of blood supply
 - e. Cornea is developed from both ectoderm and mesoderm
97. Which of the following is **true** about the hypothalamus?
- a. Supraoptic nucleus controls body temperature
 - b. Hypothalamus regulates the vegetative body functions
 - c. Hypothalamus is not involved in behavioral activities
 - d. Hypothalamus does not participate in sleep physiology
 - e. Posterior hypothalamic nucleus contains heat sensitive neurons
98. In case of brain disorders, such as bradykinesia, akinesia, resting tremor and rigidity of affected muscles followed by dementia are most likely manifested in:
- a. Parkinson's disease
 - b. Schizophrenia

- c. Athetosis
 - d. Alzheimer's disease
 - e. Atonia
99. Which of these sensory modalities is transmitted directly to the cerebral cortex without being relayed through the thalamus?
- a. Taste
 - b. Sight
 - c. Smell
 - d. Hearing
 - e. Touch
100. A coenzyme required for hydroxylation of dopamine into norepinephrine is
- a. Tetrahydrobiopterin
 - b. Pyridoxal pyrophosphate
 - c. Ascorbic acid
 - d. S-Adenosyl methionine
 - e. Folic acid

Exit 07 Batch (New)

1. Which lymphatic structure(s) is/are supposed to drain lymph directly from the tissue of dorsum of the foot?
 - a. Lymphoid tissue
 - b. Lymphatic capillaries
 - c. Thoracic duct
 - d. Right lymphatic duct
 - e. Lymph nodules
2. The inability of muscle tissue to return to original shape and length after contraction and stretching is loss of:
 - a. Extensibility
 - b. Excitability
 - c. Contractility
 - d. Elasticity
 - e. Conductivity
3. Mitotic inhibitor drugs such as vinblastine are used to arrest cell divisions in conditions of cancer chemotherapy. Which organelles are primarily targeted?
 - a. Rough endoplasmic reticulum
 - b. Golgi complex
 - c. Microfilament
 - d. Microtubules
 - e. Intermediate filaments
4. Which of the following nerves is commonly damaged during fracture of epicondyle of the humerus?
 - a. Median nerve
 - b. Ulnar nerve
 - c. Radial nerve
 - d. Musculocutaneous nerve

- e. Median cutaneous nerve of the forearm
5. A 28-year-old woman severely sprained her left ankle while playing tennis. When she tried to move the foot facing the sole medially, she experienced severe pain. What is the correct anatomic term for the movement that produced this pain?
- a. Pronation
 - b. Inversion
 - c. Flexion
 - d. Supination
 - e. Eversion
6. When B lymphocytes are activated, they differentiate into
- ✓ Plasma cells (the choices weren't visible)
7. Which of the following statements is **not correct** regarding implantation?
- a. Takes place at the stage of blastocyst
 - b. Occurs as a result of action of trophoblasts
 - c. Takes place after degeneration of zona pellucida
 - d. Extends deep into the level of myometrium of the uterus
 - e. Has normal site of implantation at the posterior wall of the upper part of the uterus
8. Which of the following congenital skin disorders is due to excessive keratinization of the skin?
- a. Congenital alopecia
 - b. Ichthyosis
 - c. Ectodermal dysplasia
 - d. Albinism
 - e. Athelia
9. Which of the following is correct regarding blood cells?
- a. Monocytes give rise to mast cells in the connective tissue
 - b. Neutrophils are the most common white blood cells in circulating blood
 - c. Mature erythrocytes contain peroxisomes as oxygen binding protein
 - d. Circulating monocytes increase in a patient with a parasitic infection
 - e. Basophils normally re-enter the bloodstream after leaving the circulation
- ✓ Use this Mnemonics to remember this= Never Let Monkeys Eat Banana (**N>L>M>E>B**) meaning Neutrophils>Lymphocytes>Monocytes>Eosinophils>Basophils
10. Which nerve is associated with loss of sensation on the lateral part of the forearm?
- a. Axillary
 - b. Radial
 - c. Posterior interosseous
 - d. Musculocutaneous
 - e. Median
11. Which nerve injury results in unilateral waddling gait?
- a. Superior gluteal nerve
 - b. Inferior gluteal nerve
 - c. Tibial division of sciatic nerve
 - d. Obturator nerve
 - e. Femoral nerve
12. All of the following structures are at risk of stab injury to the neck, **except**:

- a. Pleura and lung
 - b. Brachial plexus
 - c. Recurrent laryngeal nerve
 - d. –
 - e. –
- ✓ Only the above choices are visible on the exam...the answer is not among the above 3 choices so choose the one not among these.
13. Which of the following is **not** a branch of the first part of maxillary artery?
- a. Deep auricular
 - b. Anterior tympanic
 - c. Middle meningeal
 - d. Inferior alveolar
 - e. Descending palatine
- ✓ Descending palatine – is part of 3rd branch
14. Absence of myomesin in skeletal muscle will interrupt:
- a. The thick filaments in register at the M line
 - b. Stability between the neighboring myofibrils
 - c. Interaction between actin filament and laminin
 - d. Stability of actin filament
 - e. Attachment of thick filament to Z line
15. In the earliest stages of embryogenesis, blood cells **do not** arise from the
- a. Yolk sac
 - b. Liver
 - c. Spleen
 - d. Bone marrow
 - e. Heart
16. Which of the following is **not correct** regarding bronchopulmonary segments of the lung? They:
- a. Are surgically resectable
 - b. Have apices facing the hilum of the lung
 - c. Are made up of tertiary bronchi
 - d. Are more on the right than on the left
 - e. Have their own segmental pulmonary vein
17. Which of the following is **not correct** regarding the occlusion of arteries and corresponding area of myocardial infarction?
- a. Right coronary – Posterior wall of left ventricle
 - b. Left anterior descending – Anterior two thirds of interventricular septum
 - c. Right coronary artery – Posterior one third of interventricular septum
 - d. Left circumflex – Lateral wall of left ventricle
 - e. Left coronary – Wall of right atrium
18. Which of the following is **correct** regarding the respiratory system?
- a. There are seromucous glands and venous plexuses in the alveoli
 - b. The pleural cavity is lined by pseudostratified ciliated columnar epithelium
 - c. Pulmonary surfactant prevents alveolar collapse by increasing surface tension
 - d. The superior conchae are covered with stratified squamous epithelium

- e. Smooth muscles are found in respiratory bronchioles
19. With regard to the development of respiratory system, which statement **is not** correct?
- a. Alveolar dust cells develop from mesoderm
 - b. Surfactant production begins by 20 weeks
 - c. Sacs analogous to alveoli are present at 32 weeks
 - d. Epiglottis develops by proliferation of mesenchyme in the dorsal ends of the pharyngeal arches
 - e. The embryonic right main bronchus is slightly larger than the left one and is oriented more vertically
- ✓ It is in the **ventral end** not dorsal
20. The parietal pleural pain is referred to the shoulder region via phrenic nerve. This referred pain to the dermatome of C3-C5 originates from which part of pleura?
- a. Cervical
 - b. Costal
 - c. Mediastinal
 - d. Peripheral portion of the diaphragm
 - e. Suprapleural membrane
21. A 40-year-old patient with severe jaundice was diagnosed with pancreatic cancer. In which portion of the pancreas could the tumor be located?
- a. Head
 - b. Neck
 - c. Body
 - d. Tail
 - e. Uncinate process
22. Which of the following statements is not correct regarding the anomalies of the digestive system?
- a. Esophageal atresia could be caused by dorsal deviation of tracheal septum
 - b. Pyloric stenosis results primarily from hypertrophy of the circular muscularis externa
 - c. Rectourethral fistula is caused by abnormal partitioning of the cloaca
 - d. Incomplete rotation and failure of fixation of the midgut may result in volvulus
 - e. Meckel's diverticulum is usually found 2 meters proximal to the ileocecal valve
- ✓ It is **2 feet** from the ileocecal valve not 2 meters.
23. Which one of the following is **not correct** regarding renal pyramid?
- a. Visceral layer of Bowman's capsule is lined by simple squamous epithelium
 - b. Contains urinary pole on the side of the proximal convoluted tubule
 - c. Has urinary space between parietal and visceral layers
 - d. The visceral layer is responsible for formation of primary processes
 - e. Parietal layer of the capsule is involved in the formation of -----
- ✓ Parietal layer of capsule is not involved in formation. Its function is support.
24. Which of the following structures needs special attention during surgical removal of the uterus?
- a. Ovarian artery
 - b. Internal pudendal artery
 - c. Urethra
 - d. Ureter
 - e. Uterine artery

✓ Water under the bridge (Urine i.e. ureter is under uterine arteries)

25. The posterior relations of the stomach is formed by all of the following **except**:

- a. Pancreas
- b. Gall bladder
- c. Spleen
- d. Left kidney
- e. Left suprarenal

26. A 70-year-old man diagnosed with benign prostatic hyperplasia. Which lobe of the prostate gland is affected?

- a. Anterior lobe
- b. Posterior lobe
- c. Median lobe
- d. Right lateral lobe
- e. Left lateral lobe

27. Acidophilic cells making the most abundant cells of pars distalis of pituitary gland with an effect of stimulating growth of long bones are:

- a. Folliculostellate cells
- b. Somatotrophs
- c. Chromaffin cells
- d. Mammotrophs
- e. Corticotrophs

28. Which part of the female reproductive tract is derived from ectoderm and endoderm?

- a. Uterus
- b. Fallopian tube
- c. Cervix
- d. Vagina
- e. Urethra

29. The site of lesion in a patient with quadriplegia is:

- a. Lower cervical region of the spinal cord
- b. Upper cervical region of the spinal cord
- c. Upper thoracic region of the spinal cord
- d. Lower thoracic region of the spinal cord
- e. Lumbar region of the spinal cord

30. Which of the following structures is **less likely** affected by metastasis of tumor from the ischioanal fossa around the Alcock's canal?

- a. Obturator nerve
- b. Internal pudendal artery
- c. Internal pudendal vein
- d. Pudendal nerve
- e. Inferior rectal artery

31. A patient presents with an inability to identify a common object placed in right hand with his eyes closed (stereognosis). He also has difficulty determining the static position of the fingers of his right hand with his eyes closed (statoagnosis). Where is the lesion?

- a. Left inferior parietal lobule (areas 39, 40)

- b. Left cuneus (areas 18, 19)
 - c. Left supramarginal gyrus (area 40)
 - d. Left superior parietal lobule (areas 5, 7)
 - e. Right nucleus gracilis
32. All of the following signs and symptoms may be observed in a patient with medial medullary syndrome, **except**:
- a. Ipsilateral paralysis of muscles of the tongue
 - b. Contralateral discriminative touch deficits
 - c. Contralateral pain and temperature deficits
 - d. Contralateral paralysis
 - e. Ipsilateral proprioceptive deficits
33. Which of the following statements is true?
- a. Lysosomes produce catalase for antioxidant processes
 - b. Higher altitude increases hemoglobin affinity to O₂
 - c. The volume of fluid in the tissue is less than in the blood vessel
 - d. The lesser the plasma protein, the more the fluid in the tissue
 - e. More free calcium is found in the tissue than in the blood
34. Regulation by anticipating a change is:
- a. Positive feedback response
 - b. Negative feedback response
 - c. Feed-forward response
 - d. Amplification response
 - e. Opposing response
35. Which of the following is **true**?
- a. Hormones regulate blood pressure quicker than neuronal action
 - b. Neuronal action is more delayed than hormonal response
 - c. Cold exposure constricts surface blood vessels
 - d. Physiological changes are mainly predicted by cardiovascular system
 - e. Integrators take the first action before sensors
36. Age, sex, and clerking time are recorded in the patient's card. Select the **correct** concept why these variables are given importance.
- a. Aging increases the gain of homeostasis
 - b. Suprachiasmatic nucleus of the brain affects the values of the variables
 - c. Hemoglobin values of females are higher than that of males
 - d. Testosterone is less efficient than estrogen in erythropoiesis
 - e. Blood vessel's stiffness gets less with increasing age
37. What is allostasis?
- a. Insulin secretion for increased blood glucose level
 - b. Baroreceptor adaptation for long term blood pressure change
 - c. Increased breathing rate for CO₂ elevation
 - d. Parathyroid hormone secretion for low blood calcium level
 - e. Aggregation of platelets on the ruptured area of blood vessels
38. Which of the following describes the cardiac muscle?
- a. No summation due to long refractory period

- b. Striated and voluntary muscle
- c. Consists of branching chains of spindle shaped fiber
- d. Calcium comes only from sarcoplasmic reticulum
- e. Regulation is done by intrinsic (ANS) and extrinsic (hormonal) factors

✓ Both A and E are true

39. Motor units:

- a. Are found only in cardiac muscle
- b. Are largest in muscles responsible for delicate movements
- c. Consist of a muscle fiber and all the nerves that supply it
- d. Consist of a motor neuron and all the muscle fibers it supplies
- e. Are the same as neuromuscular junctions

✓ Note C and D can confuse – D is the one that is the correct answer.

40. During the contraction of smooth muscle, calcium ions bind with:

- a. Actin
- b. Calmodulin
- c. Myosin
- d. Troponin
- e. Tropomyosin

41. Which of the following is not caused by reduction in blood partial pressure of carbon dioxide?

- a. A decrease in H^+ concentration
- b. A decrease in bicarbonate concentration
- c. A rise in pH
- d. A decrease in the affinity of hemoglobin for oxygen
- e. Increase in P50 of oxyhemoglobin dissociation curve

42. Which substance is essential for maturation of erythrocytes?

- a. Amino acids
- b. Vitamin B12
- c. Lipids
- d. Erythropoietin
- e. Glucose

43. What is the approximate amount of oxygen bound to hemoglobin in a blood sample that contains 10g Hg/100mL of blood (assuming that hemoglobin is 100% saturated)?

- a. 13.4 ml O_2 /100ml blood
- b. 14.3 ml O_2 /100ml blood
- c. 13.7 ml O_2 /100ml blood
- d. 17.3 ml O_2 /100ml blood
- e. 20.1 ml O_2 /100ml blood

✓ If you face this kind of question again – multiply the given Hg concentration (in grams) by 1.34 and you will get the answer)

44. Stroke volume increases with all of the following except:

- a. Digitalis treatment of congestive heart failure
- b. Increased venomotor tone
- c. Administration of β_2 blocker
- d. Administration of β_1 agonist

- e. Administration of an arteriolar vasodilator
45. Which of the following factors **does not** increase the contractile state of the heart?
- a. β 1-sympathetic stimulation
 - b. Increased extracellular calcium concentration
 - c. Increased extracellular sodium concentration
 - d. Cardiac glycosides
 - e. Increased heart rate
46. During which phase of the cardiac cycle are **all valves closed**?
- a. Isovolumetric contraction phase
 - b. Atrial systolic phase
 - c. Ejection phase
 - d. Rapid ventricular filling phase
 - e. Atrial systolic phase
47. What is the amount of physically dissolved oxygen in a normal arterial blood with a partial pressure of 100 mm Hg?
- a. 0.3 mL O₂/100 mL of blood
 - b. 0.2 mL O₂/100 mL of blood
 - c. 0.1 mL O₂/100 mL of blood
 - d. 0.4 mL O₂/100 mL of blood
 - e. 0.5 mL O₂/100 mL of blood
- ✓ This is not a calculation. A is the only answer always.
48. Which of the following **is correct** regarding saliva secretion?
- a. It has a pH of 3.5 to 4.5
 - b. Centroacinar cells secrete hypotonic saliva
 - c. Sympathetic stimulation causes copious secretion of saliva
 - d. Potassium concentration is higher in saliva than plasma
 - e. Its secretion is facilitated by Mg²⁺
49. Which one of the following factors increases glomerular filtration rate?
- a. Increase in Bowman's interstitial capillary pressure
 - b. Renal efferent vasodilation
 - c. Decrease in glomerular oncotic pressure
 - d. Constriction of mesangial cells of the glomerulus
 - e. Sympathetic stimulation of afferent arterioles
50. One of the following **is not** the function of the kidney:
- a. Regulation of H⁺ concentration in the extracellular fluid
 - b. Regulation of Na⁺ and K⁺ concentrations in the extracellular fluid
 - c. Regulation of glucose concentration in the extracellular fluid
 - d. Regulation of arterial blood pressure
 - e. Regulation of erythropoietin secretion
51. Which of the following factors **does not** increase urine excretion rate:
- a. Diabetes insipidus
 - b. Diabetes mellitus
 - c. Increased renal arterial pressure
 - d. Osmotic diuresis

- e. Sympathetic stimulation
52. Which of the following is a **major factor** for protection of the duodenal mucosa from damage by gastric acid?
- a. Pancreatic bicarbonate secretion
- b. The endogenous mucosal barrier of the duodenum
- c. Duodenal bicarbonate secretion
- d. Bicarbonate contained in bile
- e. Bicarbonate secretion by stomach
53. The secretion of all of the following adrenal cortex hormones is controlled mainly by adrenocorticotrophic hormone from the pituitary gland **except**:
- a. Adrenal androgens
- b. Glucocorticoids
- c. Mineralocorticoids
- d. Cortisol
- e. Melanocyte stimulating hormone
54. A 34-year-old female patient has an 8-year history of menorrhagia and anemia. She complains of tiredness, cold intolerance, constipation, and weight gain. Laboratory results indicate **low plasma T₄ levels** and **elevated plasma TSH levels**. Which of the following is the most likely diagnosis?
- a. Grave's disease
- b. Thyroid adenoma
- c. Hypothyroidism of hypothalamic origin
- d. Primary hypothyroidism
- e. Primary hypoparathyroidism
55. Testicular temperature adjusting structures which is useful for optimal sperm synthesis
- a. Undescended testis
- b. Presence of adipose tissue in the scrotal skin
- c. Absence of cremaster muscle
- d. Abundant sweat glands in the scrotal skin
- e. Presence of hair in the scrotal skin
56. Important brain stem pain modulating neurons are:
- a. Superior colliculus and inferior colliculus
- b. Reticular formation and red nucleus
- c. Locus coeruleus and ventral tegmentum
- d. Raphe nucleus and periaqueductal gray area
- e. Red nucleus and substantia nigra
57. Hormones essential for lactogenesis and milk ejection, respectively, are:
- a. Prolactin and oxytocin
- b. Estrogen and progesterone
- c. Oxytocin and prolactin
- d. FSH and LH
- e. LH and FSH
58. The key limbic connections that operate as reward and addiction pathway is
- a. Ventral tegmental area and nucleus accumbens
- b. Substantia nigra

- c. Septal nuclei
 - d. Prefrontal cortex
 - e. Amygdala
59. In the brain EEG waves, the state of heightened brain activity is represented by:
- a. Alpha wave (Berger, synchronized)
 - b. Beta wave (Desynchronized)
 - c. Delta waves
 - d. Theta waves
 - e. Delta max waves
60. The neuronal components required for wakefulness excitatory activities
- a. Descending modulatory system
 - b. Paraventricular nuclei of thalamus
 - c. Ascending reticular activating system
 - d. Sensory vagus nerve
 - e. Anterior thalamic nuclei
61. The nigrostriatal dopaminergic system regulates
- a. Lactation
 - b. Voluntary movement
 - c. Cortical excitability
 - d. Sleep and wake cycles
 - e. Equilibrium
62. Which of the following is **not correct** regarding arachidonic acid? It is
- a. A 20-carbon fatty acid
 - b. Not synthesized in the body
 - c. A precursor of prostaglandins
 - d. Derived from an essential fatty acid
 - e. An unsaturated fatty acid
63. Disulfiram decreases catecholamine synthesis by:
- a. Stimulating dopamine β -hydroxylase
 - b. Inhibiting dopamine β -hydroxylase
 - c. Increasing catabolism of catecholamines
 - d. Stimulating dopa decarboxylase
 - e. Inhibiting MAO
64. Which of the following **is not** involved in oxidative phosphorylation?
- a. Oxygen serving as terminal electron acceptor
 - b. Proton pumping from matrix to intramembrane space
 - c. Electron pumping from matrix to intramembrane space
 - d. Electron transport across electron transport chain
 - e. Flow of protons through ATP synthase
65. Which one is **wrong** about conversion of pyruvate to oxaloacetate?
- a. It involves biotin as coenzyme
 - b. It takes place in the mitochondria
 - c. It requires fixation of carbon dioxide
 - d. It occurs at bypass point during gluconeogenesis

- e. It takes place in the cytosol
66. Which of the following is a donor of labile methyl group?
- a. Methionine
 - b. S-Adenosyl methionine
 - c. Histidine
 - d. Methylmalonyl CoA
 - e. Tetrahydrofolate
67. Which of the following lipoprotein plays a role in the development of atherosclerosis?
- a. HDL
 - b. LDL
 - c. VLDL
 - d. IDL
 - e. Chylomicrons
68. Familial hypercholesterolemia is caused by
- a. Defective lipoprotein lipase
 - b. Defective ACAT
 - c. Defective HMG-CoA reductase
 - d. Defective LDL receptors
 - e. Defective HDL particles
69. Enzymes that catalyze the bulk of the nitrogen flow from amino acids to ammonia are:
- a. Transaminases and dehydratases
 - b. Transaminases and glutamate dehydrogenase
 - c. Transaminases and amino acid oxidases
 - d. Dehydratases and glutamate dehydrogenase
 - e. Dehydratases and amino acid oxidases
70. Which enzyme is useful for the treatment of leukemia?
- a. α -chymotrypsin
 - b. Asparaginase
 - c. Hyaluronidase
 - d. Streptokinase
 - e. Alkaline phosphatase
71. A 71-year-old girl was brought to emergency department with complaints of severe polyuria and polydipsia. Laboratory investigation reveals ketone bodies in her urine. Which of the following is the most likely source of ketone bodies?
- a. Fatty acid breakdown
 - b. Glycogenolysis
 - c. Protein synthesis
 - d. Glycolysis
 - e. Gluconeogenesis
72. Which of the following statements **is not** correct?
- a. A female with Turner syndrome has an aneuploid karyotype
 - b. Chromosome number 22 has the lowest amount of DNA of all the autosomal chromosomes
 - c. Over a third of spontaneous abortuses contain gross chromosome abnormalities

- d. Each chromosome is a single large DNA with proteins
 - e. DNA is able to form 4-stranded molecular structures containing guanine
73. Which of the following statements is **true** about nucleotide excision repair of damaged DNA?
- a. DNA glycosylases recognize and remove the abnormal mutated nucleotide
 - b. DNA polymerases-delta and epsilon carry out nucleotide excision repair
 - c. Werner syndrome is caused by a genetic defect in the gene encoding polymerase
 - d. A defective helicase involved in nucleotide excision repair is cause of ataxia telangiectasia
 - e. Genetic defects in nucleotide excision repair enzymes cause xeroderma pigmentosum
74. The enzyme that catalyzes the conversion of L-Dopa to dopamine is
- a. Tyrosine hydroxylase
 - b. DOPA decarboxylase
 - c. Dopamine β - hydroxylase
 - d. Catechol-O-Methyl transferase
 - e. Phenylalanine hydroxylase
75. DNA oxidation by reactive oxygen species can lead to:
- a. Mutation and initiation of cancer
 - b. DNA damage and cell differentiation
 - c. Improved transcription and cell division
 - d. DNA damage and cell fusion
 - e. Improved transcription and cell differentiation
76. Maple Syrup urine disease is due to deficiency of:
- a. Branched-chain amino acid transaminases
 - b. alpha-keto acid dehydrogenase
 - c. Acyl CoA esterase
 - d. alpha-keto acid reductase
 - e. Xanthine oxidase
- ✓ They inserted branched chain in A to purposely confuse. Maple syrup urine disease is due to deficiency of (branched) **alpha keto acid dehydrogenase**. It leads to accumulation of branched chain amino acids (leucine, Isoleucine and valine) in the blood.
77. Which of the following Immunoglobulin is increased in allergy?
- a. IgA
 - b. IgG
 - c. IgD
 - d. IgE
 - e. IgM
78. Which of the following statement is **not correct** about receptors?
- a. Peptide hormone receptors are situated in the cell membrane
 - b. Receptors contain a hollow or cleft on their surface which is known as binding site
 - c. Receptors bind chemical messengers such as growth factors or hormones
 - d. Receptor binds all types of ligands during hormone action
 - e. Most cell surface receptors communicate intracellularly through second messengers
79. Which of the following are features of slow fibers?
- a. Abundant mitochondria, extensive capillary supply, low myoglobin, and fatigue resistant
 - b. Abundant mitochondria, extensive capillary supply, high myoglobin, and easily fatigued

- c. Abundant mitochondria, extensive capillary supply, high myoglobin, and fatigue resistant
 - d. Scarce mitochondria, extensive capillary supply, high myoglobin, and fatigue resistant
 - e. Scarce mitochondria, extensive capillary supply, low myoglobin and fatigue resistant
80. Muscle fatigue is due to non-availability of
- a. Calcium
 - b. ATP
 - c. Actin binding site
 - d. Mg as a co-factor
 - e. Lactic acid
- ✓ Primarily, it is due to non-availability of ATP.
81. Components of complement system are activated by:
- a. Microsomal hydroxylation
 - b. Phosphorylation
 - c. Glycosylation
 - d. Proteolysis
 - e. Ubiquitination
82. The primary fuel sources of cardiac muscle are:
- a. Fatty acid and ketone bodies
 - b. Glycogen and glucose
 - c. Protein and ketone bodies
 - d. Glucose and ketone bodies
 - e. Protein and glucose
83. The risk of ischemic heart disease is decreased by:
- a. 40% of total caloric intake as fat
 - b. Diet high in saturated fat
 - c. Food rich in cholesterol
 - d. Fish oil
 - e. Fat found in beef products
84. Which of the following statements is correct about phenylketonuria?
- a. Tyrosine accumulates in plasma due to enzyme block
 - b. Neonatal screening is done 2-3 months after birth
 - c. Benign hyperphenylalaninemia does not respond to dietary restriction
 - d. Infants born with PKU suffer from congenital abnormalities
 - e. Hyperphenylalaninemia due to defect in bipterin synthesis responds to treatment
85. Hydrophilic hormones:
- a. Bind to specific receptors on the cell surface and act through second messengers
 - b. Include thyroid and steroid hormones
 - c. Are transported in circulation in association with proteins
 - d. Directly bind to G proteins in the cell membrane
 - e. Have a long duration of action
86. Which one is the best source of energy for cardiomyocytes in resting condition?
- a. Free fatty acid
 - b. Glucose
 - c. Ketone bodies

- d. Amino acids
 - e. Glycerol
87. Which of the following statements is **correct** regarding adenylate cyclase and G-protein?
- a. G-protein alone directly activates adenylate cyclase
 - b. ATP and G α subunits activate adenylate cyclase
 - c. GTP- α subunit activates adenylate cyclase
 - d. G $\beta\gamma$ subunits dissociate from the G protein and then activates adenylate cyclase
 - e. G $\alpha\gamma$ GDP subunit activates the adenylate cyclase
88. A 56-year-old male died of myocardial infarction. Autopsy was done and microscopic examination revealed irregular fibrous bundles replacing the cardiac muscles. How old is the lesion?
- a. 4-12hrs
 - b. 1-3days
 - c. 3-7days
 - d. 1-3weeks
 - e. 3-6weeks
89. Following an intradermal injection of purified protein derivative (PPD), an area of induration with erythema developed peaking in 24-72hrs. What type of hypersensitivity reaction is this?
- a. Type I
 - b. Type II
 - c. Type III
 - d. Type IV
 - e. Type V
90. Which of the following is the **most likely** outcome of untreated myeloid leukemia?
- a. Secondary marrow fibrosis
 - b. Transformation to acute leukemia
 - c. Stable disease with no progression
 - d. Spontaneous disease regression
 - e. Aplastic anemia
91. The most important feature that helps distinguish malignant tumor is:
- a. Rate of growth
 - b. Local invasion
 - c. Metastasis
 - d. Microscopic features
 - e. Clinical and gross features
92. One of the main domains of pathology used by pathologists to make diagnosis of diseases is by the study of:
- a. Etiology
 - b. Pathogenesis
 - c. Structural changes
 - d. Functional changes
 - e. Clinical manifestations of disease
93. Next to breast cancer, the most common forms of cancer in Ethiopia is:
- a. Leukemia
 - b. Cervical cancer

- c. Prostate cancer
 - d. Esophageal cancer
 - e. Lymphoma
94. Primary bone tumor arising in the epiphysis of long bones close to the articular cartilage in patients between 20-40 years of age with no predilection, radiologically revealing characteristic "Soap bubble" appearance is:
- a. Chondroma
 - b. Clear cell chondrosarcoma
 - c. Chondroblastoma
 - d. Giant cell tumor
 - e. Chondromyxoid fibroma
95. Pathologic changes in osteoarthritis **does not** include:
- a. Vertical and horizontal fibrillation and cracking of the articular cartilage
 - b. Portions of the cartilage are sloughed off and dislodged
 - c. Grossly the surface becomes necrotic and hemorrhagic with abscess and granuloma
 - d. Rebuttredding and sclerosis of the underlying cancellous bone occurs
 - e. Matrix water increment and proteoglycans decrement
96. The histologic hallmark of viral pneumonia is:
- a. The interstitial nature of the inflammatory reaction
 - b. Edema of alveolar space
 - c. Fibrin in alveolar space
 - d. Coating of alveolar walls by pink, hyaline membrane
 - e. Pleural effusion
97. Which of the following characteristics of p53 best enables it to prevent the initiation of cancer?
- a. As a transcription factor, p53 promotes production of proteins that stimulate the cell cycle
 - b. p53 prevents proliferation of cells with damaged DNA
 - c. p53 inhibits cells from triggering apoptosis
 - d. p53 stimulates synthesis of DNA repair enzymes that replace telomere sequence lost during cell division
 - e. p53 is the upstream molecular governor preventing apoptosis
98. Which of the following conditions results in eccentric hypertrophy of the left ventricle, dilation of the left atrium, and concentric hypertrophy of the right ventricle?
- a. Aortic stenosis
 - b. Pulmonary hypertension
 - c. Mitral regurgitation
 - d. Mitral stenosis
 - e. Aortic insufficiency
99. Which of the following is **not correct** about pleomorphic adenoma?
- a. It is the commonest tumor of the parotid gland
 - b. Epithelial and mesenchymal differentiation is seen under microscope
 - c. It is painless and slowly growing mass
 - d. With time, malignant transformation is possible
 - e. It is a malignant tumor

100. A 55-year-old female came with engorged veins that radiate to the umbilicus. Which of the following is the **least expected** condition in this patient?
- a. Spider angioma
 - b. Leg edema
 - c. Jaundice
 - d. Thrombosis
 - e. Esophageal varices
- ✓ If there is a caput medusa (portal HTN), the veins radiate from umbilicus...this tells us that there is no portal HTN and hence no esophageal varices as well.
101. Perineural invasion is common feature found in:
- a. Acinic cell carcinoma
 - b. Adenoid cystic carcinoma
 - c. Muoepidermoid carcinoma
 - d. Pleomorphic adenoma
 - e. Malignant mixed tumor
102. A type of pneumoconiosis most commonly predisposing to a number of cancers is:
- a. Coal dust
 - b. Silicosis
 - c. Asbestosis
 - d. Berylliosis
 - e. Cotton dust
103. Which of the following is **correct** regarding kidney stones?
- a. They are more common in women than in men
 - b. Uric acid stones are the commonest type
 - c. Hypercalcemia is the underlying cause in majority of the cases
 - d. Urinary tract infections play role in the formation of struvite stones
 - e. Calcium stones commonly form large, 'staghorn' calculi
104. Which one of the following is correct about trypanosomiasis?
- a. Trypanosoma cruzi is the cause of sleeping sickness in the old world
 - b. The main manifestation of Chagas disease is sleeping sickness
 - c. Trypanosoma brucei gambiense infection is more acute and virulent than rhodesiense
 - d. African trypanosomiasis stage-1 disease is a systemic illness without Cranial nerve involvement
 - e. Severe myocarditis develops in the majority of acute Chagas disease
105. Cushing disease is caused by:
- a. Adrenal cortical carcinoma
 - b. Pheochromocytoma
 - c. B-cell adenoma of the anterior pituitary
 - d. Small cell carcinoma of the lung with hormone production
 - e. Tumor of the hypothalamus producing ACTH-releasing hormone
106. Which statement is **correct** about prostatic carcinoma?
- a. The incidence is high in hyper-estrogenic state
 - b. Metastatic deposits to the bones are often sclerotic
 - c. It does not occur outside the peripheral zone

- d. It occurs at periurethral zone
 - e. It follows benign prostatic hyperplasia
107. A 13-year-old boy, who had sore throat two weeks back came with complaints of reddish discoloration of urine. Which of the following is least likely to be seen in this case?
- a. Red blood cell casts in urine
 - b. Dysmorphic RBCs in the urine
 - c. Complement components in the glomeruli
 - d. Linear deposits on immunofluorescence
 - e. Decreased serum complement levels
- ✓ **Granular**, not linear deposits are seen. If there are immune complexes – granular pattern always. Linear deposits are seen in Anti GBM diseases.
108. Which of the following cells is involved in the formation of blood-brain barrier?
- a. Astrocyte
 - b. Microglia
 - c. Ependymal cell
 - d. Endothelial cell
 - e. Oligodendrocyte
109. The most malignant brain tumor is:
- a. Meningioma
 - b. Ependymoma
 - c. Glioblastoma
 - d. Anaplastic astrocytoma
 - e. Oligodendroglioma
110. If you receive a pathology report stating cervical carcinoma *in situ*:
- a. It means there is definite invasion of the stroma
 - b. It is associated with low risk HPV subtypes
 - c. Less than 1/3 of the thickness of the epithelium is involved
 - d. Chronologically, it occurs after invasive carcinoma
 - e. The whole thickness of the cervical epithelium is dysplastic
111. A 15-year-old child who was not vaccinated developed meningitis and a gram-negative rod is seen in the cerebrospinal fluid. What is the most likely causative agent?
- a. *Listeria monocytogenes*
 - b. *Escherichia coli*
 - c. *Hemophilus Influenzae*
 - d. *Neisseria meningitidis*
 - e. *Streptococcus agalactiae*
- ✓ The fact that the child is unvaccinated tells us that it's *H. Influenzae* – but the age should have been from 3 mo – 2 yr → anyways the answer is C (*H. Influenzae*).
112. Parasites are dependent on their hosts for survival, multiplication and sustaining their life cycle because,
- a. Hosts are not hostile for parasites
 - b. Parasites lack virulence factors
 - c. Parasites are capable of colonizing and invading host tissue
 - d. Hosts provide developmental stimuli for parasites

- e. Parasites are capable of evading host immune responses
113. The term variolation refers to:
- a. A type of gene therapy
 - b. Inoculation of scab material into small skin wounds
 - c. The attenuation of virulent organisms
 - d. The removal of swab material from an individual
 - e. The generation of antibody variable regions
- ✓ Variolation means deliberate inoculation of small pox OR inoculation of scab material. This came on Blood and immunity final – and the answer was A and C → but here it is only B since the choice “inoculation of small pox is not available”.
114. Not visible
115. Which of the following is properly paired?
- a. Mycetismus – systemic fungal infection
 - b. Mycetismus – caused by secondary metabolite
 - c. Mycotoxicosis – caused by secondary metabolite
 - d. Mycosis – caused by fungal toxin
 - e. Mycetismus – caused by fungal allergen
116. Which of the following is a primary opportunistic fungus?
- a. Histoplasma capsulatum
 - b. Pneumocystis jiroveci
 - c. Coccidioides immitis
 - d. Candida albicans
 - e. Malassezia fufur
117. Which serovars of Chlamydia trachomatis are responsible for causing pneumonia?
- a. Serovars A and B
 - b. Serovars D-K
 - c. Serovars L1-L3
 - d. Serovars A, Ba, D, E
 - e. Serovars D, L1, L2
118. Which of the following respiratory tract infections has a classical bull neck appearance?
- a. Pertussis
 - b. Diphtheria
 - c. Epiglottitis
 - d. Tonsillitis
 - e. Sinusitis
119. Sterile pyuria is caused by:
- a. Enterobacter aerogens
 - b. Escherichia coli
 - c. Candida albicans
 - d. Pseudomonas aeruginosa
 - e. Mycobacterium tuberculosis
120. Which of the following contributes to tissue invasiveness in streptococcal infection?
- a. Erythrogenic toxin
 - b. Exotoxin A

- c. Hyaluronidase
 - d. Coagulase
 - e. Exfoliative toxin
121. Which of the following enteric bacteria is associated with Guillain Barre Syndrome?
- a. Shigella dysenteriae
 - b. Yersinia enterocolitica
 - c. Campylobacter jejuni
 - d. Salmonella typhi
 - e. Campylobacter coli
122. Which of the following is **not correct** about CNS toxoplasmosis?
- a. It is an opportunistic infection associated with full blown AIDS stage
 - b. It often results from the reactivation of tissue cysts
 - c. Definitive diagnosis requires CT, MRI, or other radiographic testing
 - d. Effective antiretroviral therapy has no impact on infection with Toxoplasma gondii
 - e. The level of CD4+ T cell may help monitor the resolution of the disease
123. In Borrelia recurrentis infection, the relapse of fever is due to:
- a. Reinfection
 - b. Antigenic variation
 - c. Immunodeficiency state
 - d. Treatment failure
 - e. Diagnosis failure
124. Which form of tularemia is characterized by regional lymphadenopathy and cutaneous ulcer?
- a. Ulceroglandular
 - b. Glandular
 - c. Oculoglandular
 - d. Typhoidal
 - e. Pulmonary
125. Which of the following statements is **correct** about poliovirus?
- a. It consists of ten serotypes
 - b. Majority of infections cause paralytic form
 - c. Sabin vaccine is safe for HIV infected individuals
 - d. It has animal reservoirs
 - e. It is inactivated by chlorine and UV light
126. The possible explanation for splenomegaly and ascites in chronic stage of schistosomiasis is:
- a. Toxin production by adult parasite
 - b. Formation of granuloma
 - c. Penetration of cercariae
 - d. Presence of mother sporocysts
 - e. Migratory schistosomula
127. Which of the following pathological changes are pertinent in the clinical presentation of visceral leishmaniasis?
- a. Abscess formation in the liver
 - b. Splenomegaly without hepatomegaly
 - c. Pancytopenia and thrombocytopenia

- d. Neoplasm in the liver and spleen
 - e. Low parasitic load in late stages of the disease
128. Which of the following clinical manifestations of syphilis is a tertiary stage:
- a. Chancre
 - b. Mucocutaneous rash
 - c. Soft chancre
 - d. Condyloma lata
 - e. Neurosyphilis
129. *Trichomonas vaginalis*:
- a. Infects males more frequently than females
 - b. Infects only humans
 - c. Is an ecto-parasite
 - d. Is diagnosed by urine sample
 - e. Is diagnosed by presence of cysts on direct microscopy
130. The causative agent for break bone fever is:
- a. Yellow fever virus
 - b. O'nyong-nyong virus
 - c. Dengue fever
 - d. Chikungunya virus
 - e. Ross river virus
131. Which drug is effective in the treatment of both absence and clonic seizure?
- a. Ethosuximide
 - b. Phenytoin
 - c. Carbamazepine
 - d. Valproic acid
 - e. Phenobarbital
132. Not visible
133. The preferred drug for iron deficiency anemia due to malnutrition is:
- a. Oral ferrous sulfate
 - b. Parenteral cyanocobalamin
 - c. Oral folic acid
 - d. Parenteral iron
 - e. Oral cyanocobalamin
- ✓ Oral sulfate has GI upsets
134. Which of the following drugs is an antagonist of warfarin?
- a. Aspirin
 - b. Dicumarol
 - c. Protamine sulfate
 - d. Vitamin K
 - e. Folic acid
135. In zero order kinetics of drug elimination:
- a. Elimination rate is independent of the dose administered
 - b. $t_{1/2}$ is constant despite the rising drug concentration
 - c. Enzyme mediated metabolic reactions become saturated

- d. Uniform increase in dose may lead to proportional increase in plasma
 - e. Passive diffusion processes will not get saturated
136. Which of the following is amide type of local anesthetic?
- a. Cocaine
 - b. Procaine
 - c. Tetracaine
 - d. Benzocaine
 - e. Lidocaine
137. Which of the following is indirect cholinomimetic drug?
- a. Methacholine
 - b. Carbachol
 - c. Pilocarpine
 - d. Organophosphates
 - e. Acetylcholine
- ✓ Organophosphates **are irreversible indirect cholinomimetic drugs (AChE inhibitors)**. While Physostigmine and neostigmine are reversible indirect cholinomimetics (AChE inhibitors). Methacholine, carbachol, pilocarpine etc.... are direct cholinomimetics.
138. Antihypertensives drugs act by:
- a. Inhibiting reuptake of noradrenaline
 - b. Activating alpha1-receptor on blood vessels
 - c. Blocking beta-receptors
 - d. Blocking central alpha two receptors
 - e. Activating angiotensin receptors
139. Which of the following drugs belongs to class III antiarrhythmic agents?
- a. Quinidine
 - b. Lidocaine
 - c. Flecainide
 - d. Amiodarone
 - e. Procainamide
140. Which nucleoside bases in DNA double-helix strands is cross linked by alkylating agents to stop tumor growth?
- a. Guanine
 - b. Adenine
 - c. Thymine
 - d. Cytosine
 - e. Uridine
141. Which of the following is a stimulant (irritant) laxative?
- a. Bisacodyl
 - b. Methylcellulose
 - c. Mannitol
 - d. Magnesium sulphate
 - e. Mineral oil
142. Which of the following reactions is **irreversible**?
- a. Fatty liver in chronic alcoholics

- b. Respiratory problem in chronic smokers
 - c. Constipation precipitated in chronic khat chewers
 - d. Hypertension in patients who have taken large dose of amphetamine
 - e. Central depression in patients treated with large dose of phenobarbital
143. Anticholinergic toxidrome is characterized by:
- a. Tachycardia, mydriasis, constipation
 - b. Bradycardia, mioiosis, salivation
 - c. Hypothermia, muscle fasciculation followed by paralysis
 - d. Diaphoresis, tachycardia, blurred vision
 - e. Hyperactive bowel, pinpoint pupil, bradycardia
144. Which of the following psychotropic drugs must be monitored for its hematotoxic effects?
- a. Risperidone
 - b. Clozapine
 - c. Haloperidol
 - d. Lithium carbonate
 - e. Mirtazapine
145. Question number is missing for some reason
146. Which antimalarial agent is most commonly associated with acute hemolytic reaction in patients with glucose-6-phosphate dehydrogenase deficiency?
- a. Chloroquine
 - b. Quinine
 - c. Mefloquine
 - d. Artesunate
 - e. Primaquine
147. Which of the following drugs is specifically used for the treatment of lepromatous leprosy?
- a. Ethionamide
 - b. Rifampicin
 - c. Cycloserine
 - d. Clofazimine
 - e. Dapsone
148. Which of the following is a sign and symptom of chronic use of exogenous thyroxine (T4)?
- a. Bradycardia
 - b. Dry, puffy skin
 - c. Large tongue and drooling of the eyelids
 - d. Lethargy, sleepiness
 - e. Weight loss
149. Which drug will be prescribed to soften the cervix of gestation?
- a. Methylergonovine
 - b. Dinoprostone
 - c. Bethamethasone
 - d. Terbutaline
 - e. Aspirin
150. Which of the following penicillins is effective in pseudomonas infections?
- a. Ampicillin

- b. Dicloxacillin
- c. Piperacillin
- d. Penicillin V
- e. Penicillin G